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Quiz 01 2021 Fall

1. Thin Client Model is one of the configurations of _Model.

a) Data-centered

b) client server

c) Layered

d) Reference

2. STL Stands for_ _

a) standard template library

b) Standard Type Link

c) Standard Tempo Line

d) Standard Type Link

3. The three basic principles that guide maintainability are:

Clarity

Enhancement

Flexibility

Simplicity

a) (a), (b) and (c)

b) (a), (c) and (d)

c) (b), (c) and (d)

d) All of above

4. Goto statements violate the idea of

a) object-oriented code

b) structured code

c) control structure

d) repetition structure

5. A function should not be larger in any case and should not exceed in length

a) Half page

b) One page

c) Two pages

d) Three Pages

6. Data Centered Architectural Style is also called_

a) Repository model

b) Client Server model

c) Sub System model

d) Reference model

7. Global variables in C++ should always be referred to by using the

a) ::operator

b) :operator

c) Without an operator

d) None of the given

8. There are layers in OSI reference model.

a) 5

b) 6

c) 7

d) 8

9. Fat Client Model is one of the configurations of _____ Model.

a) Data-centered

b) Layered

c) Reference

d) Client Server

10. Dynamic process model shows the process _____ of the System.

a) Components

b) Objects

c) Structure

d) Linkage

11. Which of the following shows a commented statement in C++

a) # Ans = first + second

b) // Ans = first + second

c) \\ Ans = first + second

d) /# Ans = first + second

12. _____ architectural model is widely used in Mainframe application.

a) Client-Server Model

b) Repository Model

c) Filter Model

d) Layered Model

13. Variables should be initialized where they are _____ and they should be declared in the scope possible.

- a) defined and smallest
- b) declared and medium
- c) defined and medium
- d) declared and smallest**

14. Identifier names also play a significant role in enhancing the _____ of a program,

- a) Writ ability
- b) Readability**
- c) Reliability
- d) All of the given choices

15. MVC stands for_

- a) Model View Controller**
- b) Modern View Center
- c) Model View Center
- d) Modern View Controller

16. Which of the following is not a possible server in client server environment?

- a) Database Server
- b) Transaction Server
- c) File Server
- d) Time Server**

17. In case of using unrelated operators in a single expression, _____ would be the best choice to prevent the logical errors.

a) Comments

b) Indents

c) parenthesis

d) Short Cuts

18. In case of header files, construction is to avoid _____ errors, the construction should appear in the top of the file (before the file header).

a) Compilation

b) run time

c) logical

d) All of the given

19. A self-documenting code is a code that explains itself without the need of comments and extraneous documentation, like__

a) Flowcharts

b) UML diagrams

c) Process-flow state diagrams

d) All of the given choices

20. Which one of these represents the Krutchen's 4+1 architectural view model

a) Logical view, Process view. Physical view. Development view. Use case view

b) Logical view. Dynamic view. Physical view. Development view. Use case view

c) Logical view. Process view, Physical view. Development view. Sequence view

d) Dynamic view, Process view. Physical view, Development view. Use case view

21. Client server is a _____ system model.

a) Integrated

b) Distributed

c) Heterogeneous

d) Homogeneous

22. Vertical partitioning is also known as_

a) Balancing

b) Mutating

c) Parallelizing

d) Factoring

23. "is" prefix should be used for variables and methods.

a) General

b) Boolean

c) Constant

d) None of the given

24. Thin client model places a heavy processing load on _

a) Only server

b) Only network

c) Both server and network

d) Neither server not network

25. C++ header files should have the extension . Source files can have the extension_

a) .c and .h

b) .cc and .h

c) .h and .cpp

d) All of the given

26. Which of following is among ten things, which the basic template of GOF design pattern includes.

a) IQ

b) context

c) Performance

d) Requirements

27. Vertical partitioning divides the architecture application from a making perspective.

a) decision

b) Design

c) Conclusion

d) Move

28. Issues like system performance, availability, scalability, and security are analyzed in_

a) Logical View

b) physical view

c) Code View

d) Concurrency View

29. Class variables should never be declared public. Public variables violate which of the following:

a) Information hiding

b) Encapsulation

c) Information hiding and Encapsulation

d) None of the give

30. Charles Simonyi first discussed Hungarian Notation, He was of_

a) Microsoft

- b) IBM
- c) Dell
- d) Cisco

31. Patterns are devices that allow programs to share knowledge about their _

- a) Code

b) design

- c) Analysis
- d) Implementation

32. Process view in Krutchen's 4+1 architectural view model captures_

- a) Object model of the design

b) Concurrency and synchronization aspects of the design

- c) Mapping(s) of the software onto the hardware and reflects its distributed aspect

33. In the N-Tire architecture, the idea is to enhance scalability and _ by disturbing both the data and the application using multiple server machines

- a) Usability

b) performance

- c) Interoperability
- d) Integrity

34. Names representing methods and functions should be _ and written in mixed case starting with _ case

- a) Noun lower

b) Verb lowerpage

- c) Noun upper
- d) Verb upper

35. Floating point constant should always be written with decimal point and at

a) one decimal

b) Two decimals

c) Three decimals

d) none of the given

36. Static structural model shows the major system _

a) Leaks

b) Components

c) Activates

d) Objectives

37. Distributing the responsibilities to different subsystems so that we get a software system which is easy to maintain, is called _ the architecture

a) Subtracting

b) partitioning

c) Cloning

d) Balancing

38. It ensures that a class only has one instance and provides a global point of access to it

a) singleton pattern

b) Observer pattern

c) Real pattern

d) None of the given

39. N-tier architecture stems from the struggle to find a _____ between the fat-client architecture and the thin client architecture

- a) Concurrency
- b) Distribution point
- c) middle ground**
- d) Similarity

40. In client server model, the application is modeled as a set of _____ that are provided by servers.

- a) Request
- b) Protocols
- c) service**
- d) Requirements

41. Maintainability and portability issues are discussed in _____

- a) Functional view
- b) Physical view
- c) code view**
- d) Logical view

42. In case of a file server, client request selected records from a _____ and the server transmits recorded to client over the network

- a) Local memory
- b) Network
- c) Database
- d) file**

43. Physical view in Krutchen's 4+1 architectural view model captures _____.

a) Object model of the design

b) Concurrency and synchronization aspects of the design

c) Mapping(s) of the software onto the hardware and reflects its distributed aspect

44. Logical view in Krutchen's 4+1 architectural view model captures _____.

a) Concurrency and synchronization aspects of the design

b) Mapping(s) of the software onto the hardware and reflects its distributed aspect

c) Object model of the design

d) Static organization of the software in its development environment

45. When programmers use shortcuts and cryptic codes in their program, _____ becomes a major problem.

a) Maintenance

b) Reusability

c) Availability

d) Readability

46. Which one is correct?

a) double total = 0.5;

b) double total = .5;

c) double total = .50;

d) all of the given

47. "Description of communicating objects and classes that are customized to solve a general design problem in a particular context" is called _____.

a) Design Pattern

b) System Pattern

c) System Design

d) Design Method

48. Development view in Krutchen's 4+1 architectural view model captures

- a) Object model of the design
- b) Concurrency and synchronization aspects of the design
- c) Mapping(s) of the software onto the hardware and reflects its distributed aspect
- d) Static organization of the software in its development environment**

49. Client-server model tries to _ data and processing.

a) Distribute

- b) Merge
- c) Clone
- d) Proceed

50. Which indent size eliminates the chance of code lines splitting?

a) 2

- b) 4
- c) 3
- d) 6

51. Which one of these is a correct short cut used by the programmer?

a) $x*=a$

- b) $x=*a$
- c) $x*a=$
- d) $=a*x$

52. _ was the first pure Object-Oriented language in which observer pattern was used in implementing its Model View Controller pattern

a) Smalltalk

- b) PASCAL
- c) JAVA
- d) C++

53. Identify the TRUE statement

- a) Portability effects Security
- b) Size of data types vary from one machine to other**
- c) Size of control structures vary from one machine to other
- d) None of the given options

54. which of the following statements are same in output 1) $a = a \gg 2$

2) $a = a/4$

3) $a = a * 2$

a) (1) and (3) only

b) (2) and (3) only

c) (1) and (2) only

d) All produce the same result

56. Code should not be:

a) Commented

b) indented

c) cryptic

d) aligned

57. If $!(\text{block} < \text{activeBlock})$ is equivalent to:

a) if $(\text{block} < \text{activeBlock})$

b) if $((\text{block} == \text{activeBlock}))$

c) if $((\text{block} \geq \text{activeBlock}))$

d) None of the given

58. The form for (; ;) should be used for

a) Nested loop

b) Empty loop

c) More than 1000 iteration

d) Less than 1000 iterations

59. The clients and servers, in a client server environment, are connected through ____.

a) Framework

b) Interface

c) Middleware

d) Groupware

60. provides a unified interface to a set of intelligence in a sub-system

a) Observer pattern

b) Singleton pattern

c) Façade pattern

d) All of the above

61. Complex expressions:

a) Make the code easy to modify

b) Make the code difficult to modify

c) Make the code easy to understand

d) Does not affect the understandability

62. Which of the following is used for multi-level commenting?

- a) // Comment
- b) /*Comment*/**
- c) (Comment)
- d) */Comment/*

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Question No: 1 (Marks: 1) - Please choose one

By following modern system engineering practices simulation of reactive systems is no longer necessary

- ▶ True
- ▶ **False** [click here for detail](#)

Question No: 2 (Marks: 1) - Please choose one

The state transition diagram

- ▶ depicts relationships between data objects
- ▶ depicts functions that transform the data flow
- ▶ indicates how data are transformed by the system
- ▶ **indicates system reactions to external events** [click here for detail](#)

Question No: 3 (Marks: 1) - Please choose one

Control flow diagrams are

- ▶ **needed to model event driven systems.** [Click here for detail](#)
- ▶ required for all systems
- ▶ used in place of data flow diagrams
- ▶ useful for modeling user interfaces

Question No: 4 (Marks: 1) - Please choose one

A change becomes ----- because of close presence of data and functions

- ▶ **Localized** (page 81)
- ▶ Private
- ▶ Global
- ▶ Accessible

Question No: 5 (Marks: 1) - Please choose one

A complex System evolves from a

- ▶ **smaller system** page 83 [click here for detail](#)
- ▶ medium system
- ▶ bigger system
- ▶ non of the given

Question No: 6 (Marks: 1) - Please choose one

Coupling is a measure of -----of a module or component.

- ▶ **Independent** (Page 71)
- ▶ Dependent
- ▶ Closeness
- ▶ All of the given

Question No: 7 (Marks: 1) - Please choose one

Software Maintenance phase involves

- ▶ Debugging
- ▶ Adding new features
- ▶ making changes
- ▶ **all of the given** [click here for detail](#)

Question No: 8 (Marks: 1) - Please choose one

The hardest single part of building a software system is deciding precisely ----- to build.

- ▶ **what** (Page 17)
- ▶ How
- ▶ When
- ▶ Why

Question No: 9 (Marks: 1) - Please choose one

Interaction Diagrams depict the----- Behavior of the system

- ▶ Static
- ▶ **Dynamic** (Page 106)
- ▶ Active
- ▶ Non of the above

Question No: 10 (Marks: 1) - Please choose one

**In Sequence Diagrams the time required by the receiver object to process the message is denoted by an---
-----**

- ▶ **Activation Box** (Page 108)
- ▶ Simple Box
- ▶ Arrow
- ▶ Non of the above

Question No: 11 (Marks: 1) - Please choose one

A poorly designed interface can cause a user to make catastrophic errors is one of the motivations for GUI.

- ▶ **True** (Page 62)
- ▶ False

Question No: 12 (Marks: 1) - Please choose one
Identifying System Features include -----

- ▶ log important information
- ▶ conduct business
- ▶ analyze business results
- ▶ **all of the above** (Page 98)

Question No: 13 (Marks: 1) - Please choose one
Establishing responsibilities for objects includes

- ▶ Generalization Relationships
- ▶ Specialization Relationships
- ▶ **all of the above** (Page 86)
- ▶ identifying Association relationships

Question No: 14 (Marks: 1) - Please choose one
Which of the following is a fact finding method?

- ▶ Site visits
- ▶ **Prototyping**
- ▶ Study of similar systems
- ▶ All of given

Seven common fact-finding methods (Sampling, Research, Observation, Questionnaires, Interviews, Prototyping, Joint Requirements Planning) are introduced as a means to discover requirements.

Question No: 15 (Marks: 1) - Please choose one

Windows mobile is a popular mobile operating system which seen commonly on PDAs. Which of the following category pair(s) best describe(s) it?

- ▶ **Application software, embedded software**
- ▶ system software, web-based software
- ▶ application software, scientific software
- ▶ system software, embedded software

Question No: 16 (Marks: 1) - Please choose one
The Object Oriented approach

- ▶ **improves the reusability of codes.** (Page 83)
- ▶ makes objects less independent.
- ▶ increases testing time.
- ▶ complicates mapping of a system model to an implementation model.

The object-oriented approach improves the maintainability, reusability, and extensibility of the software [Click here for Detail](#)

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Question No: 1 (Marks: 1) - Please choose one

Most software continues to be custom built because

- ▶ Software is easier to build without using someone else's components.
- ▶ Off the shelf software components are not commonly available [Click here for detail](#)
- ▶ Component reuse is common in the software world
- ▶ Reusable components are too expensive to use

Question No: 2 (Marks: 1) - Please choose one

Which of the items listed below is not one of the software engineering layers?

- ▶ Tools
- ▶ Process
- ▶ Manufacturing [Click here for detail](#)
- ▶ Methods

Question No: 3 (Marks: 1) - Please choose one

Which of these people would not be likely to part of the FAST team?

- ▶ hardware and software engineers
- ▶ manufacturing representative
- ▶ marketing representatives
- ▶ senior financial officers **A PRACTITIONER'S APPROACH (Page 305)**

Question No: 4 (Marks: 1) - Please choose one

An state transition can only occur when triggered by a(n)_____

- ▶ actor
- ▶ collaboration attempt
- ▶ event
- ▶ none of the given

The evolution of the system depends on a sequence of state transitions triggered by a set of coming events.

Question No: 5 (Marks: 1) - Please choose one

A change becomes ----- because of close presence of data and functions

- ▶ Localized (page 81)
- ▶ Private
- ▶ Global
- ▶ Accessible

Question No: 6 (Marks: 1) - Please choose one

A process which does not take input is called

- ▶ **miracle process** Miracle": Process has output, but no input [Click here for detail](#)
- ▶ core process
- ▶ secondary process
- ▶ all of the above

Question No: 7 (Marks: 1) - Please choose one

Data Store notation in DFDs presents.

- ▶ Data input
- ▶ Data output
- ▶ **Data input and Data output** (Page 54)
- ▶ non of the given

Question No: 8 (Marks: 1) - Please choose one

Coupling is a measure of ----- of a module or component.

- ▶ **Independent** (Page 71)
- ▶ Dependent
- ▶ Closeness
- ▶ All of the given

Question No: 9 (Marks: 1) - Please choose one

If a problem is identified and fixed at a -----stage in the software development process, it will cost much more than if it was fixed at and ----- stage.

- ▶ eralier & lator
- ▶ after & before
- ▶ none of the above
- ▶ **later and earlier** (Page 17)

Question No: 10 (Marks: 1) - Please choose one

Once these requirements have been finalized, the -----process starts.

- ▶ **Construction** (Page 18)
- ▶ Development
- ▶ Maintenance phase
- ▶ Construction and Development

Question No: 11 (Marks: 1) - Please choose one

Identifying Whole-Part structures (Aggregations) means what are my -----

- ▶ **Components** (Page 95)
- ▶ Structures
- ▶ Objecs
- ▶ Classes

Question No: 12 (Marks: 1) - Please choose one

Data design actually begins during the creation of the analysis model, not the architectural model.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 13 (Marks: 1) - Please choose one

System Architecture is important to consider because it helps in making -----

- ▶ Mutual communication.
- ▶ Early design decisions.
- ▶ Reusable abstraction of a system.
- ▶ **All of the above** (Page 118)

Question No: 14 (Marks: 1) - Please choose one

Software architecture is "the ----- of the components of a program/system, their interrelationships, and principles and guidelines governing their design and evolution over time."

- ▶ combination
- ▶ collection
- ▶ **structure** (Page 117)
- ▶ unification

Question No: 15 (Marks: 1) - Please choose one

Cohesion is about making sure each component does _____ thing and does it well.

- ▶ three
- ▶ two
- ▶ infinite
- ▶ **one** [Click here for detail](#)

Question No: 16 (Marks: 1) - Please choose one

Following are some statements associated with data flow diagrams. Identify the correct statement from among them.

- ▶ DFDs are used to model complex interfaces.
- ▶ **DFDs are used to represent only functional processing, data stores and data movements between functions.** DFD over flow charts(Page 2)
- ▶ DFDs depict only processes which can be decomposed.
- ▶ DFDs do not show external data sources and external data sinks

MIDTERM EXAMINATION

2010

Question No: 1 (Marks: 1) - Please choose one

UML (unified modeling language) analysis modeling focuses on the_____.

- ▶ behavioral model and environment model.
- ▶ behavioral model and implementation model.
- ▶ user model and environmental model
- ▶ **user model and structural model** APRACTITIONER'S APPROACH (Page 604)

Question No: 2 (Marks: 1) - Please choose one

The data flow diagram must be augmented by descriptive text in order to describe the functional requirements for a software product.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

It is not possible to use ordinary data flow diagrams to model the functional requirements of real-time systems.

- ▶ True
- ▶ **False** [Click here for detail](#)

Question No: 4 (Marks: 1) - Please choose one

An object or class may further be classified on the basis of

- ▶ Behavior Driven attributes
- ▶ Data Driven attributes
- ▶ Responsibility Driven attributes
- ▶ **All of the above** (Page 85)

Question No: 5 (Marks: 1) - Please choose one

A public Interface provides a way for_____with other Classes.

- ▶ Communication
- ▶ **Accessibility** [Click here for detail](#)
- ▶ Reaching
- ▶ All of the above

Question No: 6 (Marks: 1) - Please choose one

DFD Notation contains

- ▶ Data Store
- ▶ Extenal Agents
- ▶ Processes
- ▶ **All of the given** (Page 51)

Question No: 7 (Marks: 1) - Please choose one

A complex System evolves from a

- ▶ **smaller system (Page 69)**
- ▶ medium sysetem
- ▶ bigger system
- ▶ non of the given

Question No: 8 (Marks: 1) - Please choose one

Requirement engineering mainly deals with the -----of the system

- ▶ **definition phase (Page 16)**
- ▶ development phase
- ▶ maintenance
- ▶ non of the above

Question No: 9 (Marks: 1) - Please choose one

Insufficient user involvement leads to -----products.

- ▶ **unacceptable (Page 19)**
- ▶ ambiguous
- ▶ dissatisfaction of customer to
- ▶ non of the above

Question No: 10 (Marks: 1) - Please choose one

Which one is not the purpose of Interaction Diagrams ?

- ▶ Model interactions between objects
- ▶ Assist in understanding how a system (a use case) actually works
- ▶ Identify responsibilities/operations and assign them to classes
- ▶ **Identify dependencies among objecs (Page 106)**

Question No: 11 (Marks: 1) - Please choose one

Collaboration diagrams have basically two types of components: objects and -----

- ▶ **messages (Page 111)**
- ▶ methods
- ▶ classes
- ▶ non of the above

Question No: 12 (Marks: 1) - Please choose one

The scope description establishes the----- between the system we are developing and everything else in the universe

- ▶ **Boundary (Page 31)**
- ▶ Balance
- ▶ Constraint
- ▶ None of the given

Question No: 13 (Marks: 1) - Please choose one

Which one is not a type of messages which Sequence Diagrams Depict

- ▶ Synchronous
- ▶ Asynchronous
- ▶ Create
- ▶ Update (Page 108)

Question No: 14 (Marks: 1) - Please choose one

What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?

- ▶ Security
- ▶ Reliability
- ▶ Usability
- ▶ Accuracy (not sure)

Question No: 15 (Marks: 1) - Please choose one

Given below are some statements associated with data flow diagrams. Identify the correct statement from among them.

- ▶ Data flow is made use of to model what systems do.
- ▶ Flows of data can take place from a process to a sink.
- ▶ All processes have to be levelled or decomposed.
- ▶ Context diagram shows the major system processes. (Page 54)

Question No: 16 (Marks: 1) - Please choose one

What would be the most suitable architecture to develop a commercial web page to do business transactions over the internet?

- ▶ Client server model (p129)
- ▶ Island model
- ▶ RAD model
- ▶ Repository model

MIDTERM EXAMINATION 2010

Question No: 1 (Marks: 1) - Please choose one

The first step in any OOA process model is to_____

- ▶ build an object-relationship model (p90)
- ▶ define collaborations between objects
- ▶ elicit customer requirements
- ▶ select a representation language

Question No: 2 (Marks: 1) - Please choose one

Dynamic components of an OOA model are _____

- ▶ not reusable
- ▶ **sensitive to timing and event processing**
- ▶ stable throughout the operational life of an application
- ▶ structural in nature

Dynamic components:

Focus on control and are sensitive to timing and event processing.

Dynamic view of Communication,

Dynamic view of Control and Time.

Question No: 3 (Marks: 1) - Please choose one

Events occur whenever a(n) _____

- ▶ **actor and the OO system exchange information** APRACTITIONER'S APPROACH (Page 622)
- ▶ class operation is invoked
- ▶ messages are passed between objects
- ▶ all of the given

Question No: 4 (Marks: 1) - Please choose one

By following modern system engineering practices simulation of reactive systems is no longer necessary

- ▶ True
- ▶ **False** [Click here for detail](#)

Question No: 5 (Marks: 1) - Please choose one

Information flow represents the manner in which data and control

- ▶ are related to one another.
- ▶ **changes as each moves through the system.** APRACTITIONER'S APPROACH (Page 312)
- ▶ will be implemented in the final design.
- ▶ none of the given

Question No: 6 (Marks: 1) - Please choose one

Which view should be consider first during software requirements analysis?

- ▶ actor view
- ▶ data view
- ▶ essential view
- ▶ **implementation view** APRACTITIONER'S APPROACH (Page 316)

Question No: 7 (Marks: 1) - Please choose one

State Transition Diagram is helpful in determining

- ▶ **Business Understanding (Page 52)**
- ▶ Process Flow
- ▶ Data store
- ▶ Non of the given

Question No: 8 (Marks: 1) - Please choose one

A better Design has an objective achieve

- ▶ High Cohesion
- ▶ Low Cohesion
- ▶ Low Coupling
- ▶ **High cohesion and Low coupling** **APRACTITIONER'S APPROACH (Page 316)**

Question No: 9 (Marks: 1) - Please choose one

The process of defining attributes is called

- ▶ **What I know** **(Page 95)**
- ▶ Who know me
- ▶ Whom I know
- ▶ Non of the above

Question No: 10 (Marks: 1) - Please choose one

Which of the following is not an objective for building an analysis model?

- ▶ define set of software requirements
- ▶ describe customer requirements
- ▶ **develop an abbreviated solution for the problem** **APRACTITIONER'S APPROACH (Page 329)**
- ▶ establish basis for software design

Question No: 11 (Marks: 1) - Please choose one

The output of this design process is a description of the

- ▶ **Software Architecture** **(Page 115)**
- ▶ Software Code
- ▶ Software
- ▶ Non of the above

Question No: 12 (Marks: 1) - Please choose one

Which of these characteristics are true of a data warehouse, but not a typical database?

- ▶ business level orientation and large size
- ▶ **currency and correctness of information** [Click here for detail](#)
- ▶ integration and non volatility
- ▶ all of the given

Question No: 13 (Marks: 1) - Please choose one

Identifying System Features include -----

- ▶ log important information
- ▶ conduct business
- ▶ analyze business results
- ▶ **all of the above** **(Page 98)**

Question No: 14 (Marks: 1) - Please choose one

The two main documents produced during this phase are Requirement Statement and Requirement Specification. They are also called Requirement Definition and -----

▶ **Functional Specification (Page 25)**

- ▶ Mathematical specification
- ▶ System Specification
- ▶ None of the given

Question No: 15 (Marks: 1) - Please choose one

Project----- defines the concept and range of the proposed solution, and limitations identify certain capabilities that the product will not include

▶ **Scope (Page 30)**

- ▶ Agreement
- ▶ Plan
- ▶ None of the given

Question No: 16 (Marks: 1) - Please choose one

Windows mobile is a popular mobile operating system which seen commonly on PDAs. Which of the following category pair(s) best describe(s) it?

▶ **application software, embedded software (not sure)**

- ▶ system software, web-based software
- ▶ application software, scientific software
- ▶ system software, embedded software

MIDTERM EXAMINATION 2010

Question No: 1 (Marks: 1) - Please choose one

Control flow diagrams are.

- ▶ **Needed to model event driven systems. [Click here for detail](#)**
- ▶ Required for all systems.
- ▶ Used in place of data flow diagrams.
- ▶ Useful for modeling user interfaces.

Question No: 2(Marks: 1) - Please choose one

The Data dictionary contain description of each software.

- ▶ Configuration item.
- ▶ **Data object A PRACTITIONER'S APPROACH (Page 356)**
- ▶ Diagram
- ▶ Notation

Question No: 3(Marks: 1) - Please choose one

A cohesion class is one which emphasize -----unit of----

- ▶ Single and multiple.
- ▶ Multiple and functionality
- ▶ Functional and single
- ▶ **Single and functional (Page 72)**

Question No: 4(Marks: 1) - Please choose one

Flow charts represent.

- ▶ **Sequence. (Page 50)**
- ▶ Random
- ▶ Parallel
- ▶ Non of above

Question No: 5(Marks: 1) - Please choose one

----- gives OO the ability to handle essential

- ▶ **Decentralization (Page 80)**
- ▶ Centralization
- ▶ Decentralization and Centralization
- ▶ Non of above

Question No: 6(Marks: 1) - Please choose one

In sequence Diagram events are organized in a-----time line

- ▶ **vertical (Page 106)**
- ▶ horizontal
- ▶ Vertical and Horizontal
- ▶ Non of above

Question No: 7(Marks: 1) - Please choose one

Asynchronous messages are denoted

- ▶ **Half Arrow (Page 109)**
- ▶ Simple Line
- ▶ Full Arrow
- ▶ Non of above

Question No: 8(Marks: 1) - Please choose one

Software crisis came in 1960 what is the main reason to for the crisis

- ▶ **Software development technique (Page 4)**
- ▶ Hardware

- ▶ Software
- ▶ Non of above

Question No: 10 (Marks: 1) - Please choose one

central role of the software requirement Document in entire development process is related to

- ▶ Project planning
- ▶ Project tracing
- ▶ Change control
- ▶ **All of above (Page 19)**

Question No: 11 (Marks: 1) - Please choose one

The scope description establishes the ----- between the system we are developing and everything else in the universe. Project planning

- ▶ **Boundary (Page 31)**
- ▶ System
- ▶ limitation
- ▶ None of above

Question No: 12 (Marks: 1) - Please choose one

which one is not the type of message which sequence diagram depict

- ▶ Synchronous
- ▶ asynchronous
- ▶ Create
- ▶ **Update (Page 108)**

Question No: 13 (Marks: 1) - Please choose one

cohesion is about making sure each component does ----- thing and does it well

- ▶ Three
- ▶ infinite
- ▶ **one A PRACTITIONER'S APPROACH (Page 381)**
- ▶ two

Question No: 14 (Marks: 1) - Please choose one

what is the most important software quality of a web based transaction

- ▶ efficiency
- ▶ Modularity
- ▶ **usability [click here for detail](#)**
- ▶ Security

CS504 Subjective

Midterm Examination 2011

Question No: 1 (Marks: 3)

Define Asynchronous Messages and Synchronous Messages.

Answer:- (Page 109)

Asynchronous Messages:-

Asynchronous messages are “signals,” denoted by a half arrow. They do not block the caller.

Asynchronous messages typically perform the following actions:

- ❖ Create a new thread
- ❖ Create a new object

Communicate with a thread that is already running

Synchronous Messages:-

Synchronous messages are “call events” and are denoted by the full arrow. They represent nested flow of control which is typically implemented as an operation call. In case of a synchronous message, the caller waits for the called routine to complete its operation before moving forward.

Question No: 2 (Marks: 5)

Law of balancing act in software

Answer:- (Page 7)

The Balancing Act!

Software Engineering is actually the balancing act. You have to balance many things like cost, user friendliness, Efficiency, Reliability etc. You have to analyze which one is the more important feature for your software is it reliability, efficiency, user friendliness or something else. There is always a trade-off among all these requirements of software. It may be the case that if you try to make it more user-friendly then the efficiency may suffer. And if you try to make it more cost-effective then reliability may suffer. Therefore there is always a trade-off between these characteristics of software. These requirements may be conflicting. For example, there may be tension among the following:

- ❖ Cost vs. Efficiency
- ❖ Cost vs. Reliability
- ❖ Efficiency vs. User-interface

A Software engineer is required to analyze these conflicting entities and tries to strike a balance.

Midterm Examination 2011

Question No: 3 (Marks: --)

HOW DO YOU DETERMINE THAT AN OBJECTIVE BELONGS TO CERTAIN CLASS?

Answer:- (Page 85)

The basic unit of object oriented design is an object. An object can be defined as a tangible entity that exhibits some well defined behavior. The structure and behavior of similar objects are defined in their common class. A class specifies an interface and defines an implementation.

Question No: 4 (Marks: --)

What is meant by "System's Static View"?

Answer:- (Software engineering ---Page 608)

Static view of semantic classes.

Requirements are assessed and classes are extracted (and represented) as part of the analysis model. Static view of attributes. Every class must be explicitly described. The attributes associated with the class provide a description of the class.

Static view of relationships.

Objects are "connected" to one another in a variety of ways. The analysis model must represent these.

Static view of behaviors.

The relationships just noted define a set of behaviors that accommodate the usage scenario (use-cases) of the system

Question No: 5 (Marks: --)

What is behavior driven perspective of an objective?

Answer:- (Page 85)

Behavior is how an object acts and reacts in terms of its state changes and message passing. The behavior of an object is completely defined by its actions. A message is some action that one object performs upon another in order to elicit a reaction. The operations that clients may perform upon an object are called methods.

Question No: 6 (Marks: --)

What is Textual Analysis? Explain it

Answer:- (Page 90)

Textual analysis was developed by Abbot and then extended by Graham and others. In this technique different parts of speech are identified within the text of the specification and these parts are modeled using different components.

Question No: 7 (Marks: --) What is

Software Architecture? Answer:-

(Page 116)

Architecture is the organizational structure of a system. Architecture can be recursively decomposed into parts that interact through interfaces, relationships that connect parts, and constraints for assembling parts. Parts that interact through interfaces include classes, components and subsystems.

Question No: 8 (Marks: --)

What is the Cardinality of Objective?

Answer :- (Software engineering ---Page 334)

Cardinality. The data model must be capable of representing the number of occurrences Objects in a given relationship. Tillmann [TIL93] defines the *cardinality* of an object/relationship pair in the following manner:

- ❖ One-to-one (1:1)—An occurrence of [object] 'A' can relate to one and only one occurrence of [object] 'B,' and an occurrence of 'B' can relate to only one occurrence of 'A.'
- ❖ One-to-many (1:N)—One occurrence of [object] 'A' can relate to one or many occurrences of [object] 'B,' but an occurrence of 'B' can relate to only one occurrence of 'A.' For example, a mother can have many children, but a child can have only one mother.
- ❖ Many-to-many (M:N)—An occurrence of [object] 'A' can relate to one or more occurrences of 'B,' while an occurrence of 'B' can relate to one or more occurrences of 'A.' For example, an uncle can have many nephews, while a nephew can have many uncles.

Midterm Examination 2011

Question No: 9 (Marks: 2)

Define abstraction?

Answer:- (Page 79)

An abstraction is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.

Question No: 10 (Marks: 2)

Is the design of software architecture a creative process?

Answer:- (Page 120)

Design of software architecture is a creative and iterative process. This involves performing a number of activities, not necessarily in any particular order or sequence.

Question No: 11 (Marks: 2)

Suppose you are working as a software engineer involved in the development of an e-commerce website. What are the 2 most important characteristics your software must have?

Answer:- (Software engineering ---Page 797)

Two most important characteristics are that software should be reliable and useable.

Question No: 12 (Marks: 3)

What is the purpose of collaboration diagrams?

Answer:- (Page 111)

Collaboration diagrams can also be used to depict the dynamic behavior of a system. They show how objects interact with respect to organizational units (boundaries!). Collaboration diagrams can also show synchronous, asynchronous, create, and destroy message using the same notation as used in sequence diagrams.

Question No: 13 (Marks: 3)

What is the difference between Aggregation and Association?

Answer:- (Page 87)

As compared to association, aggregation implies a tighter coupling between the two objects which are involved in this relationship. Therefore, one way to differentiate between aggregation and association is that if the two objects are tightly coupled, that is, if they cannot exist independently, it is an aggregation, and if they are usually considered as independent, it is an association.

Question No: 14 (Marks: 5)

what parameters are used to measure and analyze design quality?

Answer:- (Page 71)

A software design can be looked at from different angles and different parameters can be used to measure and analyze its quality. These parameters include efficiency, compactness, reusability, and maintainability. A good design from one angle may not seem to be suitable when looked from a different perspective. For example, a design that yields efficient and compact code may not be very easy to maintain. In order to establish whether a particular design is good or not, we therefore have to look at the project and application requirements.

Midterm Examination 2011

Question No: 15 (Marks: 5)

Discuss some of the purpose of interacting diagram?

Answer:- [Click here for detail](#)

The purposes of interaction diagrams are to visualize the interactive behaviour of the system. Now visualizing interaction is a difficult task. So the solution is to use different types of models to capture the different aspects of the interaction.

That is why sequence and collaboration diagrams are used to capture dynamic nature but from a different angle.

So the purposes of interaction diagram can be describes as:

- ❖ To capture dynamic behaviour of a system.
- ❖ To describe the message flow in the system.
- ❖ To describe structural organization of the objects.
- ❖ To describe interaction among objects.

Question No: 16 (Marks: 5)

What should be consideration for maintain design?

Answer:- (Page 71)

In order to make a design that is maintainable, it should be understandable and the changes should be local in effect. That is, it should be such that a change in some part of the system should not affect other parts of the system. This is achieved by applying the principles of modularity, abstraction, and separation of concern. If applied properly, these principles yield a design that is said to be more cohesive and loosely coupled and thus is easy to maintain.

**Question No: 17 (Marks: 3) Purpose
of collaborating diagram? Answer:-**

(repeated)

Question No: 18 (Marks: 3)

It is fact that good design makes maintenance easier. Which design principle help this to be achieved?

Answer:- [\(Page 71\)](#)

A good design from one angle may not seem to be suitable when looked from a different perspective. For example, a design that yields efficient and compact code may not be very easy to maintain. In order to establish whether a particular design is good or not, we therefore have to look at the project and application requirements.

Midterm Examination 2011

Question No: 20 (Marks: --)

To manage the complexity of the system we need to apply the principle of abstraction. Discuss briefly?

Answer:- [\(Page 79\)](#)

An abstraction is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details. The principle of abstraction also helps us in handling the inherent complexity of a system by allowing us to look at its important external characteristic, at the same time, hiding its inner complexity. Hiding the internal details is called encapsulation.

Question No: 21 (Marks: 2)

Differentiate between architectural design and system architecture in a single line?

Answer:- [Click here for detail](#)

Architecture faces towards strategy, structure and purpose, towards the abstract while Design faces towards implementation and practice, towards the concrete.

Question No: 5 (Marks: 3)

What are architectural designs Process, explain briefly?

Answer:- [\(Page 79\)](#)

System structuring: -

System structuring is concerned with decomposing the system into interacting sub-systems. The system is decomposed into several principal sub-systems and communications between these sub-systems are identified.

Control modeling:-

Control modeling establishes a model of the control relationships between the different parts of the system.

Modular decomposition:-

During this activity, the identified sub-systems are decomposed into modules. This design process is further elaborated in the following section where architectural views are discussed.

Question No: 5 (Marks: 5)

What should be the consideration for maintainable design?

Answer:- (rep)

Question No: 5 (Marks: 5)

Discuss the relationship between sequence diagram and logical complexity?

Answer:- (Page 110)

It is important to understand that the diagrams are meant to make things clear. Therefore, in order to keep them simple, special attentions should be paid to the conditional logic. If it is simple then there is no harm in adding it to the diagram. On the other hand if the logic is complex then we should draw separate diagrams like flow charts.

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Question No: 17 (Marks: 2)

To manage the complexity of the system we need to apply the principles of abstraction. Discuss briefly.

Answer: rep

Question No: 18 (Marks: 2) p2chapter 5

Define Motivation of GUI

Answer:

Motivation for GUI

- ❖ System users often judge a system by its interface rather than its functionality
- ❖ A poorly designed interface can cause a user to make catastrophic errors
- ❖ Poor user interface design is the reason why so many software systems are never used

Question No: 19 (Marks: 2)

Is the design of software architecture a creative process?

Answer: rep

Question No: 20 (Marks: 3)

Keeping in mind the Connie's case study, what rule of thumbs was identified, list them down.

Answer:- (Page 101)

Who I Know - Rules of Thumb

- ❖ an actor knows about its participants person knows about cashier
- ❖ a transaction knows about its participants a session knows about its register and cashier
- ❖ A transaction contains its transaction line items sale contains its sales line items
- ❖ A transaction knows its sub transactions session knows about its sales sale knows about its payments
- ❖ A place knows about its transactions store knows about its sessions
- ❖ A place knows about its descriptive objects store knows about its tax categories
- ❖ A container knows about its contents a store knows about its cashiers, items, and registers

Question No: 21 (Marks: 3)

The CPU of a computer consists of an ALU and a CU. Intel and AMD are two types of CPUs, which are widely used in computers. Users of computers can be either novices or experts.

Consider the following classes.

- (i) CPU and AMD
- (ii) User and Computer
- (iii) CPU and ALU

Answer:

CPU and AMD:-

There is relationship between CPU and AMD is “A kind of” type as AMD is a kind of CPU.

User and Computer:-

The relationship between User and Computer is “simple Association” type as there is weak relationship between user and computer.

CPU and ALU:-

Relationship between CPU and ALU is “composition” type as ALU is a part of CPU.

Question No: 22 (Marks: 5)

What is action-oriented approach for Software Design?

Answer: (Page 80)

In the case of action-oriented approach, data is decomposed according to functionality requirements. That is, decomposition revolves around function. In the OO approach, decomposition of a problem revolves around data. Action-oriented paradigm focuses only on the functionality of a system and typically ignores the data until it is required. Object- oriented paradigm focuses both on the functionality and the data at the same time. The basic difference between these two is decentralized control mechanism versus centralized control mechanism respectively. Decentralization gives OO the ability to handle essential complexity better than action-oriented approach.

Question No: 23 (Marks: 5)

Collaboration Diagrams depict Dynamic behavior of the system, explain it.

Collaboration diagrams:- Answer: (Page 110)

Collaboration diagrams can also be used to depict the dynamic behavior of a system. They show how objects interact with respect to organizational units (boundaries!). Since a boundary shapes communication between system and outside world e.g. user interface or other system, collaboration diagrams can be used to show this aspect of the system. The sequence of messages determined by numbering such as 1, 2, 3, 4, This shows which operation calls which other operation.

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Question No: 17 (Marks: 2)

To manage the complexity of the system we need to apply the principles of separation of concern. Discuss briefly

Answer: (Page 69)

Separation of concern allows us to deal with different individual aspects of a problem by considering these aspects in isolation and independent of each other. A complex system may be divided into smaller pieces of lesser complexity called modules.

Question No: 18 (Marks: 2)

What is elaborated Use case? Explain it

Answer: (Page 36)

After the derivation of the use case model, each use is elaborated by adding detail of interaction between the user and the software system. An elaborated use case has the following components: Use Case Name, actors, summary, precondition, post-condition, extend, uses, normal course of events, alternative path, exception, assumption.

Question No: 19 (Marks: 2)

What is Software Architecture?

Answer: rep

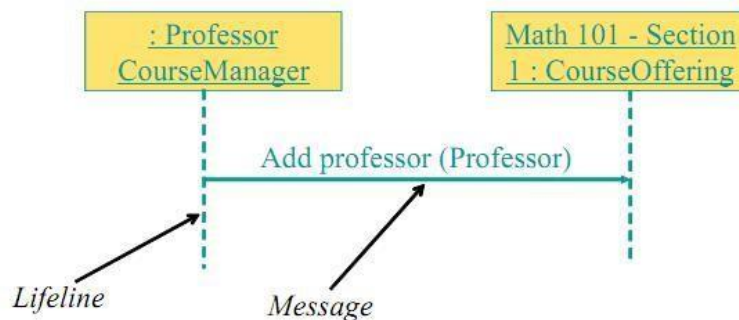
Question No: 20 (Marks: 3)

What notation is used for Sequence Diagrams? Draw it graphically.

Answer: (Page 106)

The Notation

Following diagram illustrates the notation used for drawing sequence diagrams.



Question No: 21 (Marks: 3)

How can we decrease Coupling explain it.

Answer: (Page 113)

That is, we can reduce the coupling of a system by minimizing the number of messages in the protocol of a class.

Question No: 22 (Marks: 5)

Narrate some of the architectural design guidelines that can help in addressing non-functional requirements challenges.

Answer: (Page 120)

Software architecture must address the non-functional as well as the functional requirements of the software system. Following are some of the architectural design.

Performance –Performance can be enhanced by localizing operations to minimize sub-system communication.

That is, try to have self-contained modules as much as possible so that inter-module communication is minimized.

Security –Security can be improved by using a layered architecture with critical assets put in inner layers.

Safety –Safety-critical components should be isolated

Availability –Availability can be ensured by building redundancy in the system and having redundant components in the architecture.

Maintainability–Maintainability is directly related with simplicity. Therefore, maintainability can be increased by using fine-grain, self-contained components.

Question No: 23 (Marks: 5)

What is the importance of Classification in identifying Classes and objects? **Answer:**

(Page 763)

Consider a large university library. Tens of thousands of books, periodicals, and other information resources are available for use. But to access these resources, a categorization scheme must be developed. To navigate this large volume of information, librarians have defined a classification scheme that includes a Library of Congress classification code, keywords, author names, and other index entries. All enable the user to find the needed resource quickly and easily.

Question No: 17 (Marks: 2)

Define abstraction

Answer: rep

Question No: 18 (Marks: 2)

What is Software Architecture?

Answer: rep

Question No: 19 (Marks: 2)

Suppose you are working as a software engineer involved in the development of an e-commerce website. What are the 2 most important characteristics your software must have?

Answer: rep

Question No: 20 (Marks: 3)

What is the purpose of collaboration diagrams?

Answer: rep

Question No: 21 (Marks: 3)

It is a fact that a good design makes maintenance easier. List a design principle which helps this to be achieved.

Answer: rep

Question No: 22 (Marks: 5)

Discuss the relationship between Sequence diagrams and logical complexity Answer: rep

Question No: 23 (Marks: 5)

What is the difference between Association and composition?

Answer:- (OOP, PAGE 51-52)

In Association, interacting objects have no intrinsic relationship with other object. It is the weakest link between objects. While in Composition An object may be composed of other smaller objects, the relationship between the “part” objects and the “whole”.

MIDTERM EXAMINATION

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Question No: 23 (Marks: 3)

How can we decrease coupling? Explain it.

Answer: rep

Question No: 23 (Marks: 5)

What is meant by “system’s static View”? Discuss briefly.

Answer: rep

Question No: 23 (Marks: 5)

How the objects are identified in peter codd’s technique?

Answer: (Page 93)

Objects are identifying in the following way.

Select actors

Actors are people and organizations that take part in the system under consideration.Examples of actors are: person, organization (agency, company, corporation, foundation).

Select Participants

A participant is a role that each actor plays in the system under consideration. Examples of participants are: agent, applicant, buyer, cashier, clerk, customer, dealer, and distributor. Etc.

Select Places

Places are where things come to rest or places that contain other objects. Examples of places are: airport, assembly-line, bank, city, clinic, country, depot, garage and hospital etc.

Select Transactions

Transactions are the “events”. These transactions usually come from a window (GUI), some object which monitors for significant event and logs that information, or a another system that interacts with the system under consideration and logs some information. Examples of transactions are: agreement, assignment, authorization, contract, delivery, deposit, incident, inquiry, order, payment, problem report, purchase and sales etc.

Select Container Objects

Containers are objects that hold other objects. e.g. bin, box, cabinet, folder, locker, safe, shelf, etc. Therefore a place is also a container but every container need not be a place.

Select Tangible things

Take a “walk” through the system and select “tangible” things around you used in the problem domain. These may be characterized as all the remaining (not yet selected) “nouns” that make up the problem domain.

Examples are: account, book, calendar, cash box, cash drawer, item, plan, procedure, product, schedule, skill, tool, etc

Question No: 23 (Marks: 3)

What is the purpose of interaction diagram?

Answer: rep

Question No: 23 (Marks: 2)

Define cohesion.

Answer: (Page 72)

Cohesion is an internal property of a module. Cohesion describes the intra-component linkages while couple shows the inter-component linkages. Cohesion measures the independence of a module.

Question No: 23 (Marks: 2)

Keeping connie’s case study in mind, as discussed in lecture, list down whole parts structures which were identified.

Answer: (Page 100)

Identify Whole-Part Structures

- ❖ A store as a whole is made up of cashiers, registers, and items.
- ❖ A register contains a cash drawer.
- ❖ A sale is constituted of sale line items.

Question No: 23 (Marks: --)

Define data flow diagram

Answer: (Page 100)

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the

system which can later be elaborated.

MIDTERM EXAMINATION - Spring 2010

Question No: 17 (Marks: 2)

Define abstraction?

Answer: rep

Question No: 18 (Marks: 2)

Is the design of software architecture a creative process?

Answer: rep

Question No: 19 (Marks: 2)

Suppose you are working as a software engineer involved in the development of an e-commerce website. What are the 2 most important characteristics your software must have?

Answer: rep

Question No: 20 (Marks: 3)

What is the purpose of collaboration diagrams? Repeated

Answer: rep

Question No: 21 (Marks: 3)

What is the difference between Aggregation and Association? P5 chapter 7

Answer: rep

Question No: 22 (Marks: 5)

What parameters are used to measure and analyze design quality?

Answer: rep

Question No: 23 (Marks: 5)

How the objects are identified in Peter codd's technique? Repeated

Answer: rep

Question No: 24 (Marks: 5)

comparison of software architecture and Building architecture

Answer: (Page 115)

When building a house, the architect, the general contractor, the electrician, the plumber, the interior designer, and the landscaper all have different views of the structure. Although these views are pictured differently, all are inherently related: together, they describe the building's architecture. The same is

true with software architecture. Architectural design basically establishes the overall structure of a software system.

Question No: 25 (Marks: 10) Code

example of High Coupling Answer:

[Click here for detail](#)

Tightly Coupled Example:

```
public class CartEntry
{
    public float Price;
    public int Quantity;
}

public class CartContents
{
    public CartEntry[] items;
}

public class Order
{
    private CartContents cart;
    private float salesTax;

    public Order(CartContents cart, float salesTax)
    {
        this.cart = cart;
        this.salesTax = salesTax;
    }

    public float OrderTotal()
    {
        float cartTotal = 0;
        for (int i = 0; i < cart.items.Length; i++)
        {
            cartTotal += cart.items[i].Price * cart.items[i].Quantity;
        }
        cartTotal += cartTotal*salesTax;
        return cartTotal;
    }
}
```

MIDTERM EXAMINATION

Spring 2010

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Question No: 16 (Marks: 2)

Adding user interface detail in the SRS is controversial a creative process i.e by adding GUI detail to the SRS document, focus e shift from what to how (analysis design) do you agree ?

Answer: (Page 115)

Yes I am agree because Client appreciates more the contents of the SRS document if our SRS document contains the GUI details than if we don't have them there.

Question No: 17 (Marks: 2)

Is the design software of architecture a creative process? Answer: rep

Question No: 18 (Marks: 3)

What is the architecture design process explain briefly. Answer: rep

Question

No: 19 (

Marks: 3

) What is

textual

analysis?

Answer: rep

Question No: 20 (Marks: 5)

What should be the consideration for the maintainable design? Answer: rep

Question No: 21 (Marks: 5)

Discusses the relation between sequence diagram & logical complexity. Answer: rep

CS504 GRAND QUIZ

- 1) Modules with high cohesion and low coupling can be treated and analysed as
 - i. White boxes
 - ii. **black boxes**
 - iii. grey boxes
 - iv. none of these
- 2) While establishing the services for an object, the following fundamental questions should be asked:
 - i) Why does the system need this object anyway?
 - ii) What useful questions can it answer?

- iii) What useful action can it perform?
 - iv) **All of the given options.**
- 3) is a role that each actor plays in the system under consideration.
 - i) An act
 - ii) **A participant**
 - iii) A function
 - iv) None of the given
- 4) Any Engineering approach must be founded on organizational commitment to ------.
 - i. Cost
 - ii. Scheduling
 - iii. **Quality**
 - iv. **Performance**
- 5) Return values in synchronous messages are:
 - i) Compulsory
 - ii) **May not used when response is obvious**
 - iii) Not used at all
 - iv) Represented by solid lines
- 6) According to Caper Jhones analysis of project activities, coding only has ----- affect part in system development.
 - i. **13-14%**
 - ii. **36-40%**
 - iii. **50-60%**
 - iv. **70-80%**
- 7) Normally a system is more easy to modify if its modules have
 - i. High coupling and high cohesion
 - ii. High coupling and low cohesion
 - iii. **Low coupling and high cohesion**
 - iv. **Low coupling and Low cohesion**
- 8) In multi-threaded or multiprocessing applications where different execution threads may pass information to one another by sending ----- to each other.

- i. Interrupt calls
 - ii. Synchronous messages
 - iii. Asynchronous messages
 - iv. System Calls
- 9) Which of the following is not among the four layers of the object-oriented pyramid?
- i. The subsystem layers
 - ii. The class and object layer
 - iii. The abstract layer
 - iv. The message layers
- 10) System models include:
- i. User business processes
 - ii. User activities for conducting the business process
 - iii. Processes that need to be automated
 - iv. All of the given options
- 11) In the architecture trade-off analysis method the architectural style should be described using the
- i. Data flow view
 - ii. Module view
 - iii. Process view
 - iv. All of the given
- 12) is concerned with decomposing the system into interacting sub-systems.
- i. System structuring
 - a) Control Modelling
 - b) Molecular Decomposition
 - c) None of the given
- 13) A use case represents:
- i. A class, its attributes and operations
 - ii. An operation's interface and signature
 - iii. The role a user plays when interacting with the system
 - iv. The system's functionality for a particular purpose
- 14) External entity may be:
- i. Source of input data only
 - ii. Source of input data and destination of results
 - iii. Destination of results
 - iv. Repository of data
- 15) The process of utilizing our knowledge of computer science in effective production of
- .
- i. Chemical Engineering
 - ii. Electrical Engineering
 - iii. Computer Engineering
 - iv. System Engineering
- 16) Coupling is a measure of ----- of a component.
- i. Independence
 - ii. Dependence
 - iii. Aggregation
 - iv. Composition
- 17) has become a standard notation for object-oriented system modelling:

- i. UML
 - ii. C++
 - iii. OCL(Object Constraint Language)
 - iv. None of the given option
- 18) An arrow in data flow diagram represents:
- i. Direction of flow of data
 - ii. Processing of data
 - iii. External agent
 - iv. Internal Agent
- 19) diagrams does not capture control flow information, it just shows the flow of data in a system.
- i. Sequence
 - ii. Data Flow
 - iii. Activity
 - iv. Class
- 20) In -----the analyst determines the source of requirements and where do these requirements consume:
- a) Data flow analysis
 - b) Source and sink analysis
 - c) Down parsing
 - d) Up parsing
- 21) Data cannot flow from one external entity to other external entity because:
- a) It will get corrupted
 - b) It is not allowed in DFD
 - c) An external entity has no mechanism to read or write
 - d) Both are outside the context of the system
- 22) In the functional design, the structure of the system resolves around:
- a) Objects
 - b) Properties
 - c) Functions
 - d) All of the given options
- 23) is one of the techniques to document domain knowledge
- a) State transition diagram
 - b) Feasibility matrix
 - c) System matrix
 - d) None of the given options
- 24) In case of ----- approach , decomposition of a problem revolves around data.
- a) Object-Oriented
 - b) Action-Oriented
 - c) Event-Oriented
 - d) Process-Oriented
- 25) The-----relationship is a kind of a generalization specialization relationship:
- a) Bit-Byte
 - b) Uses
 - c) Binary
 - d) Extends
- 26) Which statement is not true about system software?

- a) Change request has direct impact on software
 - b) Passes through a constant process of evolution
 - c) Change requests have direct impact in the form of defects
 - d) None of the given
- 27) Strong cohesion implies that:
- a) All parts of a component have a close logical relationship with each other
 - b) All parts of a component don't have a close relationship with each other
 - c) Component is dynamic in nature
 - d) Component is static in nature
- 28) The intent of Object-Oriented Analysis(OOA) is to define:
- a) All classes
 - b) Relationships among classes
 - c) Behaviour of classes
 - d) All of the given options
- 29) Requirement engineering focuses on----- aspect of the software development process.
- a) Both what and how
 - b) What
 - c) How
 - d) Why and how
- 30) relationship is concerned with classes, not with class instantiates.
- a) Association
 - b) Inheritance
 - c) Aggregation
 - d) Composition
- 31) Which of the following statements are true in context of the object model deviation through the Coad methodology?
- a) A place is also a contains
 - b) Every container needs to be a place
 - c) Same person may play different times in the system.
 - i. A only
 - ii. A and b
 - iii. A and c
 - iv. All of the given
- 32) The goal of ----- is to translate the customer's desire for a set of defined capabilities into a working product.
- i. Electrical engineering
 - ii. Product engineering
 - iii. Hardware engineering
 - iv. Mechanical engineering
- 33) In case of a ----- message, the called routine that handles the message is completed before the caller resumes execution.
- a) Synchronous
 - b) Asynchronous
 - c) Bidirectional
 - i. A only
 - ii. B only

- iii. C only
 - iv. All of the given
- 34) In multiprocessing applications, different execution threads may pass information to one another by sending----- to each other.
- i. Interrupt calls
 - ii. Synchronous messages
 - iii. Asynchronous messages
 - iv. System calls
- 35) A car is made up of a body, three or four wheels, a steering mechanism, a breaking mechanism, and a power engine”

The above statement is example of:

Whole-part relationship

- i. Inheritance
 - ii. Specialization
 - iii. Generalization
- 36) To help separate an object’s external behaviour from its implementation, the technique used is called -----
- i. Generalization
 - ii. Association
 - iii. Composition

Abstraction

- 37) Sequences of messages can be present in:
- (a) Use case diagram
 - (b) Sequence diagram
 - (c) Collaboration diagram
- 1) a only
 - 2) b only
 - 3) c only
 - 4) b and c
- 38) Which of the following strategies lead to good software design:
- i. Separation of concerns
 - ii. Modularity
 - iii. Divide-and-conquer
 - iv. All of the given options
- 39) Data flow model:
- i. Captures the flow of data in a system
 - ii. Helps in developing an understanding of system’s functionality
 - iii. Describes data origination, transformations and consumption in a system
 - iv. All of the given options
- 40) requirements are often called product features.
- i. Functional
 - ii. Non-functional
 - iii. Developer
 - iv. User
- 41) In the functional design, the structure of the system revolves around.
- i. Objects
 - ii. Properties

- iii. **Functions**
 - iv. All of the given option
- 42) The first step in any OOA process model is to **i. Build an object-relationship model.**
- ii. Define collaborations between objects.
 - iii. Elicit customer requirements
 - iv. Select a representation language
- 43) The----- relationship is kind of a generalization specialization relationship.
- i. Bit-byte
 - ii. Uses
 - iii. Binary
 - iv. Extends**
- 44) Regarding data flow model, which of the following statement(s) is true:
- i. It captures the transformation of data between processes/functions of a system **ii.**
 - Processes on a data flow can operate in parallel**
 - iii. Only those processes are represented which we need to automate
 - iv. All of the given option
- 45) is an extremely powerful technique for dealing with complexity.
- i. Aggregation
 - ii. Abstraction**
 - iii. Inheritance
 - iv. Association
- 46) In “point of sale system”. the term “payment” represents
- i. Actor
 - ii. Participant
 - iii. Transaction**
 - iv. Container**
- 47) The architecture components for product engineering are
- i. Data, hardware, software, people**
 - ii. Data, documentation, hardware, software
 - iii. Data, hardware, software, procedures
 - iv. Documentation, hardware, people, procedures
- 48) An object model encompasses the principle(s) of
- i. Abstraction
 - ii. Encapsulation
 - iii. Hierarchy or inheritance
 - iv. All of the given option**
- 49) Prototyping is used when there is----- regarding requirements.
- i. **Uncertainty**
 - ii. Confirmation
 - iii. Conflict
 - iv. Consensus
- 50) In ----- phase of software development, requirement analyst focuses on possible design of the **proposed solution.**
- i. Maintenance
 - ii. Development
 - iii. Definition**
 - iv. Vision**
- 51) At which stage of software development loop, results are delivered?
- i. Problem definition
 - ii. Solution integration
 - iii. Technical development
 - iv. Status quo**

- 52) A class will be cohesive if:
- Class does not implement complex interfaces
 - Class does not have complex methods
 - If most of the methods do not use most of the data members most of the time
 - If most of the methods use most of the data members most of the time.
- 53) A DFD is normally levelled (adding more levels of abstraction) as
- It is a good idea in design
 - It is recommended by many experts
 - It is easy to do it
 - It is easier to read and understand a number of smaller DFDs than one large DFD
- 54) Identify the true statement(s)
- An attribute that may have a number of values should be replaced by a new class and an object connection
 - An attribute that varies over time, e.g. price of an item, should be replaced by an additional class with an affective data and value
 - Replace “yes/no” type attribute with “status” type attributes for flexibility
 - All of given option
- 55) Is a technique in which we construct a model of an entity based upon its essential **characteristics and ignore the inessential details.**
- Inheritance
 - Polymorphism
 - Aggregation
 - Abstraction
- 56) A structure is a manner of an organization which expresses a -----strong organization within the **problem domain.**
- Semantically
 - Syntactically
 - Graphically
 - None of the given
- 57) An object model of a system captures the----- structure of a system.
- Static
 - Dynamic
 - Iterative
 - Hierarchical
- 58) To determine the architectural style or combination of styles that best fits the proposed system, requirements engineering is used to uncover
- Algorithmic complexity
 - Characteristics and constraints
 - Control and data
 - Design patterns

- 59) Which statement is not according to the software engineering principles? Software engineering is a(n) _____
- Balancing act
 - Disciplined approach
 - Unsystematic approach**
 - Quantifiable approach**
- 60) In abbot's textual analysis technique, different part of speech is identified within the text of the specification and these part are modelled using different _____
- Event
 - Process
 - Operations
 - Components**
- 61) Quantitative methods for assessing the quality of proposed architectural designs are readily available.
- True
 - False**
- 62) In order to determine the role and responsibilities of the identified objects, we need to consider which of the following step(s):
- Who I am?
 - What I know?
 - Who I know?
 - What I do?
- A only
 - A and b
 - B ,c and D
 - All of the given**
- 63) In object oriented design _____ layer contains the details that enable each object to communicates with its collaborators.
- Subsystem
 - Responsibility
 - Message**
 - Object
- 64) In sequence diagram, the boxes denote:
- Objects (or classes)**
 - Messages, sent from one object to other
 - Life-time of objects
 - None of the given option
- 65) In " railway tickit reservation system" the roles such as enquiry. Reservation and ticketing and **cancellation are to be performed by the user called:**
- Passenger**
 - System analyst
 - System developer
 - System designer
- 66) Class responsibilities are defined by _____
- Its attributes only
 - Its collaborators
 - Its operations only

- iv. Both its attributes and operations
- 67) Requirement engineering mainly deals with the _____ of the system
- i. Vision phase
 - ii. **Definition phase**
 - iii. Development phase
 - iv. Maintenance phase
- 68) In UML based object oriented model of a system, a composition relation between two objects is shown by a _____ sign on the whole side of a relation line.
- i. An unfilled diamond
 - ii. A filled diamond
 - iii. A half diamond
 - iv. A dot
- 69) _____ analysis educates the analyst on business domain complexity and shows a way to deal with it.
- i. **Domain**
 - ii. Use case
 - iii. Object collaboration
 - iv. None of the given options
- 70) In this case of _____ intra component linkages are stronger while inter component linkages are weak.
- i. High cohesion
 - ii. Low coupling
 - iii. Low cohesion
 - iv. **High coupling**
- 71) An architectural style encompasses which of the following elements?
- i. Constraints
 - ii. Set of components
 - iii. Semantic models
 - iv. **All of the given**
- 72) In software engineering paradigm, any engineering approach must be founded on organizational commitment to _____
- i. Cost
 - ii. Scheduling
 - iii. **Quality**
 - iv. Performance
- 73) Identify the true statement:
- i. **Normally object oriented design is more maintainable than functional oriented.**
 - ii. Software with functional oriented design does not fulfil non functional requirements.
 - iii. Object oriented design can not implement “separation of concerns” strategy
 - iv. Function oriented design does not lead to an efficient product
- 74) A process in data flow diagram (DFD) represents
- i. Flow of data
 - ii. **Transformation of data**
 - iii. Storage of data
 - iv. An external agent
- 75) A maintainable design is a design , which supports
- i. Change

- ii. Debugging
 - iii. Adding new features
 - iv. All of the given
- 76) Whole part structure is also called _____
- i. Generalization
 - ii. Aggregation
 - iii. Specialization
 - iv. Association
- 77) _____ are kind of umbrella activities that are used to smoothly and successfully perform the construction activities.
- i. Design activities
 - ii. Management activities
 - iii. Testing activities
 - iv. Maintenance activities
- 78) When you encounter both transform flow in the same DFD the flow is partitioned and the appropriate mapping technique is used on each part of the DFD.
- i. True
 - ii. False
- 79) Software architecture must address ----- requirements of a software system.
- i. Functional
 - ii. Non-functional
 - iii. User Interface Requirements
 - iv. Both functional and non-functional.
- 80) To construct a system model the engineer should consider one of the following restraining factors.
- i. Assumptions and constraints
 - ii. Budget and expenses
 - iii. Data objects and operations
 - iv. Schedule and milestones
- 81) A cohesive class is one which emphasizes on ----- unit of functionality.
- i. Single
 - ii. Multiple
 - iii. Static
 - iv. None
- 82) The best way to conduct a requirement validation review is to
- i. Examine the system model for errors
 - ii. have the customer look over the requirements
 - iii. Send them to the design team and see if they have any concerns
 - iv. use a checklist of the questions to examine each requirement
- 83) Defining the services of an object means:
- i. What it does?
 - ii. What it knows?
 - iii. Who knows it?
- 84) Which one of the following is the external quality of a software product?
- i. Correctness
 - ii. Concision
 - iii. Cohesion

- iv. **Low coupling**
- 85) In Data Flow Diagram, the entity or system, outside the boundary of this system is called:
- i. Process
 - ii. Data flow
 - iii. **External agent**
 - iv. Data store
- 86) GUI stands for:
- i. Genaric user Interface
 - ii. **Graphical user interface**
 - iii. Genaric user interaction
 - iv. Graphical user interaction
- 87) Specialization means:
- i. Calling the same method with the object of child object
 - ii. Hiding the data
 - iii. **Creating new subclasses for an existing class**
 - iv. None of the given options
- 88) In a use case diagram, an ellipse signifies a(n):
- i. Actor
 - ii. Class
 - iii. **Use case**
 - iv. System boundary
- 89) Software development is a step-by-step process, and in -----phase of software development **Business objective of an organization get cleared**
- i. Maintenance
 - ii. Development
 - iii. Definition
 - iv. **Vision**
- 90) If you try to make software more user-friendly then the----- may suffer.
- i. Reliability
 - ii. Software
 - iii. **Efficiency**
 - iv. **Cost**
- 91) In object-oriented design, the structure of the system revolves around. **i. Objects**
- ii. Properties
 - iii. Methods
 - iv. All of the given option
- 92) In -----relationship, a class shares the structure and behaviour defined in another class:
- i. Aggregation
 - ii. Composition
 - iii. **Inheritance**
 - iv. **Uses**
- 93) In Object Oriented Design, combining the services offered by an object with the attributes they work on, results in:
- (a) Lower coupling and strong cohesion
 - (b) Lower cohesion and strong coupling
 - (c) Increased likelihood of reuse

(d) Decrease the modularity of the system

- i. A only
- ii. B and c only
- iii. A and c only

94) A change becomes ----- because of close presence of data and functions.

- i. Accessible
- ii. Global
- iii. Private
- iv. Localized

95) Software engineering is a-----approach.

- i. Systematic
- ii. Disciplined
- iii. Scheduled
- iv. All of the given options

96) An external entity that interacts with the system is called a(n):

- i. Use case
- ii. Actor
- iii. Stakeholder
- iv. Association

97) More powerful hardware resulted into the development of -----powerful and -----
software.

- i. Less, complex
- ii. More , complex
- iii. More, simple
- iv. Less, simple

98) A context diagram is used:

- i. As a first step in developing a detailed DFD of a system
- ii. In systems analysis of very complex systems
- iii. As an aid to system design
- iv. As an aid to programmers

99) “System should maintain transaction log of every system”

The above statement is an example of:

- i. Functional requirement
- ii. Non-functional requirement
- iii. Pseudo requirement
- iv. None of these

100) The architectural model provides the software engineer with the view of the system as a whole:

- i. True
- ii. False

101) As per Peter Coad’s methodology , which of the following may not be a perfect candidate for
being an object?

- i. Zone
- ii. Recipient
- iii. Garage
- iv. Password

- 102) In the case of-----approach , data is decomposed according to functionality **requirements.**
- Object-oriented
 - Action-oriented**
 - Event-oriented
 - Process-oriented
- 103) In UML based Object Oriented Model of a system, the diamond sign is used to depict-----
--- relations between two objects/classes.
- Aggregation and Association
 - Inheritance and Association
 - Composition and Aggregation**
 - iv. Composition, Aggregation and Association**
- 104) The system specification describes the:
- Function and behavior of a computer-based system**
 - Implementation of each allocated system element
 - Algorithmic detail and data structures
 - Time required for system simulation
- 105) In object oriented approach, ----- are the people and organizations that take part in the **system under consideration: i.**
- Actors**
 - Places
 - Participants
- 106) Software Design discusses ----- aspect of software development.
- What
 - ii. How**
 - Who
 - When
- 107) ----- requirements cause frequent modifications in user interface.
- Functional
 - Non-functional
 - iii. Unstable**
 - iv. User**
- 108) By levelling a DFD (adding more levels of abstraction) we mean:
- Splitting it into different levels**
 - Make its structure uniform
 - Expanding a process into one with more sub-processes giving more detail
 - Summarizing a DFD to specify only these essentials
- 109) A “register” in “Point of Sale system” is an example of:
- Actor
 - Participant
 - Tangible thing**
 - Transaction
- 110) is a set of processes and tools to develop software.
- Software engineering**
 - Information
 - Software
 - None of the given

- 111) The-----on which program operates is also considered as part of the software.
- Data**
 - Information
 - Program
 - None of the given
- 112) diagram provides a time-based view and collaboration diagrams which provide an **organization based view of the system's dynamics.**
- Data flow diagram
 - Entity relationship diagram
 - Class diagram
 - Sequence diagram**
- 113) Synchronous messages are "call events" and are denoted by -----
- Full arrow**
 - Half arrow
 - <<create>>
 - <<destroy>>
- 114) Which of the following are the components of system engineering software?
- Process
 - Methods
 - Tools
 - All of the given**
- 115) Identifying system features include -----
- Log important information
 - Conduct business
 - Analyze business results
 - All of the above**
- 116) is yet another technique that is used to reduce customer dissatisfaction at the **requirement stage.**
- Study of similar systems
 - Site visits
 - Prototyping**
 - All of the above
- 117) Data store notation in DFD represents:
- Data input
 - Data output
 - Data input and data output**
 - None of the above
- 118) The process of defining attributes is called -----
- Who know me?
 - What I know?**
 - Whom I know?
 - All of the above
- 119) The output of the design process is a description of the:
- Software architecture**
 - Software Code
 - Software
 - All of the above

- 120) Which of the following are the levels of software requirements?
- Business requirements
 - User requirements
 - Functional requirements
 - All of the above**
- 121) Given below are some statements associated with data flow diagrams. Identify the correct statement among them.
- Data flow is made used of to model what systems do
 - Flows of data can take place from a process to a sink
 - Context diagrams shows the major system processes**
 - All processes have to be labelled or decomposed**
- 122) In which of the following diagram the actors and attributes are represented with system boundary?
- Data flow diagram
 - Entity relationship diagram
 - Class diagram
 - Use case diagram**
- 123) _____ is real looking mock_up of what would be eventually delivered and might not do anything useful.
- Study of similar system
 - Site visits
 - Prototyping**
 - All of the above**
- 124) _____ is blueprint for software construction.
- Object oriented design**
 - Sequence design
 - Software design
 - All of the above
- 125) _____ requirements lead to ill-spent time and rework.
- Unacceptable
 - Ambiguous**
 - Dissatisfaction of customer
 - None of the above
- 126) Which type of diagram is used to depict the dynamic behaviour of a system.
- ERD diagram
 - DFD diagram
 - Class diagram
 - Collaborations diagram**
- 127) What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?
- Security
 - Reliability
 - Easy
 - usability **iv. Accuracy**
- 128) A better design has an objective achieve
- High cohesion
 - Low cohesion

- iii. Low coupling
 - iv. **High cohesion and low coupling**
- 129) Which of the following are the components of software engineering framework is combine the three remaining components?
- i. Process
 - ii. Method
 - iii. **Tools**
 - iv. **All of the above**
- 130) In sequence diagrams the time required by the receiver object to process the message is denoted by an _____
- i. **Activation box**
 - ii. Message line
 - iii. Life line
 - iv. All of the above
- 131) How many types of OOD modes have_____.
- i. One
 - ii. **Two**
 - iii. Three
 - iv. Four
- 132) Which notation is used to represent the process of the system in DFD model.
- i. **Process**
 - ii. External agent
 - iii. Data flow
 - iv. Data store
- 133) Requirement engineering mainly deals with the_____ of the system
- i. Process
 - ii. Maintenance
 - iii. Development phase
 - iv. **Definition phase**
- 134) Insufficient user involvement leads to_____products.
- i. **Unacceptable**
 - ii. Ambiguous
 - iii. Dissatisfaction of customer
 - iv. None of the above
- 135) Collaboration diagrams have basically two types of components: objects and _____
- i. Messages
 - ii. Method
 - iii. Classes
 - iv. **None of the above**
- 136) In object-oriented analysis how many number of tasks must occurs_____.
- i. 1
 - ii. **2**
 - iii. 3
 - iv. None of the above
- 137) State transition diagram is helpful in determining_____.
- i. Data store
 - ii. Process flow

- iii. **Business understanding**
 - iv. None of the above
- 138) In sequence diagram events are organized in a _____ time life line
- i. **Vertical**
 - ii. Horizontal
 - iii. Both A and B
 - iv. All of the above
- 139) Asynchronous messages are “signals,” denoted by _____.
- i. **Full arrow**
 - ii. Half arrow
 - iii. <<create>>
 - iv. <<destroy>>
- 140) When we write a program for computer and then we named it as _____.
- i. Data
 - ii. Information
 - iii. **Software**
 - iv. None of the given
- 141) Context level diagram present in which of the following document.
- i. **SRS-software requirement specification**
 - ii. Design document
 - iii. Test phase
 - iv. All of the above
- 142) _____ is diagram in which objects are interact with each other and these are arranged in a sequence.
- i. ERD diagrams
 - ii. Inheritance diagrams
 - iii. Class diagrams
 - iv. **Sequence diagrams**
- 143) Which of the following layers are include in object-oriented design?
- i. The subsystem layers
 - ii. The class and object layer
 - iii. **All of the above**
 - iv. **The message layers**
- 144) Which notation is used to represent the boundary of the system in DFD model?
- i. Process
 - ii. **External agent**
 - iii. Data flow
 - iv. Data store
- 145) Identifying whole-part structures (aggregations) means what are my _____.
- i. **Components**
 - ii. Structures
 - iii. Class
 - iv. Object
- 146) An object or class may further be classified on the basis of _____.
- i. Behaviour driven attributes
 - ii. Data driven attributes
 - iii. Responsibility driven attributes

- iv. All of the above
- 147) DFD notation contains
 - i. Process
 - ii. External agent
 - iii. Data flow
 - iv. All of the above
- 148) The dotted lines in sequence diagram are called _____.
 - i. Life line
 - ii. Message line
 - iii. Entities line
 - iv. All of the above
- 149) An object may create another object via a _____ message .
 - i. Full arrow
 - ii. half arrow
 - iii. <<create>>
 - iv. <<destroy>>
- 150) How many levels of software requirements are _____?
 - i. One
 - ii. Two
 - iii. Three
 - iv. Four
- 151) Which of the following diagram has iterative activities?
 - i. Data flow diagram
 - ii. Entity relationship diagram
 - iii. Class diagram
 - iv. Use case diagram
- 152) Which of the items listed below is not one of the software engineering layers?
 - i. Tools
 - ii. Manufacturing
 - iii. Process
 - iv. Methods
- 153) Coupling is a measure of _____ a module or component.
 - i. Independent
 - ii. Dependent
 - iii. Closeness
 - iv. All of the given
- 154) Software maintenance phase involves
 - i. Debugging
 - ii. Adding new features
 - iii. Making changes
 - iv. All of the given
- 155) The hardest single part of building a software system is deciding precisely _____ to build.
 - i. When
 - ii. What
 - iii. Why
 - iv. All of the given
- 156) Interaction diagrams depict the _____ behaviour of the system.

- i. Static
 - ii. Active
 - iii. **Dynamic**
 - iv. **None of the given**
- 157) A _____ can be used to describe the dynamic behaviour of an object-oriented system.
- i. ERD diagrams
 - ii. Inheritance diagrams
 - iii. Class diagrams
 - iv. **Series diagrams**
- 158) The Use case diagram shows that which _____ interact with each use case.
- i. Use case
 - ii. **Actor**
 - iii. Component
 - iv. Relation
- 159) Transactions are the _____ that must be remembered through time.
- i. **Events**
 - ii. Action
 - iii. Triggers
 - iv. Methods
- 160) A necessary supplement to transform or transaction mapping needed to create a complete architectural design is
- i. Entity relationship diagrams
 - ii. The data dictionary
 - iii. **Processing narratives for each module**
 - iv. Best cases for each module

CS 504-Software Engineering-I

Grand Quiz December 2020

IF there is any correction needed do it
accordingly

1. Whole part structure is also called _____
 - a. Generalization
 - b. **Aggregation**
 - c. Specialization
 - d. Association
2. A prototype is not the real product. It is rather just a real looking _____ of what would be eventually delivered and might not do anything useful.
 - a. **Mock-up** page 68
 - b. Ad-hoc
 - c. Design
 - d. Structure
3. In "Railway ticket reservation system" the roles such as inquiry, reservation, ticketing and cancellation are to be performed by the user called:-
 - a. **Passenger**
 - b. System analyst
 - c. System developer
 - d. System designer
4. A useful technique for evaluating the overall complexity of a proposed architecture is to

look at the component's_____.

- a. number and size of components
 - b. flow dependencies and sharing dependencies
 - c. size and cost
 - d. function points
5. _____ relationship is concerned with classes not with the class instantiates.
- a. Association
 - b. Inheritance
 - c. Aggregation
 - d. Composition
6. Software development is a step-by-step process, and in_____phase of software development Business Objective of an organization gets cleared.
- a. Maintenance
 - b. Development
 - c. Definition
 - d. Vision
7. The data on which the program operates is also considered as part of the_____.
- a. Important Data
 - b. Software
 - c. Logical Data
 - d. Utility Software

8. From the following which is/are not the example(s) of illegal data flow in Data Flow Diagram (DFD)
- External Agents directly communicating with each other
 - External Agent updating information in a DataStore
 - External Agent accessing information from a Data Store
 - There must be an intermediate process which should transform data received from one external entity and then send the transformed data to the other external entity
9. In sequence diagram, the objects are organized in a _____ line and the events in a _____ time line.
- Horizontal, straight
 - Vertical, straight
 - Horizontal, vertical
 - Vertical, horizontal
10. Consider the following piece of code:
- ```
public class Square extends Shape {
// some code
}
```
- The above code is an example of:
- Part-Whole relationship
  - Generalization/Specialization
  - Data Sharing
  - Data encapsulation
11. Requirement engineering mainly deals with the \_\_\_\_\_ of the system
- Vision Phase
  - Definition Phase page 16
  - Development Phase
  - Maintenance Phase
12. A \_\_\_\_\_ relationship indicates that one entity is composed of one or more parts which are themselves instances of that or another entity.
- Inheritance
  - Whole-part
  - Generalization
  - Specialization
13. In UML Object Model Notation \_\_\_\_\_
- C++ language-specific programs are constructed
  - mathematical problems are visualised
  - relationships among classes and sub-classes are expressed page 92
  - Graphs and tables are used to explain Software Engineering principles
14. In the case of \_\_\_\_\_, intra component linkages are stronger while inter component linkages are weak.
- high cohesion
  - low coupling page 73
  - low cohesion
  - high coupling

15. Transactions are the \_\_\_\_\_ that must be remembered through time.

**a. Events** page 93

- b. Actions
- c. Triggers
- d. Methods

16. The modules interacting with each other through message passing have \_\_\_\_\_ between them.

- a. low cohesion
- b. high cohesion

**c. low coupling** page 74

**d. high coupling**

17. In UML based Object Oriented model of a system, the inheritance relation between two classes is shown by a \_\_\_\_\_ sign towards the super-class side on the association line between the two.

- a. A filled diamond
- b. An unfilled diamond
- c. A half arrowhead

**d. An arrowhead** page 92

18. Which statement is not according to the software engineering principles? Software engineering is a(n)\_\_\_\_\_.

- a. Balancing act
- b. Disciplined approach
- c. **Unsystematic approach** page 5
- d. Quantifiable approach

19. A \_\_\_\_\_ is not the real product but just a real looking mock-up of what would be eventually delivered.

- a. Software
- b. Program
- c. **Prototype(page 68)**
- d. Test Case

20. The best way to conduct a requirements validation review is to

- a. examine the system model for errors
- b. have the customer look over the requirements
- c. send them to the design team and see if they have any concerns
- d. **use a checklist of questions to examine each requirement**

21. More powerful hardware resulted into the development of \_\_\_\_\_ powerful and \_\_\_\_\_ software.

- a. less, complex

**b. more, complex** page 4

- c. more, simple
- d. less, simple

22. \_\_\_\_\_ pointed out the elegant conceptual integrity exhibited by layered organization of software systems, with the resulting gains in development and maintenance ease.
- a. David Parnas
  - b. Shaw and Garlan
  - c. Barry Boehm
  - d. **EdsgerDijkstra** **page 115**
23. The architecture components for product engineering are
- a. **data, hardware, software, people**
  - b. data, documentation, hardware, software
  - c. data, hardware, software, procedures
  - d. documentation, hardware, people, procedures
24. In Object oriented design, combining the services offered by an object with the attributes they work on, results in:
- a) Lower coupling and stronger cohesion
  - b) Lower cohesion and stronger coupling
  - c) Increased likelihood of reuse
  - d) Decreases the modularity of the system
- a. a only
  - b. band c
  - c. **a and c**
  - d. b and d
25. Architectural design process involves performing a number of activities which includes system structuring, control modeling and\_\_\_\_\_.
- a. System Analysis
  - b. **Modular Decomposition** **page 120**
  - c. Testing
  - d. Graphical User Interface
26. In use case diagram, the scope of the system is defined by:
- a. Actor
  - b. Entity
  - c. **System Boundary**
  - d. "Extends" relationship
- 27.Specialization means
- a. Calling the same method with object of child object
  - b. Hiding the data
  - c. **Creating new subclasses from an existing class** **page 34 , 86**
  - d. Abstraction
28. Which of the following sentence is true regarding user interface design?
- a. GUI interfaces are good for all tasks which a user needs to perform at an interface.
  - b. The higher the response time, the better is the interface
  - c. **The simpler the interface, the efficient is the system**
  - d. Command-line interfaces are faster for some tasks which the user needs to perform

29. In the architecture trade-off analysis method the architectural style should be described using the
- a. data Flow view and process view
  - b. data Flow view and module view
  - c. module view and process view
  - d. data Flow view, module view and process view page 136
30. In Data Flow Diagram, the entity or system, outside the boundary of this system is called
- a. Process
  - b. Data Flow
  - c. External Agent
  - d. Data Store
31. A cohesive Class is one which emphasizes on \_\_\_\_\_ unit of functionality
- a. Single
  - b. Multiple page 76
  - c. Static
  - d. Both Single and Multiple
32. The goal of \_\_\_\_\_ is to translate the customer's desire for a set of defined capabilities into a working product.
- a. Electrical Engineering
  - b. Product Engineering
  - c. Hardware Engineering
  - d. Mechanical Engineering
33. In Object Oriented Design, \_\_\_\_\_ layer contains the details that enable each object to communicate with its collaborators
- a. Subsystem
  - b. Responsibility
  - c. Message page 89
  - d. Object
34. A DFD is normally leveled (adding more levels of abstraction) as
- a. it is a good idea in design
  - b. it is recommended by many experts
  - c. it is easy to do it
  - d. it is easier to read and understand a number of smaller DFDs than one large DFD
35. As per Peter Coad's methodology, which of the following may NOT be a perfect candidate for being an object?
- a. Zone
  - b. Recipient
  - c. Garage
  - d. Password page 93 94

36. Which of the given component of software engineering framework provides different techniques that can be used to perform a particular task?
- a. Processes
  - b. Tools
  - c. Methods page 12**
  - d. Quality Focus**
37. To determine the architectural style or combination of styles that best fits the Proposed system requirements engineering is used to uncover
- a. algorithmic complexity
  - b. characteristics and constraints page 126**
  - c. control and data
  - d. design patterns
38. In \_\_\_\_\_ relationship, a class shares the structure and behavior defined in another class.
- a. Aggregation
  - b. Composition
  - c. Inheritance page 86**
  - d. Uses
39. Prototyping is used when there is regarding requirements.
- a. Uncertainty(page 68)**
  - b. Confirmation
  - c. Conflict
  - d. Consensus
40. The process of utilizing our knowledge of computer science in effective production of software systems is called
- a. Chemical Engineering
  - b. Electrical Engineering
  - c. Computer Engineering
  - d. Software Engineering page 2**
41. In Abbot's Textual Analysis technique, different parts of speech are identified within **the text of the specification and these parts are modeled using different\_\_\_\_\_**
- a. Event
  - b. Process
  - c. Operations
  - d. Components page 90**
42. To help separate an object's external behavior from its implementation, the technique used is called \_\_\_\_\_
- a. Generalization
  - b. Association
  - c. Composition
  - d. Abstraction page 86**

43. A Process in Data Flow Diagram (DFD) represents
- a. Flow of data
  - b. Transformation of data (page 49)**
  - c. Storage of data
  - d. An external agent
44. In data flow diagram (DFD). Create, Update, Delete and Read operations are normally called:
- a. CRUD operations page 53**
  - b. DURC operations
  - c. RUDC operations
  - d. CDUR operation
45. Collaboration diagram can show \_\_\_\_\_
- a) Binary messages
  - b) Asynchronous messages
  - c) Synchronous messages
- a. a only
  - b. b only
  - c. c only
  - d. both b and c page 111**
46. Data cannot flow from one external entity to other external entity because
- a. It will get corrupted**
  - b. It is not allowed in DFD page 59**
  - c. An external entity has no mechanism to read or write
  - d. Both are outside the context of the system
47. In \_\_\_\_\_ Phase of software development, Requirement Engineer focuses on realizing the Business Object of an under developed product.
- a. Maintenance**
  - b. Development page 16**
  - c. Definition
  - d. Vision
48. Which of the following is not part of software architecture?
- a. Databases
  - b. data design
  - c. program structure
  - d. algorithm details**
49. Selecting objects in a domain) include:
- a. Actors. Participants and Places**
  - b. Only Participants
  - c. Only Actors
  - d. Only Actors and Places
50. Defining the services of an object means:
- a. What it does? Page 96**
  - b. What it knows?
  - c. Who knows it?
  - d. Whom it knows?

51. In sequence diagram, the solid lines depict:
- Objects (or classes)
  - Messages being sent from one object to the other** page 107
  - Life-time of an object
  - state of the object
52. Sequence diagrams:
- Provide the static behavior
  - Provide Data Flow
  - Provide a time-based view** 106
  - Provide parallel data flow**
53. \_\_\_\_\_ requirements cause frequent modifications in user interface
- Functional
  - Non-functional
  - Unstable** page 62
  - User
54. , security and maintainability are the types of \_\_\_\_\_ requirements.
- Non-functional** page 120
  - Domain
  - Functional
  - Business
55. A change becomes \_\_\_\_\_ because of close presence of data and functions
- Accessible
  - Global
  - Private
  - Localized** page 81
56. System \_\_\_\_\_ are built to allow the system engineer to evaluate the system components in relationship to one another.
- Requirements
  - Documents
  - Models** page 42
  - Test cases**
57. In \_\_\_\_\_ phase of software development, requirement analyst focuses on possible design of the proposed solution.
- Maintenance
  - Development** (page 16)
  - Definition
  - Vision
58. The focus of sequence diagrams is:
- On objects/classes and messages exchanged among them** (page 106)
  - On static Model of system
  - On object constraints
  - On the flow of Control

59. In Data Flow Diagram (DFD), data flow can:
- Only originate from an external entity
  - Only terminate in an external entity
  - Originate and terminate in an external entity**
  - Either originate or terminate in an external entity but not both
60. In \_\_\_\_\_ the analyst determines all the sources of requirements and where do these requirements consume
- Data Flow Analysis
  - Source and Sink Analysis (page 40)**
  - Down Parsing
  - Up Parsing
61. In a top-down system analysis, a/an \_\_\_\_\_ required to develop high level view of the system at first.
- Analyst page 54**
  - Designer
  - Tester
  - Developer
62. The Use case diagram shows that which \_\_\_\_\_ interact with each use case.
- Use case
  - Actor page 32**
  - Component
  - Relation
63. The context diagram is used as the top level abstraction in a \_\_\_\_\_ developed according to principles of structured analysis.
- Dataflow diagram (page 31)**
  - Activity Diagram
  - State Transition Diagram
  - Use Case Diagram
64. Different messages in sequence diagrams includes:
- Simple
  - Asynchronous
  - Notify
  - Both Simple and Asynchronous page 108**
65. A software requirement document describes all the \_\_\_\_\_ provided by the system along with the constraints under which it must operate.
- Conditions
  - Services page 23**
  - Events
  - Processes
66. In case of a \_\_\_\_\_ message, the called routine that handles the message is completed before the caller resumes execution.
- Synchronous
  - Asynchronous
  - Bidirectional
  - a only page 108**
- b only
  - c only
  - both b and c

67. In object oriented design, \_\_\_\_\_ layer contains the data structures and algorithmic design for all attributes and operations for each object.
- Subsystem
  - Class and Object
  - Message
  - Responsibility** page 89
68. Asynchronous messages:
- are implemented as operation call
  - These block caller before response
  - occurs in multi-threaded applications** (page 109)
  - are shown by dotted line**
69. In UML based object oriented model of a system, a composition relation between two objects is shown by a \_\_\_\_\_ sign on the whole side of a relation line.
- An unfilled diamond
  - A filled diamond**
  - A half diamond
  - A dot
70. In UML based Object Oriented Model of a system, the diamond sign is used to depict \_\_\_\_\_ **relations between two objects/classes.**
- Aggregation and Association
  - Inheritance and Association
  - Composition and Aggregation**
  - Composition, Aggregation and Association
71. A “register” in “Point of sale system” is an example of:
- Actor
  - Participant
  - Tangible thing** page 100
  - Transaction
72. A use case represents
- A class, its attributes and operation
  - An operation’s interfaces and signature
  - The role a user plays when interaction with the system** page 32
  - The system’s functionality for a particular purpose
73. In the case of \_\_\_\_\_ approach, data is decomposed according to functionality requirements.
- Object-oriented
  - Action-oriented** page 80
  - Event-oriented
  - Process-oriented
74. To construct a system model the engineer should consider one of the following restraining factors?
- Assumptions and constraints**
  - Budget and expenses
  - Data object and operation
  - Schedule and milestone

75. When two components of a system are using the same global data area, they are related as
- a. Data coupling
  - b. Content coupling
  - c. Common coupling** google
  - d. External coupling**
76. \_\_\_\_\_ Structure represents the internal organization of the various data and control items.
- a. **Data**
  - b. Value
  - c. Information
  - d. Conceptual
77. The method of dividing and assigning different portions of a large system to different groups for construction is called \_\_\_\_\_
- a. **Work Breakdown Structure** (page 119)
  - b. Working Boundary Structure
  - c. Work Basic Structure
  - d. Work Breakdown System
78. "A car is made up of a body, three or four wheels, a steering mechanism. a breaking mechanism and a power-engine"
- The above statement is example of :**
- a. **Whole-Part relationship** page 95
  - b. Inheritance
  - c. Specialization
  - d. Generalization
79. In software engineering paradigm, any engineering approach must be founded on organizational commitment to \_\_\_\_\_.
- a. Cost
  - b. Scheduling
  - c. Quality** (page 15)
  - d. Performance**
80. \_\_\_\_\_ is a technique that can be used to reduce customer dissatisfaction at requirement stage.
- a. Analysis
  - b. Negotiation
  - c. Prototyping** (page 71)
  - d. GUI**
81. Which of the following activities are included in the design process of a software architecture \_\_\_\_\_?
- a. System Development and Deployment
  - b. Architectural Analysis and Testing
  - c. Requirement Specifications of the system
  - d. System Structuring and Modular Decomposition

82. In UML based Object Oriented model of a system, the association relation between two objects is depicted by\_\_\_\_\_.
- a. **A straight line**
  - b. A filled diamond sign on the whole side of the line
  - c. An unfilled diamond sign on the whole side of the line
  - d. Any arrowhead sign on one side of the line
83. \_\_\_\_\_are kind of umbrella activities that are used to smoothly and successfully perform the construction activities
- a. Designee activities
  - b. Management activities (page 14)**
  - c. Testing activities
  - d. Maintenance activities
84. Software Design discusses \_\_\_\_\_aspect of software development
- a. What
  - b. **How**
  - c. Who
  - d. When
85. Include and extend relationship is used in UML notation of a/an\_\_\_\_\_.
- a. Activity Diagram
  - b. Data Flow Diagram
  - c. Entity Relationship Diagram
  - d. **Use Case Model**
- 86.External Entity may be
- a. source of input data only
  - b. **source of input data or destination of results**
  - c. destination of results only
  - d. repository of data
87. An object model of a system captures the \_\_\_\_\_structure of a system.
- a. Static page 93**
  - b. dynamic
  - c. iterative
  - d. Hierarchical
88. The criteria used to assess the quality of an architectural design should be based on system
- a. accessibility and reliability
  - b. **data and control**
  - c. functionality
  - d. implementation details
89. In case of \_\_\_\_\_approach, decomposition of a problem revolves around data
- a. Object-oriented
  - b. **Action-oriented page 80**
  - c. Event-oriented
  - d. Process-oriented

90. How can we implement generalization in Object Oriented programming languages?
- Polymorphism
  - Encapsulation
  - Abstraction
  - Inheritance**
91. Which of the following is NOT among one of the four layers of the Object Oriented (OO) design pyramid
- The subsystem layer
  - The class and object layer
  - The Abstract layer** **page 89**
  - The message layer**
92. \_\_\_\_\_ is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.
- Inheritance
  - Polymorphism
  - Aggregation
  - Abstraction** **page 79**
93. The project manager would need \_\_\_\_\_ document to monitor and track the progress of the project.
- Design
  - Project
  - Requirement** **page 18**
  - Planning**
94. In the case of action-oriented approach data is decomposed according to:
- Object requirements
  - Functionality requirements**
  - Corresponding domain model
  - Compatibility with object interface
95. An architectural style encompasses which of the following elements?
- Constraints, Set of Components and Semantic Models** **page 126**
  - Set of Components and Semantic Models
  - Semantic Models and Constraints
  - Set of Components and Constraints
96. In order to determine the role and responsibilities of the identified objects, we need to consider which of the following step(s):
- Who I am?
  - What I know?
  - Who I know?
  - What I do?
- a only
  - a and b
  - b, c and d** **page 102**
  - c and d

97. In sequence diagram, the boxes denote:

**a. Objects (or classes) page 106**

- b. Messages, sent from one object to other
- c. Life-time of Objects
- d. Relationships

98. \_\_\_\_\_ is a system component that provides services to other components but would not normally be considered as a separate system.

a. Method

**b. Module page 121**

- c. Message
- d. Relationship

99. A tangible entity in the real life is called \_\_\_\_\_.

a. Functions

**b. Object page 85**

- c. Class
- d. Sub-Class

100. Sequence of messages can be present in:

- a) Use case diagram
- b) Sequence diagram
- c) Collaboration diagram

a. a only

b. b only

c. c only

d. **b and c**

101. The system model template contains which of the following elements

a. Input

b. Output

c. System Out

d. **Input/Output**

102. Identify the TRUE statement:

a. **Normally Object Oriented design is more maintainable than functional oriented.**

b. Software with Functional oriented design does not fulfill non functional requirements.

c. Object Oriented design cannot implement "Separation of concerns" strategy

d. Function Oriented design does not lead to an efficient product

103. An external entity that interacts with a system is called a(n):

a. Use case

b. **Actor**

c. Stakeholder

d. Association

104. By levelling a DFD (adding more levels of abstraction) we mean

a. Splitting it into different levels

b. Make its structure uniform

c. **Expanding a process into one with more sub-processes giving more detail**

d. Summarizing a DFD to specify only the essentials

105. Software Engineering is the combination of tools, techniques and\_\_\_\_\_.
- a. Testing
  - b. Processes page 6
  - c. Maintenance
  - d. Design
106. \_\_\_\_\_ is concerned with decomposing a system into interacting sub-systems.
- a. System Structuring
  - b. Control Modeling
  - c. Modular Decomposition page 121
  - d. Work Breakdown Structure
107. There are \_\_\_\_\_most important characteristics of an object.
- a. Six
  - b. Four
  - c. Two
  - d. Three
108. “System should maintain transaction log of every transaction”
- The above statement is an example of
- a. Functional requirement
  - b. Non-functional requirement
  - c. Pseudo requirement
  - d. Both Non-functional and Pseudo requirements
109. In object oriented approach,\_\_\_\_\_are the people and organizations that take part in the system under consideration.
- a. Actors
  - b. Places
  - c. Participants
  - d. Tangible things
110. The modules that interact with each other through message passing have \_\_\_\_\_.
- a. Low coupling page 73
  - b. High coupling
  - c. Low cohesion
  - d. High cohesion
111. Identifying Whole-part structures (Aggregations) means what are my \_\_\_\_\_
- a. Components page 94
  - b. Attributes
  - c. Methods
  - d. Messages
112. In “point of sale system “ the term payment represents
- a. Actor
  - b. Participant
  - c. Transaction (page 99)
  - d. Container

113. Return values in synchronous messages are:
- Compulsory
  - May not used when response is obvious (page 109)
  - Not used at all
  - Represented by solid lines
114. GUI stand for
- Generic user interface
  - Graphic user interface
  - Generic user interaction
  - Graphical user interaction
115. Which of the following is not the object model principle?
- Abstraction
  - Encapsulation
  - Hierarchy or inheritance
  - Exposure page 86
116. The system specification describes the
- function and behavior of a computer-based system
  - implementation of each allocated system element
  - algorithmic detail and data structures
  - time required for system simulation
117. The \_\_\_\_\_ provides the software engineer with a view of the system as a whole.
- Process Model
  - Architectural Model
  - Business model
  - Requirements Model
118. Requirement engineering focuses on \_\_\_\_\_ aspect of the software development process.
- Both what and how
  - What
  - How
  - why and how
119. An arrow in Data Flow Diagram (DFD) represents
- Direction of flow of data
  - Processing of data
  - External agent
  - Internal agent
120. The \_\_\_\_\_ relationship is kind of a generalization specialization relationship.
- Bit-Byte
  - Uses
  - Binary
  - Extends

121. In the case of \_\_\_\_\_ in a system, module boundaries are not well defined.
- low cohesion
  - high coupling
  - low coupling
  - high cohesion
122. \_\_\_\_\_ Component of software engineering framework provides automated or semi-automated support in a software development.
- Processes
  - Methods
  - Quality Focus
  - Tools
123. In Data Flow Diagram (DFD), one data store cannot directly copy the data from another \_\_\_\_\_.
- Agent
  - Process
  - Data store
  - Flow
124. All the documents related to the software are also considered as part of the \_\_\_\_\_.
- Physical Document
  - Logical Document
  - Relational Database
  - Software
125. Which elements of business processing engineering are the responsibilities of the software engineer?
- business area analysis
  - business system design
  - product planning
  - information strategy planning
126. A class will be cohesive if:
- Class does not implement Complex interfaces
  - Class does not have complex methods
  - If most of the methods do not use most of the data members most of the time
  - If most of the methods use most of the data members most of the time
127. A context diagram is used
- As the first step in developing a detailed DFD of a system
  - In systems analysis of very complex systems
  - As an aid to system design
  - As an aid to programmers
128. \_\_\_\_\_ is an extremely powerful technique for dealing with complexity.
- Aggregation
  - Abstraction
  - Inheritance
  - Association

129. In multiprocessing applications, different execution threads may pass information to one another by sending\_\_\_\_\_to each other.
- Interrupt calls
  - Synchronous messages
  - Asynchronous messages
  - System calls
130. Normally a system is more easy to modify if its modules have
- High coupling and high cohesion
  - High coupling and low cohesion
  - Low coupling and high cohesion
  - Low coupling and Low cohesion
131. UML is a language for
- High-level Programming
  - Low-level Programming
  - Modeling and Design
  - Creating diagrams only
132. \_\_\_\_\_is not the part of Peter Coad methodology
- Select actors
  - Select participants
  - Select places
  - . Select actions
133. Which of the following is the external quality of software product?
- Correctness
  - Concision
  - cohesion
  - Low coupling
134. While establishing the services for an object, the goal is to keep data and action together for\_\_\_\_\_coupling and\_\_\_\_\_cohesion
- Lower, Higher
  - Higher, Lower
  - Lower, Lower
  - Higher, Higher
135. A necessary supplement to transform or transaction mapping needed to create a complete architectural design is
- entity relationship diagrams
  - the data dictionary
  - processing narratives for each module
  - test cases for each module
136. Construction activities are directly related to software\_\_\_\_\_.
- Management
  - Planning
  - Quality Assurance
  - Development

137. OOD results in a design that achieves a number of different levels of \_\_\_\_\_.
- a. Operation
  - b. Event page 89
  - c. Modularity
  - d. Process
138. At which stage of software development loop, results are delivered?
- a. Problem definition
  - b. Solution integration
  - c. Technical development
  - d. Status quo
139. \_\_\_\_\_ Requirements are often called product features.
- a. Functional
  - b. Non-functional
  - c. Developer
  - d. User
140. Strong cohesion implies that
- a. All parts of component have a class logical relationship with each other
  - b. All part of component do not have a close logical relationship with each other
  - c. Component is dynamic in nature
  - d. Component is static in nature
141. \_\_\_\_\_ Diagram does not capture control flow information; it just shows the flow of the data in a system.
- a. Sequence
  - b. Data Flow
  - c. Activity
  - d. Class
142. A life line represents the object's life during the interaction in a sequence diagram while its notation is depicted by \_\_\_\_\_
- a. Solid Lines
  - b. Dotted Lines
  - c. Full Arrow
  - d. Curved Lines
143. An attribute that varies over time, e.g. price of an item, should be replaced by a/an \_\_\_\_\_ with an effective date and value
- a. Additional Class
  - b. Additional Method/Function
  - c. Interface
  - d. Structure
144. Which of the given component of software engineering framework demands rational and Timely development of a software?
- a. Tools
  - b. Methods
  - c. Quality Focus
  - d. Processes
145. A context diagram
- a. describes detailed design of a system
  - b. is a DFD which gives an overview of the system
  - c. is a detailed description of a system

- d. is not used in drawing a detailed DFD
146. Which one is not a part of Software Development phase?
- a. Construction
  - b. **Scope**
  - c. Project Vision
  - d. Definition
147. Software architecture must address \_\_\_\_\_ requirement of a software system
- a. Functional
  - b. Non Functional
  - c. User interface Requirements
  - d. **Both Functional and Non Functional**
148. If Cat is derived from Mammal Class, and Mammal is derived from Animal Class, then:
- a. **Cat will inherit Animal's functions and data**
  - b. Cat will not inherit Animal's functions and data
  - c. Cat will not be able to access any class
  - d. Cat is allowed to access only the Mammal's Class
149. Which of the following is not supported by a maintainable design?
- a. Change
  - b. debugging
  - c. Adding new features
  - d. **Higher maintenance cost**
150. The condition that must be met before the use case can be invoked, is called:
- a. **Pre-Condition**
  - b. Post-Condition
  - c. Pre-Assertion
  - d. Post-Assertion

1. Choose the correct option in terms of Issues related to professional responsibility
  - a) Confidentiality
  - b) Intellectual property rights
  - c) Both Confidentiality & Intellectual property rights**
  - d) Managing Client Relationships
2. Efficiency in a software product does not include \_\_\_\_\_
  - a) responsiveness
  - b) licensing**
  - c) memory utilization
  - d) processing time
3. As per an IBM report, “31% of the project get cancelled before they are completed, 53% overrun their cost estimates by an average of 189% and for every 100 projects, and there are 94 restarts”. What is the reason for these statistics?
  - a) Lack of adequate training in software engineering**
  - b) Lack of software ethics and understanding
  - c) Management issues in the company
  - d) All of the mentioned

4. The reason for software bugs and failures is due to
- a) Software companies
  - b) Software Developers
  - c) Both Software companies and Developers**
  - d) All of the mentioned
5. Build & Fix Model is suitable for programming exercises of \_\_\_\_\_ LOC (Line of Code).
- a) 100-200**
  - b) 200-400**
  - c) 400-1000**
  - d) above 1000**
6. RAD stands for
- a) Relative Application Development
  - b) Rapid Application Development**
  - c) Rapid Application Document
  - d) None of the mentioned
7. Which one of the following models is not suitable for accommodating any change?
- a) Build & Fix Model
  - b) Prototyping Model
  - c) RAD Model
  - d) Waterfall Model**

8. Which is not one of the types of prototype of Prototyping Model?
  - a) Horizontal Prototype
  - b) Vertical Prototype
  - c) Diagonal Prototype**
  - d) Domain Prototype
9. Which one of the following is not a phase of Prototyping Model?
  - a) Quick Design
  - b) Coding**
  - c) Prototype Refinement
  - d) Engineer Product
10. Which of the following statements regarding Build & Fix Model is wrong?
  - a) No room for structured design
  - b) Code soon becomes unfixable & unchangeable
  - c) Maintenance is practically not possible
  - d) It scales up well to large projects**
11. RAD Model has
  - a) 2 phases
  - b) 3 phase
  - c) 5 phases**
  - d) 6 phases

**Explanation: RAD Model consists of five phases namely: Business modeling, Data modeling, Process modeling, Application generation and Testing & Turnover.**

12. What is the major drawback of using RAD Model?
  - a) Highly specialized & skilled developers/designers are required
  - b) Increases reusability of components
  - c) Encourages customer/client feedback
  - d) Increases reusability of components, Highly specialized & skilled developers/designers are required**
13. 9. SDLC stands for
  - a) Software Development Life Cycle**
  - b) System Development Life cycle
  - c) Software Design Life Cycle
  - d) System Design Life Cycle
14. 10. Which model can be selected if user is involved in all the phases of SDLC?
  - a) Waterfall Model
  - b) Prototyping Model
  - c) RAD Model**
  - d) both Prototyping Model & RAD Model
15. What are the types of requirements?
  - a) Availability
  - b) Reliability
  - c) Usability
  - d) All of the mentioned**
16. Select the developer-specific requirement?
  - a) Portability
  - b) Maintainability

- c) Availability
  - d) Both Portability and Maintainability**
17. Which one of the following is not a step of requirement engineering?
- a) Elicitation
  - b) design**
  - c) analysis
  - d) documentation

**Explanation: Requirement Elicitation, Requirement Analysis, Requirement Documentation and Requirement Review are the four crucial process steps of requirement engineering. Design is in itself a different phase of Software Engineering.**

18. FAST stands for
- a) Functional Application Specification Technique
  - b) Fast Application Specification Technique
  - c) Facilitated Application Specification Technique**
  - d) None of the mentioned
19. QFD stands for
- a) quality function design
  - b) quality function development
  - c) quality function deployment**
  - d) none of the mentioned
20. A Use-case actor is always a person having a role that different people may play.
- a) True
  - b) False**

**Explanation: Use-case Actor is anything that needs to interact with the system, be it a person or another (external) system.**

21. The user system requirements are the parts of which document ?
- a) SDD
  - b) SRS**
  - c) DDD
  - d) SRD
22. A stakeholder is anyone who will purchase the completed software system under development.
- a) True
  - b) False**
23. Conflicting requirements are common in Requirement Engineering, with each client proposing his or her version is the right one.
- a) True**
  - b) False
24. Which is one of the most important stakeholder from the following ?
- a) Entry level personnel
  - b) Middle level stakeholder
  - c) Managers
  - d) Users of the software**
25. Which of the following property does not correspond to a good Software Requirements Specification (SRS)?
- a) Verifiable
  - b) Ambiguous**
  - c) Complete
  - d) Traceable

26. The SRS document is also known as \_\_\_\_\_ specification.  
**a) black-box**  
b) white-box  
c) grey-box  
d) none of the mentioned
27. Which of the following is included in SRS?  
**b) Design Constraints**  
c) Staffing  
d) Delivery Schedule
28. The Unified Modeling Language (UML) has become an effective standard for software modelling. How many different notations does it have?  
a) Three  
b) Four  
c) Six  
**d) Nine**

**Explanation: The different notations of UML includes the nine UML diagrams namely class, object, sequence, collaboration, activity, state-chart, component, deployment and use case diagrams.**

29. Which model in system modelling depicts the dynamic behavior of the system?  
a) Context Model  
**b) Behavioral Model**  
c) Data Model  
d) Object Model
30. Which model in system modelling depicts the static nature of the system?  
a) Behavioral Model  
b) Context Model  
c) Data Model  
**d) Structural Model**
31. Which perspective in system modelling shows the system or data architecture?  
**a) Structural perspective**  
b) Behavioral perspective  
c) External perspective  
d) All of the mentioned
32. Activity diagrams are used to model the processing of data.  
**a) True**  
b) False
33. Model-driven engineering is just a theoretical concept. It cannot be converted into a working/executable code.  
**a) True**  
**b) False**
34. The UML supports event-based modeling using \_\_\_\_\_ diagrams.  
a) Deployment  
b) Collaboration  
**c) State chart**  
d) All of the mentioned
35. Which is the first step in the software development life cycle?  
a) Analysis  
b) Design

**c) Problem/Opportunity Identification**

d) Development and Documentation

36. Which tool is use for structured designing ?

a) Program flowchart

**b) Structure chart**

c) Data-flow diagram

d) Module

**Explanation: A Structure Chart (SC) in software engineering and organizational theory, is a chart which shows the breakdown of a system to its lowest manageable levels.**

37. A step by step instruction used to solve a problem is known as

a) Sequential structure

b) A List

c) A plan

**d) An Algorithm**

38. In the Analysis phase, the development of the \_\_\_\_\_ occurs, which is a clear statement of the goals and objectives of the project.

a) documentation

b) flowchart

**c) program specification**

d) design

39. Actual programming of software code is done during the \_\_\_\_\_ step in the SDLC.

a) Maintenance and Evaluation

b) Design

c) Analysis

**d) Development and Documentation**

40. Who designs and implement database structures.

a) Programmers

b) Project managers

c) Technical writers

**d) Database administrators**

41. \_\_\_\_\_ is the process of translating a task into a series of commands that a computer will use to perform that task.

a) Project design

b) Installation

c) Systems analysis

**d) Programming**

42. Debugging is:

a) creating program code

**b) finding and correcting errors in the program code**

c) identifying the task to be computerized

d) creating the algorithm

43. In Design phase, which is the primary area of concern?

a) Architecture

b) Data

c) Interface

**d) All of the mentioned**

44. The importance of software design can be summarized in a single word which is:

- a) Efficiency
- b) Accuracy
- c) Quality**
- d) Complexity

45. Cohesion is a qualitative indication of the degree to which a module

- a) can be written more compactly
- b) focuses on just one thing**
- c) is able to complete its function in a timely manner
- d) is connected to other modules and the outside world

**Explanation: Cohesion of a single module/component is the degree to which its responsibilities form a meaningful unit.**

46. Coupling is a qualitative indication of the degree to which a module

- a) can be written more compactly
- b) focuses on just one thing
- c) is able to complete its function in a timely manner
- d) is connected to other modules and the outside world**

47. How many diagrams are there in Unified Modelling Language?

- a) six
- b) seven
- c) eight
- d) nine**

**Explanation: The nine UML diagrams include use-case, sequence, collaboration, activity, state- chart, deployment, class, object and component.**

48. What can static analysis NOT find?

- a. The use of a variable before it has been defined.
- b. Unreachable (“dead”) code.
- c. Memory leaks.**
- d. Array bound violations.

49. Which plan describes how the skills and experience of the project team members will be developed ?

- a. HR Plan
- b. Manager Plan
- c. Team Plan
- d. Staff Development Plan**

50. Alpha and Beta Testing are forms of\_\_\_\_\_.

- a. Acceptance testing**
- b. Integration testing
- c. System Testing
- d. Unit testing

51. What is the Interaction diagram?

- a) Interaction diagrams are the UML notations for dynamic modeling of collaborations
- b) Interaction diagrams are a central focus of engineering design
- c) All of the mentioned**
- d) None of the mentioned

52. What are the different interaction diagram notations does UML have?

- a) A sequence diagram
- b) A communication diagram

c) An interaction overview diagram

**d) All of the mentioned**

53. What is a sequence diagram?

**a) A diagram that shows interacting individuals along the top of the diagram and messages passed among them arranged in temporal order down the page**

b) A diagram that shows messages superimposed on a diagram depicting collaborating individuals and the links among them

c) A diagram that shows the change of an individual's state over time

d) All of the mentioned

54. Which of the following is true about Sequence frames?

a) A sequence diagram has a frame consisting of a rectangle with a pentagon in its upper left-hand corner

b) The pentagon is its name compartment; the interaction is represented inside the rectangle

c) The string in the name compartment has the form sd interaction Identifier where interaction Identifier is either a simple name or an operation specification with the same format as in a class diagram

**d) All of the mentioned**

55. What is a lifeline?

a) It is a frame consisting of a rectangle with a pentagon in its upper left-hand corner

**b) It is a rectangle containing an identifier with a dashed line extending below the rectangle**

c) It is a name compartment; the interaction is represented inside the rectangle

d) None of the mentioned

56. What does a message mean?

**a) It Passes all communications from one object to another and are represented by message arrows in sequence diagrams**

b) The message goes from the sending object's lifeline to the receiving object's lifeline

c) It is a rectangle containing an identifier with a dashed line extending below the rectangle

d) All of the mentioned

57. What are the three different types of message arrows?

a) Synchronous, asynchronous, asynchronous with instance creation

b) Self, Multiplied, instance generator

**c) Synchronous, Asynchronous, synchronous with instance creation**

d) None of the mentioned

58. Which of these are true with respect to the message arrows?

a) The synchronous message arrow is used when a sending individual continues execution after sending the message

b) The asynchronous message arrow is used when a sending individual suspends execution after sending the message

**c) The dashed arrow is used either to show the return of control from a synchronous message or to create a new entity**

d) All of the mentioned

59. When is the operation executing, suspended and active?

a) An operation is executing when some process is actually running its code

b) An operation is suspended when it sends a synchronous message and it is waiting for the message to return

c) An operation is active when it is either executing or suspended

**d) All of the mentioned**

60. What is the interaction fragments?
- a) A fragment which is a rectangular frame with a pentagonal operation compartment in the upper left-hand corner
  - b) A fragment which has a marked part of an interaction specification
  - c) The regions resulting from these divisions will not hold the interaction fragment operations
  - d) All of the mentioned**
61. What are the heuristics which the sequencing diagram follows?
- a) Put pairs of lifelines that interact heavily next to one another
  - b) Position lifelines to make message arrows as short as possible
  - c) Position lifelines to make message arrows go from left to right
  - d) All of the mentioned**
62. \_\_\_\_\_ is yet another technique that can be used to reduce customer dissatisfaction at the requirement stage.
- a. Design
  - b. UML
  - c. RAD
  - d. Prototype**
63. How to get vision of the product and get early feedback from user to ensure that the development team understands requirements.
- a. Using prototype**
  - b. Using sequence diagram
  - c. Using Design Document
  - d. None of given
64. A \_\_\_\_\_ is not the real product. It is rather just a real looking mock-up of what would be eventually delivered and might not do anything useful.
- a. Design
  - b. UML
  - c. Prototype**
  - d. Standard
65. During the \_\_\_\_\_, the focus shifts from what to how.
- a. Implementation
  - b. Analysis
  - c. Requirement
  - d. Design**
66. \_\_\_\_\_, modularity, and abstraction are different but related principles.
- a. Separation of concern**
  - b. Layering
  - c. Substitution
  - d. Cohesion
67. A complex system may be divided into smaller pieces of lesser complexity called \_\_\_\_\_
- a. Pieces
  - b. Fragmentation
  - c. Modules**
  - d. Abstraction
68. \_\_\_\_\_ provides a road map for implementation by clearly describing how the software system is to be realized.

- a. Requirement Engineering
  - b. Implementation process
  - c. Software design
  - d. Testing process
69. How many software design strategies are there \_\_\_\_\_
- a. 3
  - b. 2**
  - c. 4
  - d. 5
70. In the \_\_\_\_\_ software design strategy, the structure of the system revolves around functions.
- a. object oriented design
  - b. Class oriented design
  - c. Module oriented Design
  - d. functional or structured design**
71. A \_\_\_\_\_ is the one in which cost of system change is minimal and is flexible enough so that it can be easily adapted to modify existing functionality and add new functionality.
- a. Software Design Qualities
  - b. Maintainable Design**
  - c. Coupling and Cohesion
  - d. Software Design Strategies
72. \_\_\_\_\_ is a measure of independence of a module or component.
- a. Cohesion
  - b. Coupling**
  - c. Mentality
  - d. Cohesive
73. \_\_\_\_\_ means that different system components have loose or less reliance upon each other
- a. Loose coupling**
  - b. Tight coupling
  - c. Loose cohesion
  - d. Tight cohesion
74. \_\_\_\_\_ are contrasting concepts but are indirectly related to each other.
- a. Coupling and cohesion**
  - b. Object Oriented & Function oriented
  - c. Flowchart and data flow
  - d. None of given
75. \_\_\_\_\_ is an internal property of a module whereas \_\_\_\_\_ is its relationship with other modules.
- a. Coupling, cohesion
  - b. Cohesion, coupling**
  - c. Object, function
  - d. Module, layering
76. \_\_\_\_\_ describes the intra-component linkages while \_\_\_\_\_ shows the inter-component linkages.
- a. Coupling, cohesion
  - b. Cohesion, coupling**
  - c. Object, function
  - d. Module, layering

77. Coupling measures the interdependence of two modules while cohesion measures the independence of a module.
- a. Coupling, cohesion**
  - b. Cohesion, coupling
  - c. Object, function
  - d. Module, layering
78. \_\_\_\_\_ is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.
- a. Encapsulation
  - b. Abstractions**
  - c. Modularization
  - d. Object Design
79. The technique that allowing us to look at its important external characteristic, at the same time, hiding its inner complexity.
- a. Abstractions**
  - b. De-encapsulation
  - c. Encapsulation
  - d. Hidden
80. Hiding the internal details is called \_\_\_\_\_
- a. Abstractions
  - b. De-encapsulation
  - c. Encapsulation**
  - d. Hidden
81. function oriented Design is also called \_\_\_\_\_
- a. Class oriented
  - b. Scope Oriented
  - c. Object oriented
  - d. Action oriented**
82. Briefly, inheritance denotes a
- a. "kind of" relationship**
  - b. "part of" relationship
  - c. "has a" relationship
  - d. None of given
83. In -----approach, data is decomposed according to functionality requirements.
- a. Object Oriented
  - b. Function oriented
  - c. Action oriented
  - d. Both B & C**
84. In the \_\_\_\_\_ approach, decomposition of a problem revolves around data.
- a. Class oriented
  - b. Function oriented
  - c. Object Oriented**
  - d. Action oriented
85. Briefly, aggregation denotes a
- a. "kind of" relationship
  - b. "part of" relationship**
  - c. "has a" relationship
  - d. None of given

86. \_\_\_\_\_ Paradigm focuses only on the functionality of a system and typically ignores the data until it is required.
- a. Object Oriented
  - b. class oriented
  - c. Module Oriented
  - d. **Action Oriented**
87. \_\_\_\_\_ paradigm focuses both on the functionality and the data at the same time.
- a. **Object-oriented**
  - b. Action oriented
  - c. Function oriented
  - d. Inheritance
88. Which one of the following the oldest techniques to identify objects and their relationships.
- a. The Coad Methodology
  - b. **Abbot's Textual Analysis**
  - c. Dijkstra's Approach
  - d. None of given
89. The \_\_\_\_\_ diagram can be used to model simple sequential flow, branching, iteration, recursion and concurrency.
- a. Use case
  - b. Dataflow
  - c. Class
  - d. **Sequence**
90. In the sequence diagram object are shown on the \_\_\_\_\_ axis.
- a. Y-axis
  - b. Z-Axis
  - c. **X-Axis**
  - d. Object are not shown in the sequence diagram
91. Sequence diagrams can depict \_\_\_\_\_ different types of messages.
- a. 2
  - b. **4**
  - c. 6
  - d. 8
92. Collaboration diagrams have basically two types of components:
- a. Class & Object
  - b. Use case & user
  - c. **Objects and messages.**
  - d. Function & Data
93. Sequence diagrams are best to see the flow of time. On the other hand, static object connections are best represented by \_\_\_\_\_ diagrams.
- a. Class diagram
  - b. Sequence diagram
  - c. **Collaboration diagram**
  - d. Object design
94. UML provides \_\_\_\_\_ different mechanisms to document the dynamic behavior of the system.
- a. 3
  - b. 5
  - c. 4
  - d. **2**

95. In SRS \_\_\_\_\_ means that two different readers of the same document interpret the requirement differently.
- a. **Ambiguity**
  - b. Clearness
  - c. Self-deceptive
  - d. None of given
96. Use case model component are
- a. Message, function
  - b. Actor, message
  - c. Cases, data
  - d. **Cases, actor**
97. In the past, software was considered as a collection of \_\_\_\_\_ to transform that information from input to the output format.
- a. Code & Programming
  - b. Data and sequence
  - c. Algorithms and code
  - d. **information and procedures**
98. The process of productive use of scientific knowledge is called \_\_\_\_\_
- a. Software engineering
  - b. Requirement engineering
  - c. **Engineering.**
  - d. Science
99. "The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; is called
- a. Computer Science
  - b. Engineering
  - c. **Software engineering**
  - d. Requirement engineering
100. After reaching at a reasonable level of quality we do not try to improve the quality of software any further this law is called
- a. Balance law
  - b. Saturated law
  - c. **Law of diminishing returns**
  - d. None of given
101. There are \_\_\_\_\_ basic phases of software development
- a. 3
  - b. 5
  - c. **4**
  - d. 2
102. In \_\_\_\_\_ SE phase we control the change in system, whether that change is in the form of enhancements or defect removal.
- a. Development
  - b. **Maintenance**
  - c. Definition
  - d. Analysis
103. Following the Software Engineering Phases except the
- a. **Mission**
  - b. Vision
  - c. Maintenance
  - d. Development
104. \_\_\_\_\_ as a statement of needs by a user that triggers him development of a program or system.
- a. Requirements analysis

- b. Requirement engineering
  - c. **Requirement**
  - d. Flow chart
105. \_\_\_\_\_ add further detail to the business requirements.
- a. Functional requirement
  - b. **User requirements**
  - c. Non-Functional requirement
  - d. None of given
106. Following all are the Requirements statement Characteristics except the:
- a. Correct
  - b. Verifiable
  - c. Prioritized
  - d. **Polite**
107. \_\_\_\_\_ collected from multiple sources might conflict.
- a. **Business requirements**
  - b. Functional Requirement
  - c. User Requirements
  - d. Non-Functional requirement
108. All are the typical non-functional requirements are **except** the
- a. Reliability.
  - b. **Feasible**
  - c. Availability
  - d. Scalability.
109. A \_\_\_\_\_, sometimes called a level 0 data-flow **diagram**, is drawn in order to define and clarify the boundaries of the software system.
- a. Data through
  - b. **context diagram**
  - c. Conceptual diagram
  - d. Structure diagram
110. In a \_\_\_\_\_ model, boundaries of the system are defined by **functionality that is handled by the system.**
- a. In OOP Design
  - b. In Function oriented
  - c. **use case**
  - d. None of given.

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## CS504 Solved Grand Quiz 2020

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Which of the following is not the object model principle?

**Select Correct  
Option**

- ✓ Abstraction
- ✓ Encapsulation
- ✓ Hierarchy or inheritance
- ✓ **Exposure** .....

Question No 2

Return values in Synchronous messages are represented by:

**Select Correct  
Option**

- ✓ A solid line
- ✓ A dotted line with label
- ✓ **A solid line with label** .....
- ✓ Double line

Question No 3

The \_\_\_\_\_ relationship is kind of a generalization specialization relationship.

**Select Correct**

### Option

- ✓ Bit=byte
- ✓ Uses
- ✓ Binary
- ✓ **Extends**.....(

#### Question No 4

The architecture components for product engineering are

### Select Correct Option

- ✓ **Data, hardware, software, people**..... (
- ✓ Data, documentation, hardware, software
- ✓ Data, hardware, software, procedures
- ✓ Documentation, hardware, people, procedures

### Question No 5

A context diagram is used

**Select Correct Option**

- ✓ As the first step in developing a detailed DFD of a system
- ✓ **In systems analysis of very complex systems**
- ✓ As an aid to system design
- ✓ As an aid to programmer

### Question No 6

Consider the following piece of code:

```
Public class Square extends Shape {
// some code
}
```

The above code is an example of:

**Select Correct Option**

- ✓ Part-Whole relationship
- ✓ Generalization/Specialization
- ✓ **Data Sharing**..... (
- ✓ Data encapsulation

### Question No 7

When two components of a system are using the same global data area, they are related as

**Select Correct Option**

- ✓ Data Coupling
- ✓ Content Coupling
- ✓ **Common Coupling**.....(
- ✓ External Coupling

### Question No 8

The context diagram is used as the top level abstraction in a \_\_\_\_\_ developed according to principles of structures analysis.

Select Correct Option

- ✓ **Dataflow diagram**.....(
- ✓ Activity Diagram
- ✓ State Transition Diagram
- ✓ Use Case Diagram

### Question No 9

In the case of \_\_\_\_\_ in a system, module boundaries are not well defined.

Select Correct Option

- ✓ Low cohesion
- ✓ **High coupling** ..... (
- ✓ Low coupling
- ✓ High cohesion

### Question No 10

In UML based Object Oriented model of a system, the association relation between two objects is depicted by \_\_\_\_\_.

Select Correct Option

- ✓ A straight line
- ✓ A filled diamond sign on the whole side of the line
- ✓ **An unfilled diamond sign on the Whole side of the line**
- ✓ An arrowhead sign on one side of the line

### Question No 11

External Entity may be

Select Correct Option

- ✓ Source of input data only
- ✓ **Source of input data or destination or results**
- ✓ Destination of results only

✓ Repository of data

### Question No 12

In UML based Object Oriented model of a system, the inheritance between two classes is shown by a \_\_\_\_\_ sign towards the super-class side on the association line between the two.

**Select Correct Option**

- ✓ A filled diamond
- ✓ An unfilled diamond
- ✓ A half arrowhead
- ✓ **An arrowhead**..... (

### Question No 13

\_\_\_\_\_ are kind of umbrella activities that are used to smoothly and successfully perform the construction activities.

**Select Correct Option**

- ✓ Design Activities
- ✓ **Management Activities**.....(
- ✓ Testing Activities
- ✓ Maintenance Activities

### Question No 14

\_\_\_\_\_ is not the part of Peter Coad methodology.

**Select Correct Option**

- ✓ Select Actors
- ✓ Select participants
- ✓ Select places
- ✓ **Select actions** .....(

### Question No 15

An object model of a system captures the \_\_\_\_\_ structure of a system.

**Select Correct Option**

- ✓ **Static**..... (
- ✓ Dynamic

- ✓ Iterative
- ✓ Hierarchal

### Question No 16

As per Peter Coad's methodology, which of the following may NOT be a perfect candidate for being an object?

Select Correct Option

- ✓ Zone
- ✓ Recipient
- ✓ Garage
- ✓ **Password**.....(

### Question No 17

The system model template contains which of the following elements

Select Correct Option

- ✓ Input
- ✓ Output
- ✓ System Out
- ✓ **Input / Output**.....(

### Question No 18

A \_\_\_\_\_ is not the real product but just a real looking mock-up of what would be eventually delivered.

Select Correct Option

- ✓ Software
- ✓ Program
- ✓ **Prototype**.....(
- ✓ Test Case

### Question No 19

USL is a language for\_\_\_\_\_.

Select Correct Option

- ✓ High-level programming

- ✓ Low-level Programming
- ✓ **Modeling and Design** .....(
- ✓ Creating Diagrams only

### Question No 20

In multiprocessing applications, different execution threads may pass information to one another by sending\_\_\_\_\_to each other.

Select Correct Option

- ✓ Interrupt calls
- ✓ Synchronous messages
- ✓ **Asynchronous messages** ..... (
- ✓ Systems calls

### Question No 21

To construct a system model the engineer should consider one of the following restraining factors?

Select Correct Option

- ✓ **Assumptions and constrains**.....(
- ✓ Budget and expenses
- ✓ Data objects and operations
- ✓ Schedule and milestones

### Question No 22

A software requirement document describes all the\_\_\_\_\_provided by the system along with the constraints under which it must operate.

Select Correct Option

- ✓ Conditions
- ✓ **Services** ..... (
- ✓ Event
- ✓ Processes

### Question No 23

In “Point of Sale system”, the term “Payment” represents

Select Correct Option

- ✓ Actor

- ✓ Participant
- ✓ **Transaction**.....
- ✓ Container

### Question No 24

A \_\_\_\_\_ relationship indicates that one entity is composed of one or more parts which are themselves instances of that or another entity

Select Correct Option

- ✓ Inheritance
- ✓ **Whole-part**.....(
- ✓ Generalization
- ✓ Specialization

### Question No 25

An arrow in Data Flow Diagram (DFD) represents

Select Correct Option

- ✓ Direction of flow of data
- ✓ **Processing of data** .....(
- ✓ External agent
- ✓ Internet agent

### Question No 26

In sequence diagram, the boxes donates

Select Correct Option

- ✓ **Objects (or classes)**..... (
- ✓ Messages, sent from one object to other
- ✓ Life-time of objects
- ✓ Relationships

### Question No 27

Which of the given component of software engineering framework provides different techniques that can be used to perform a particular task?

Select Correct Option

- ✓ **Processes** .....(
- ✓ Tools



- ✓ Methods
- ✓ Quality Focus



**Question  
No 28**

**Which of the following is not supported by a maintainable design?**

**Select  
Correct  
Option**

- ✓ **Change**..... (
- ✓ Debugging
- ✓ Adding new features
- ✓ Higher maintenance cost

**Question  
No 29**

**A necessary supplement to transform or transaction mapping needed to create a complete architectural design is**

**Select  
Correct  
Option**

- ✓ Entity relationship diagrams
- ✓ The data dictionary
- ✓ **Processing narratives for each module**
- ✓ Test cases for each module

**Question  
No 30**

**A prototype is not the real product. It is rather just a real looking \_\_\_\_\_ of what would be eventually delivered and might not do anything useful.**

**Select  
Correct  
Option**

- ✓ **Mock-up** .....
- ✓ Ad-hoc
- ✓ Design
- ✓ Structure

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**AND Many MoRe**

1. In a....., each program module is represented by a rectangular box.

**Use case diagram 33**

**Class diagram**

**Document flow**

**diagram Data flow**

**diagram**

2..... is a diagramming technique used to identify the types of objects in the system

and the static relationships that exist among

them? **Class Diagram**

**Document flow**

**diagrams Data flow**

**diagrams**

**Flow charts**

3. Which one is NOT a type of messages which Sequence Diagrams Depict \_\_\_\_\_

Synchronous

Synchronous

Create

**Update 108**

4. Identifying Whole-Part structures (Aggregations) means, what are my -----

**Components 95**

**Structures**

**Modules**

**Interaction Protocols**

5. The design process for identifying the sub-systems making up a system and the framework for sub-system control and communication is:

**Architectural Design**

**115 Interface Design**

**Component Design**

**Data Design**

6. A complex System evolves from a

Smaller system

**Simpler system 69**

**Bigger system**

**Medium system**

7. A context diagram \_\_\_\_\_  
**Describes the context dependencies of a system**

**Is a DFD which gives an overview of the system**

**31 Is a detailed description of a system?**

**Is not used in drawing a detailed DFD**

8. When measure of independence of a module or component is low to the other,  
How would changes in one component have effect on other component?

**No Low**

**71 High**

**Equal**

9. What is meant by the term 'software crisis'? Ability  
to deliver software on time with good quality

**A situation in which experienced developers leave the  
company Inability of new software to inter-operate with  
existing software**

**A situation in which large scale software projects took more time and budget  
than was planned**

10. Which is not included in test criteria applied in a phase of testing?  
Functional validity

**Interface**

**integrity**

**Correctness**

**Programing**

**Logic**

11. For inspections,-----are prepared that contain information regarding defects.

**Tables**

**Checklists 210**

**Lists**

**Frames**

12. Static analyzers are software tools for ----- processing.

**Analysis text**

**Source text 211**

**Design text**

**Maintenance**

**Text**

13. Holistic medicine, concerns itself with the state of the body as a whole, not the-----  
**that is currently attacking**

**it. Target**

**Reason**

**Disease 224**

**Source**

14. What factor has no precipitation in more sophisticated and complex computer-based  
systems?

**Vast use of personal computers.**

**Vast increases in computer memory and storage  
capacity. Greater variety of exotic input/output  
options.**

**Profound changes in computer architectures.**

15. Software Bugs have multiple names. Which one of the below is not the name of Software  
Bugs?

**Bugs**

**Defects**

**Errors**

**Mistakes 213**

16. Unit testing is roughly equivalent to \_\_\_\_\_ testing for hardware in which each chip is tested thoroughly after manufacturing.

**Circuit level**

**Chip Level 207**

**Component level**

**System level**

17. Equivalence\_\_\_\_\_help you in designing test cases to test the system effectively and efficiently.

**Functions**

**Classes 202**

**Objects**

**Interfaces**

18-\_\_\_\_\_are used to describe flow of data or control in an application. Code structures

**String**

**matching. Paths**

**Flow graphs 203**

19. Writing test cases and generating test data are processes that demand\_\_\_\_\_building capabilities.

**Domain**

**System**

**Scenario 198**

**Code**

20. The raising of the imaginary error flag is simply called raising or\_\_\_\_\_an error.

**Throwing 187**

**Sending**

**Casting**

**Handling**

21. \_\_\_\_\_requirements are often called product features.

Functional

**Business 22**

**User**

**Non-functional**

22. Many compilers limit the maximum number of bits in the bit field to the size of a(n)

\_\_\_\_\_.  
**Integer 183**

**Float**

**Character**

**Double**

23. In Java, ">>" is used for\_\_\_\_\_shift and ">>>" for\_\_\_\_\_shift.

**Arithmetic, Logical 181**

**Mathematical, Logical**

**Incremental,**

**Arithmetic Logical,**

**Arithmetic**

24. When large amount of data is to be shared, repository model is used. This model has been extensively used in the\_\_\_\_\_based application.

**Mainframe 130**

**Super Computers**

**Personal  
Computers Real  
time**

25. There are four basic coding structures: sequence, if statement, case statement, and

\_\_\_\_\_.

**For loop**

**While loop 203**

**Switch statement**

**Logical**

**operations**

26. \_\_\_\_\_ is only a reference for defining protocols and designing and implementing systems developed by different parties.

**OSI model**

**Reference architecture 137**

**Layered architecture**

**N-tier architecture**

27. The pattern movement became very quiet until \_\_\_\_\_ when patterns appeared again at OOPSLA conference.

**1987 141**

**1988**

**1962**

**1995**

28. STL Stands for \_\_\_\_\_.

Standard Temporary Line

Standard Temporary Library

**Standard Template Library 141**

**Standard Type Link**

29. MVC stands for \_\_\_\_\_.

**Modern View Center**

**Model View Controller 143**

**Modern View**

**Controller Model View**

**Center**

30. \_\_\_\_\_ ensures that a class only has one instance and provides a global point of access to it.

**Singleton Pattern 145**

**Observer Pattern**

**Behavioral**

**Pattern Joint**

**Pattern**

31. Which design pattern provides a unified interface to a set of interfaces in a sub-system?

**Façade 146**

**Singleton**

**Observer**

**Joint**

32. Testing activities require destructive instincts in \_\_\_\_\_ for the purpose of breaking system to discover loopholes into its functionality.

**Bug Fixers**

**Tester 198**

**Developer**

**Requirement**

**Engineer**

33. Camel Case is now the official convention for file names and identifiers in the \_\_\_\_\_

**programming**

language. C#

C++

Java

152

## Visual Basic.Net

34. In presence of \_\_\_\_\_ bug in a program, the results are the opposite of what is expected.

## Memory Leak

## Memory Over- runs Syntax Error

## Logical Error 221

35. Split lines occur when a statement exceed the \_\_\_\_\_ column limit.

80 158

90

95

85

36. Identifier names play a significant role in enhancing the \_\_\_\_\_ of a program.

Writability

## Readability 151

## Reliability

## Usability

37. "is" prefix should be used for \_\_\_\_\_ variables and methods.

Static

## General

## Boolean 155

## Constant

38. Comments should be indented relative to their position in the \_\_\_\_\_.

## Code 165

## Design

## Analysis

## Requirements

39. Modularity is a tool that can help us in \_\_\_\_\_ the size of individual functions.

Stabilizing

## Reducing 173

## Increasing

## Strengthening

40. When planning for performance, one should always remember the \_\_\_\_\_ rule.

80/20

180

70/20

60/20

100/20

41. Suppose you are working as a software engineer involved in the development of an e-commerce website.

## What are the 2 most important characteristics your software must have? 2 marks

42. Name any two standards which are being used by C/C++ for writing programs. 2 marks

43. Which other technique along with testing should we use for increasing the effectiveness of defect removal efficiency in a project ? 2marks

44. What do you understand by term "self-documenting code"? 2 marks

45. Briefly describe a business process diagram (at least 3 lines): 3marks

46. How does software differ from the artifacts produced by other engineering disciplines ? 3 marks

47. What is Exception handling? 3marks

48. In which scenario, short circuiting is used? 3marks

49. Rewrite the following code by following General naming conventions for C++

- 1) int fileprefix;
- 2) const int max\_iterations;
- 3) int getmaxiterations() { ... }
- 4) template<class t>
- 5) private int length; 05 marks

50. "A manufacturing company assembles the fans. For this purpose it purchases the raw material containing wires, motors, wings, and magnets. It then assembles the fans from these products".

**From above case study, identify different objects that, you think, should be present in final implementation of system.**

**Note: you can use any methodology to identify the objects. 5marks**

51. Discuss the importance of software debugging. 5marks
52. Exceptions do increase the code complexity. Explain this with the help of an example. 5marks

**1. Which of the items listed below is NOT one of the software engineering layers? Tools**

**Process**

**Manufacturing Methods**

02. Which of the following is a non-functional requirement of a website which sells songs? A catalogue of the stock needs to be available for the users to choose from

**Customer information should be retained to allow future transactions easier Users should be able to choose from a set of different languages for the interface Time taken to download songs in the catalogue should not irritate users**

03. The state transition diagram \_\_\_\_\_ Depicts relationships between data objects Depicts functions that transform the data flow

**Indicates how data are transformed by the system Indicates system reactions to external events**

04. Control flow diagrams are:

**Needed to model event driven systems Required for all systems Used in place of data flow diagrams Useful for modeling user interfaces**

05. Prototyping \_\_\_\_\_

**Ensures getting the design right the first time Is the execution of the standard systems development cycle using CASE tools. Is used to verify if a key process is feasible Involves an iterative development process with minimum end use involvement.**

06. Identifying Whole-Part structures (Aggregations) means, what are my -----

**Components**

**Structures 98**

**Modules**

**Interaction Protocols**

07. Software architecture is "The \_\_\_\_\_ of the components of a program/system, their interrelationships, and principles and guidelines governing their design and evolution over

time".  
**Combination  
Collection**

## Structure 120

### Unification

08. The object-behavior model indicates how the system \_\_\_\_\_

**Functions in the operating environment**  
**Objects collaborate with one another**  
**Responds to external stimuli**

**Responds to internal stimuli**

09. Perry and Wolfe proposed the following formula for software architecture:

Software architecture = {Forms, Rationale}

**Software architecture = {Elements, Forms, Rationale}**

**125 Software architecture = {Parts, Rationale}**

**Software architecture = {Parts, Elements}**

10. Repository model also provides the ----- view of the system.

**Global 130**

**Local**

**General**

**Private**

11. In the case of zero-install, the network environment is used to \_\_\_\_\_ server side processing by adding a number of servers which share processing load.

**Distribute 133**

**Centralize**

**Shift**

**De-Synchronize**

12. Requirement Statement and Requirement Specification are also called Requirement Definition and \_\_\_\_\_

**Functional Specification 28**

**Mathematical**

**specification System**

**Specification**

**Cost Specification**

13. Which of the following is not a fundamental structured programming construct?

Recursion

**Condition**

**Repetition**

**Sequence**

14. For equivalence partitions, we divide the problem in-----obvious categories of

**equal strings**

**4**

**3**

**2**

**202**

**5**

15. Which is not included in test criteria applied in a phase of testing?

Functional validity

**Interface**

**integrity**

**Correctness**

**Programing**

**Logic**

16. Static analyzers are software tools for ----- processing.

**Analysis text**

**Source text      214**

**Design text**

**Maintenance**

**Text**

17. The first “bug” was actually a moth, which flew through an open window and into one of the Mark-----'s relays.

**II 216**

**I**

**III**

**IV**

18. Holistic medicine, concerns itself with the state of the body as a whole, not the -----  
**that is currently attacking**  
**it. Target**

**Reason**

**Disease**

**Source**

19. Bugs Fixing is done by which of the teams in Software Development lifecycle?  
Development Team

**Testing Team not**

**sure Analysis & Design Team**

**Process Team**

20. Which of the following testing involve purely black box testing?

Unit testing, Beta testing

**Acceptance testing, Interface**

**testing Beta testing, Acceptance**

**testing Integration testing,**

**Interface testing**

21. A memory leak bug is one in which memory is somehow allocated from either the operating system or an----- "pool", but never deallocated when the memory is finished

**being used.**

**External**

**memory Mixed**

**memory**

**Internal Memory 219**

**Mutually exclusive memory**

22. Equivalence\_\_\_\_\_help you in designing test cases to test the system effectively and efficiently.

**Functions**

**Classes 202**

**Objects**

**Interfaces**

23\_\_\_\_\_are used to describe flow of data or control in an application. Code structures

**String**

**matchings Paths**

**Flow graphs 203**

24. \_\_\_\_\_requirements are often called product features.

Functional

**Business**

**User**

**Non-functional**

25. In Java, ">>" is used for\_\_\_\_\_shift and ">>>" for\_\_\_\_\_shift.

**Arithmetic, Logical**

**Mathematical, Logical  
Incremental,  
Arithmetic Logical,  
Arithmetic**

26. When large amount of data is to be shared, repository model is used. This model has been extensively used in the \_\_\_\_\_based application.

**Mainframe 130**

**Super Computers**

**Personal**

**Computers Real**

**time**

27. Idea of zero install architecture is to develop a system where no installation on the \_\_\_\_\_ is needed.

**Client side 133**

**Server Side**

**Client & Server**

**Side Network**

28. \_\_\_\_\_ is only a reference for defining protocols and designing and implementing systems developed by different parties.

**OSI model**

**Reference architecture 137**

**Layered**

**architecture N-tier**

**architecture**

29. Description of communicating objects and classes that are customized to solve a general design in a particular context is called\_\_\_\_\_.

**Design Pattern 140**

**System Pattern**

**System Design**

**Design System**

30. Which of the following is the correct formula for calculating cyclomatic complexity of a program ?

**V - N + 2**

**E - V + 2**

**E - N + 2 208**

**E + N - 2**

31. In unit testing, developers test\_\_\_\_\_code units (modules, classes, etc.) during implementation.

**Other's**

**Their own 210**

**Mixed code of**

**people Hidden**

32. Performance & Usability are examples of\_\_\_\_\_requirements.

**Business**

**Functional**

**Non-Functional 213**

**User**

33. Which of the following is the correct definition for software testing?

The process of demonstrating that errors are not present

**The process of establishing confidence that a program does what it is supposed to do**  
**The process of executing a program to show it is working as per specifications**

**The process of executing a program with the intent of finding errors**

34. The code is misbehaving in a way that cannot be easily explained.  
The above statement is the symptom of\_\_\_\_\_.

**Runtime Error**

**Logical Error 221**

**Memory**

**Leak**

**Exception**

35. Secondary private classes can be declared as \_\_\_\_\_ and reside in the file of the class they belong to.

**Asynchronous Classes**

**Inner classes 158**

**Outer classes**

**Synchronous**

**classes**

36. Split lines occur when a statement exceed the \_\_\_\_\_ column limit.

**80 158**

**90**

**95**

**85**

37. Identifier names play a significant role in enhancing the \_\_\_\_\_ of a program.

Writability

**Readability 151**

**Reliability**

**Usability**

38. Names representing methods and functions should be \_\_\_\_\_ and written in mixed case starting with \_\_\_\_\_ case.

**Noun, Lower**

**Noun, Upper 153**

**Verb,**

**Upper**

**Verb,**

**Lower**

39. "is" prefix should be used for \_\_\_\_\_ variables and methods.

Static

**General**

**Boolean 155**

**Constant**

40. If you are initializing a variable at the time of declaration, do not declare another \_\_\_\_\_ in the same

statement. **Variable 179**

**Object**

**Class**

**Module**

41. Consider the following statement:

**"A basket contains 2 Apples and 3 bananas".**

**What is the relationship between basket and fruits (apples and bananas) 02marks**

42. What issue can arise if code is written without keeping the portability factor in mind ?02

43. Which four techniques should be applied in parallel to attain maximum defect removal efficiency ? 02

44. How names representing abbreviations and acronyms should be described in coding style guide? 02

45. What is Textual Analysis? Explain it.03

46. Discuss the symptoms of Memory and Resource Leak bug class. 03
47. Write any two guidelines for developing equivalence partitioning. 03

48. When a small set (a couple of functions) of functions which use each other is so overwhelmingly the bottleneck, there are two alternatives. Write down at least one of them.03
49. Using the architecture of a house or building as a metaphor, draw a comparison with software architecture.
50. Discuss the symptoms and an example of Loop errors bug class. 5
51. In Intel architecture, the lowest byte is the most significant byte while in Motorola architecture the highest byte of a word is the most significant one.

**Discuss how the above mentioned factor affects portability of program 05**

52. Which of the following is/are example of cryptic code? x

\*= a;

**subkey = subkey >> (bitoff – (bitoff >> 3) << 3)); a = a >> 2;**

### **FINALTERM EXAMINATION**

1. A change becomes----- because of close presence of data and functions

**Localized**

**Private**

**Global**

**Accessible**

2. Which of the following is NOT an objective for building an analysis model?

Develop an abbreviated solution for the problem

**Establish basis for software**

**design Define set of software**

**requirements Describe customer**

**requirements**

3. Data flow diagrams are used to handle processes involved in \_\_\_\_\_

**Business Process**

**Transformation Flow of data**

**53**

**Flow of**

**information Cost**

**estimation**

4. The first most important study in developing any system is to identify

Purpose of the system

**Implementation of the**

**system Design of the system**

**Architecture of the system**

**not sure ☺**

5. Software Architecture is all about \_\_\_\_\_

**Requirements gathering**

**Design**

**not conform**

**Coding**

**Testing**

**06. Dynamic components of an OOA model are \_\_\_\_\_**

**Not reusable**

**Sensitive to timing and event processing**

**Stable throughout the operational life of an**

**application Structural in nature**

7. UML (unified modeling language) analysis modeling focuses on the \_\_\_\_\_.

**Behavioral model and environment model.**  
**Behavioral model and implementation**  
**model. User model and environmental**  
**model**

## User model and structural model

8. Perry and Wolfe proposed the following formula for software architecture:

Software architecture = {Forms, Rationale}

**Software architecture = {Elements, Forms, Rationale}**  
**Software architecture = {Parts, Rationale}**

**Software architecture = {Parts, Elements}**

9. Repository model also provides the----- view of the system

**Global 130**

**Local**

**General**

**Private**

10. Which of the below is NOT a part of Architectural styles?

Data Center architecture

**Client Server Architecture**

**Hierarchical Architecture 129**

**Layered Architecture**

11. Consider the following scenario:

**“Student logs in the system and checks his/her lecture schedule”**

**Keeping in mind the use case diagram, and above scenario the one of the post conditions might be:**

**Student should have a**

**login System should be**

**available**

**Student should log-off after checking**

**schedule Lecture schedule should be**

**displayed date wise**

12. When measure of independence of a module or component is low to the other,

How would changes in one component have effect on other component?

**No**

**Low**

**High**

**Equal**

13. One of the most powerful features of exception handling is that an error can be -----

**over function**

**boundaries. Thrown 187**

**Called**

**back**

**Caught**

**Sent**

14. What is the normal order of activities in which software testing is organized?

Unit, integration, system, validation

**System, integration, unit,**

**validation Unit, integration,**

**validation, system Integration,**

**unit, system, validation**

15. Two tests are considered to be equivalent if it is believed that: if one discovers a defect, the other probably will too, and if one does not discover a defect, -----.

**The other probably  
may The other  
probably can**

**The other probably won't either**

**202**

**The other probably will**

16. For equivalence partitions, we divide the problem in-----obvious categories of  
**equal strings**

4  
3  
2  
5

17. In ----- testing we test the structure of the program.

**White Box**      **201**

**Black Box**

**System**

**Regression**

18. Static analyzers are software tools for ----- processing.

**Analysis text**

**Source text**      **214**

**Design text**

**Maintenance**

**Text**

19. Unit testing is roughly equivalent to \_\_\_\_\_ testing for hardware in which each chip is tested thoroughly after manufacturing.

**Circuit level**

**Chip Level**      **210**

**Component**

**level System**

**level**

20. A SRS is said to be \_\_\_\_\_ if and only if, every requirement stated therein has only one interpretation.

**Compact**

**Unambiguous**

**Consistent**

**Detailed**

21. In Java, ">>" is used for \_\_\_\_\_ shift and ">>>" for \_\_\_\_\_ shift

**Arithmetic, Logical**

**Mathematical, Logical**

**Incremental,**

**Arithmetic Logical,**

**Arithmetic**

22. A(n) \_\_\_\_\_ is a variance from a desired product attribute

**Error**

**Exception**

**Defect**      **195**

**Mistake**

23. st cases should be generated for unit testing and system testing during \_\_\_\_\_.

**Testing and maintenance**

**respectively Requirement analysis**

**Design and requirement analysis**

**respectively Coding and design**

**respectively**

24. a of zero install architecture is to develop a system where no installation on the \_\_\_\_\_ is needed.

**Client side**      **133**

**Server Side**

## **Client & Server**

### **Side Network**

25. \_\_\_\_\_ is NOT a layer of operating system.

Core layer

### **Application layer**

## **Presentation layer 136**

### **UI layer**

26. patterns are devices that allow programs to share knowledge about their\_\_\_\_\_.

**Design 140**

### **Code**

### **Analysis**

### **Testing**

27. Which design pattern provides a unified interface to a set of interfaces in a sub-system?

**Façade 146**

### **Singleton**

### **Observer**

### **Joint**

28. In unit testing, developers test\_\_\_\_\_code units (modules, classes, etc.) during implementation.

### **Other's**

**Their own 210**

### **Mixed code of**

### **people Hidden**

### **29Inspections cannot check**

\_\_\_\_\_characteristics. **Non-**

**Functionanl. 213**

### **Business**

### **User**

### **Functional**

30. Performance & Usability are examples of\_\_\_\_\_requirements.

Business

### **Functional**

**Non-Functional 213**

### **User**

31. Inspections may be applied to any representation of the system which may include all except\_\_\_\_\_.

### **Requirements**

### **Design**

### **Testing**

**Analysis 213**

32. Debuggers can be used to display values of\_\_\_\_\_.

### **local variables only**

### **global variables**

### **only static variables**

### **only**

**both local and global variables 229**

33. The code is misbehaving in a way that cannot be easily explained.

The above statement is the symptom of\_\_\_\_\_.

### **Runtime Error**

**Logical Error 221**

### **Memory**

### **Leak**

### **Exception**

34. Ideally speaking, a function should not be larger than\_\_\_\_\_lines of code and in any

case should not exceed one page in length.

**20**

**150**

**15**

**30**

25

35. CamelCase is now the official convention for file names and identifiers in the \_\_\_\_\_

**programming**

**language. C#**

**C++**

**Java 152**

**Visual Basic.Net**

36. Split lines occur when a statement exceed the \_\_\_\_\_ column limit.

80

90

95

85

37. The construction should appear at the \_\_\_\_\_ of the header file

Top 160

**Bottom**

**Left**

**Right**

38. Identifier names play a significant role in enhancing the \_\_\_\_\_ of a program.

Writability

**Readability**

**Reliability**

**Usability**

39. One of the guidelines to avoid common mistakes is to never use \_\_\_\_\_ except for declar

, **lectr 33or 34**

;

=

||

40. When planning for performance, one should always remember the \_\_\_\_\_ rule.

80/20

70/20

60/20

100/20

41. E = Number of Edges = 8

N = Number of Nodes = 6

**Using the above measures, calculate the cyclomatic complexity, V(G) of Graph G. 02**

42. What do you understand by the term System Downtime ? 02

43. If a class uses most of the methods and most of the data members most of the time defined in that class. Then class is having high cohesion or high coupling ? 2

44. Conditional expression with negation is always difficult to understand while writing code. Give an example. 2

45. Define (Be specific) the following following terms:

1) Program

2) Data

3) Documentation 03

46. Describe three coverage schemes related to white box testing. 3

47. Why Special characters like TAB and Page Break must be avoided.? Explain 3

48. Do you agree or not that the use of *do - while* loops should be avoided. Give solid reason? 3

49. Consider the following examples:

After playing games/watching shows/engaging in your hobby for X number of hours, your enjoyment level decreases.

After eating X amount of food, your satisfaction of the additional food becomes lesser. Every additional unit of fertilizer generates a different increase in yield. The first few units probably gives a burst in yield compared to without fertilizer. After that, adding more fertilizer will still give more yield, but at a decreasing rate.

From the above examples, which Law of Economics comes to your mind? Discuss Software Engineering case by applying that law. 05.

50. Discuss the symptoms and an example of logical error bug class. 5.

51. Being a software tester, what do you think would be the software testing objectives? Also which test will you call a successful test ? 05

52. Below are two programming terms. You are required to write their uses with at least one example.

*get/set*

*is prefix* 05.

## FINAL TERM EXAMINATION

### CS-504-Software Engineering

1. Following are some statements associated with data flow diagrams. Identify the correct statement from among them.

DFDs are used to model complex interfaces.

DFDs are used to represent only functional processing, data stores and data movements between functions.

DFDs depict only processes which can be decomposed.

DFDs do not show external data sources and external data sinks.

2. A project is considered successful if:

The system was delivered in time and within budget.

The system meets at least some of the customer's requirements.

The system development process has a maximum impact on the ongoing business process. Minimum time was spent for requirement gathering and designing.

3. A process which does not take input is called \_\_\_\_\_

**Miracle process**

Core process

Secondary

process Zombie

process

4. Data Flow Diagrams \_\_\_\_\_

Replace the job of systems analysts

Do not allow every function to work at the same time.

Can be automatically generated by computer aided software engineering tools **Describes how data flows from one logical processing unit to another**

5. In a ....., each program module is represented by a rectangular box.

Use case diagram

Document flow

diagram Class

diagram

### **Data flow diagram**

6. Identify the correct description.

**The collection of use cases for a system constitutes all the defined ways the system may be used.**

**UML stands for Unified Markup Language**

**State transition diagrams are drawn for objects with no significant dynamic behavior. Object Model depicts the dynamic behavior of system**

7. Dynamic components of an OOA model are \_\_\_\_\_

**Not reusable**

**Sensitive to timing and event processing**

**Stable throughout the operational life of an application Structural in nature**

8. To determine the architectural style or combination of styles that best fits the proposed system, requirements engineering is used to uncover:

**Algorithmic complexity**

**Characteristics and**

**constraints Control and data**

**Design patterns**

9. Krutchen's 4+1 architectural view model proposes the development of \_\_\_\_\_ main views

5 125

4

3

2

10. Repository model also provides the ----- view of the system.

**Global**

**Local**

**General**

**Private**

11. Client server model tries to \_\_\_\_\_ data and processing.

Distribute 130

**Centralize**

**Process in**

**parallel Combine**

12. We can divide the whole Software Engineering process in 4 distinct phases namely vision, definition, development, and maintenance.

**Which of the following is not among "Definition related" activities of software development? Requirement verification**

**Change Control not sure**

☺ **Requirement Engineering**

**Software Requirements**

13. What is meant by the term 'software crisis'?

Ability to deliver software on time with good quality

**A situation in which experienced developers leave the company Inability of new software to inter-operate with existing software**

**A situation in which large scale software projects took more time and budget than was planned**

14. Which of the following are external qualities of a software product?

Maintainability, reusability, portability, efficiency, correctness

Correctness, reliability, robustness, efficiency, usability

**Portability, interoperability, maintainability, reusability**

**Robustness, efficiency, reliability, maintainability, reusability**

15. Which writing style is best regarding identifier role in enhancing the readability of a

program?

**If (Flag == 0)**

**If (Flag == START\_NUMBER)**

If (Z = =  
START\_NUMBER) If  
(Z = 0)

16. A useful technique for evaluating the overall complexity of a proposed architecture is to look at the component

**Number and size of components**

**Flow dependencies and sharing**

**dependencies Size and cost**

**Algorithms used**

17. For equivalence partitions, we divide the problem in-----obvious categories of  
**equal strings**

4

3

2

5

18. In ----- testing we test the structure of the program.

**White Box**

**Black Box**

**System**

**Regression**

19. Which is not included in test criteria applied in a phase of testing?

Functional validity

**Interface**

**integrity**

**Correctness**

**Programing**

**Logic**

20. Inspections can check conformance with a specification but not conformance with the---  
-----**requirements.**

**Customer's real 213**

**Developer**

**Tester**

**Manger**

21. Static analyzers are software tools for ----- processing.

**Analysis text**

**Source text**

**Design text**

**Maintenance**

**Text**

22. Holistic medicine, concerns itself with the state of the body as a whole, not the-----  
**that is currently attacking**

**it. Target**

**Reason**

**Disease**

**Source**

23. Software deteriorates rather than wears out because

Software suffers from exposure to hostile environments

**Defects are more likely to arise after software has been used**  
**often Multiple change requests introduce errors in component**

**interactions Software spare parts become harder to order**

24. Testing is an intellectually demanding activity and has a lifecycle ----- to software **development.**

**Equal**

**Parallel 198**

## **Sequential**

### **In contrast**

25. A memory leak bug is one in which memory is somehow allocated from either the operating system or an----- "pool", but never deallocated when the memory is finished

**being used.**

### **Mixed memory**

#### **External**

#### **memory**

### **Internal Memory 219**

### **Mutually exclusive**

#### **memory**

26. \_\_\_\_\_are used to describe flow of data or control in an application.

Code structures

## **String**

### **matchings Paths**

### **Flow graphs 203**

27. Writing test cases and generating test data are processes that demand\_\_\_\_\_building capabilities.

## **Code**

### **Domain**

### **System**

### **Scenario**

28. What types of errors are missed by black-box testing and can be uncovered by white- box testing?

### **Runtime errors**

### **Logic errors**

### **Performance**

### **errors Input errors**

29. The cyclomatic complexity metric provides the designer with information regarding the number of\_\_\_\_\_.

### **Statements in the**

### **program Cycles in the**

### **program Errors in the**

### **program**

### **Independent logic paths in the program 207**

30. The best reason for using Independent software test teams is that

Software developers do not need to do any testing

### **Strangers will test the software mercilessly**

### **Testers do not get involved with the project until testing**

### **begins The conflicts of interest between developers and testers**

### **is reduced**

31. \_\_\_\_\_is intended to define a many to many relationship between objects so that when one object changes state all its dependants are notified and updated automatically.

### **Observer**

### **Pattren Facade**

### **Pattren**

### **Singleton**

### **Pattren Joint**

### **Pattren**

32. \_\_\_\_\_ ensures that a class only has one instance and provides a global point of access to it.

**Behavioral Pattern**

**Joint Pattern**

**Singleton Pattern**

**145 Observer**

**Pattern**

33. Inspections cannot check\_\_\_\_\_characteristics.

## **Non-Functionl 213**

**Business**

**User**

**Functional**

34. Secondary private classes can be declared as \_\_\_\_\_ and reside in the file of the class they belong to.

**Asynchronous Classes**

**Inner classes 158**

**Outer classes**

**Synchronous**

**classes**

35. The construction should appear at the \_\_\_\_\_ of the header file.

**Top**

**Bottom**

**Left**

**Right**

36. In the switch statement, cases should always end with \_\_\_\_\_.

**Return**

**Semi colon**

**Break 170**

**Full stop**

37. Modularity is a tool that can help us in \_\_\_\_\_ the size of individual functions.

Stabilizing

**Reducing 173**

**Increasing**

**Strengthening**

38. Abstraction and encapsulation are two important tools that can help in managing and mastering the \_\_\_\_\_ of a program.

**Usability**

**Complexity 173**

**Understandabilit**

**y Reliability**

39. One of the guidelines to avoid common mistakes is to never use \_\_\_\_\_ except for declaration.

,

;

=

||

40. Which of the following is/are NOT one of the umbrella activities?

Requirement analysis

**Architecture design**

**Test case**

**development**

**All of the given options 14**

41. Early design decision can vary in software development. Do you agree or not ? Justify your choice 2.

42. Name any two standards which are being used by C/C++ for writing programs. 2

43. Write qualitative benefits of unit testing. 2

44. Do you agree that multilevel commenting is not supported in C++ and Java? 2

45. Briefly describe a business process diagram (at least 3 lines): 3

46. What are static analyzers? 3

47. Write one advantage and one disadvantage of using bit fields in writing the code.3

48. Why the use of `do_while` loops should be avoided ? 3

49. Suppose you were analyzing a Requirement statement, How will you determine that an object belongs to a certain class? 05

50. *Discuss the symptoms and an example of Loop errors bug class.* 5

51. In Intel architecture, the lowest byte is the most significant byte while in Motorola architecture the highest byte of a word is the most significant one.

**Discuss how the above mentioned factor affects portability of program. 05**

52. Which of the following are correct with reference to the abbreviation rules of naming convention?

**We should always write: command instead of**

**cmd copy instead of cp initialize instead of init**

**html instead of HypertextMarkupLanguage cpu instead of**

**Central Processing Unit 05**

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## **CS504 Solved Grand Quiz 2020**

### **Question No 1**

**Which of the following is not the object model principle?**

**Select Correct Option**

- ☐ Abstraction
- ☐ Encapsulation
- ☐ Hierarchy or
- ☐ inheritance **Exposure**

### **Question No 2**

**Return values in Synchronous messages are represented by:**

**Select Correct Option**

- ☐ A solid line
- ☐ A dotted line with label
- ☐ **A solid line with label**
- ☐ Double line

### Question No 3

The\_\_\_\_\_relationship is kind of a generalization specialization relationship.

**Select Correct Option**

- ☐ Bit=byte
- ☐ Uses
- ☐ Binary
- ☐ **Extends**

### Question No 4

The architecture components for product engineering are

**Select Correct Option**

- ☐ **Data, hardware, software, people** Data,
- ☐ documentation, hardware, software
- ☐ Data, hardware, software, procedures
- ☐ **Documentation, hardware, people, procedures**

### Question No 5

A context diagram is used

Select Correct Option

- ☐ As the first step in developing a detailed DFD of a system
- ☐ In systems analysis of very complex systems
- ☐ As an aid to system
- ☐ design As an aid to programmer

### Question No 6

Consider the following piece of code:

```
Public class Square extends Shape {
// some code
}
```

The above code is an example of:

Select Correct Option

- ☐ Part-Whole relationship
- ☐ Generalization/Specializati
- ☐ on Data Sharing
- ☐ Data encapsulation

### Question No 7

When two components of a system are using the same global data area, they are related as

Select Correct Option

- ☐ Data Coupling
- ☐ Content Coupling
- ☐ Common Coupling
- ☐ External Coupling

### Question No 8

The context diagram is used as the top level abstraction in a \_\_\_\_\_ developed according to principles of structures analysis.

Select Correct Option

- ☐ Dataflow diagram
- ☐
- ☐ Activity Diagram
- ☐

**State Transition  
Diagram Use Case  
Diagram**

### Question No 9

In the case of \_\_\_\_\_ in a system, module boundaries are not well defined.

Select Correct Option

- ☐ Low cohesion
- ☒ High coupling
- ☐ Low coupling
- ☐ High cohesion

### Question No 10

In UML based Object Oriented model of a system, the association relation between two objects is depicted by\_\_\_\_\_.

Select Correct Option

- ☐ A straight line
- ☐ A filled diamond sign on the whole side of the line
- ☒ An unfilled diamond sign on the Whole side of the line
- ☐ An arrowhead sign on one side of the line

### Question No 11

External Entity may be

Select Correct Option

- ☐ Source of input data only
- ☒ Source of input data or destination or results
- ☐ Destination of results
- ☐ only Repository of data

### Question No 12

In UML based Object Oriented model of a system, the inheritance between two classes is shown by a \_\_\_\_\_ sign towards the super-class side on the association line between the two.

Select Correct Option

- ☐ A filled diamond
- ☐ An unfilled diamond
- ☐ A half arrowhead
- ☒ An arrowhead

### Question No 13

\_\_\_\_\_ are kind of umbrella activities that are used to smoothly and successfully perform the construction activities.

**Select Correct Option**

☐ **Design Activities**

☐ **Management Activities**

- ☐ Testing Activities
- ☐ Maintenance Activities

### Question No 14

\_\_\_\_\_ is not the part of Peter Coad methodology.

Select Correct Option

- ☐ Select Actors
- ☐ Select participants
- ☐ Select places
- ☐ **Select actions**

### Question No 15

An object model of a system captures the \_\_\_\_\_ structure of a system.

Select Correct Option

- ☐ **Static**
- ☐ Dynamic
- ☐ Iterative
- ☐ Hierarchal

### Question No 16

As per Peter Coad's methodology, which of the following may NOT be a perfect candidate for being an object?

Select Correct Option

- ☐ Zone
- ☐ Recipient
- ☐ Garage
- ☐ **Password**

### Question No 17

The system model template contains which of the following elements

Select Correct Option

- ☐ Input
- ☐ Output
- ☐ System
- ☐ Out

**Input / Output**

**Question No 18**

**A \_\_\_\_\_ is not the real product but just a real looking mock-up of what would be eventually delivered.**

### Select Correct Option

- ☐ Software
- ☐ Program
- ☒ **Prototype**
- ☐ Test Case

### Question No 19

USL is a language for\_\_\_\_\_.

### Select Correct Option

- ☐ High-level programming
- ☐ Low-level Programming
- ☒ **Modeling and Design**
- ☐ Creating Diagrams only

### Question No 20

In multiprocessing applications, different execution threads may pass information to one another by sending\_\_\_\_\_to each other.

### Select Correct Option

- ☐ Interrupt calls
- ☐ Synchronous messages
- ☒ **Asynchronous messages**
- ☐ Systems calls

### Question No 21

To construct a system model the engineer should consider one of the following restraining factors?

### Select Correct Option

- ☒ **Assumptions and constrains**
- ☐ Budget and expenses
- ☐ Data objects and operations
- ☐ Schedule and milestones

### Question No 22

A software requirement document describes all the\_\_\_\_\_provided by the system along with the constraints under which it must operate.

## Select Correct Option

- ☐ Conditions
- ☒ **Services**
- ☐ Event

- ☐ **Processes**

### Question No 23

In “Point of Sale system”, the term “Payment” represents

**Select Correct Option**

- ☐ Actor
- ☐ Participant
- ☐ **Transaction**
- ☐ Container

### Question No 24

A \_\_\_\_\_ relationship indicates that one entity is composed of one or more parts which are themselves instances of that or another entity

**Select Correct Option**

- ☐ Inheritance
- ☐ **Whole-part**
- ☐ Generalization
- ☐ n  
Specialization

### Question No 25

An arrow in Data Flow Diagram (DFD) represents

**Select Correct Option**

- ☐ Direction of flow of data
- ☐ **Processing of data**
- ☐ External agent
- ☐ Internet agent

### Question No 26

In sequence diagram, the boxes denotes

**Select Correct Option**

- ☐ **Objects (or classes)**
- ☐ Messages, sent from one object to other
- ☐ Life-time of objects
- ☐ Relationships

### **Question No 27**

**Which of the given component of software engineering framework provides different techniques that can be used to perform a particular task?**

### Select Correct Option

- ☐ **Processes**
- ☐ Tools
- ☐ Methods
- ☐ Quality Focus

### Question No 28

Which of the following is not supported by a maintainable design?

#### Select Correct

- ☐ **Option**
- ☐ **Change**
- ☐ Debugging
- ☐ Adding new features
- ☐ Higher maintenance cost

### Question No 29

A necessary supplement to transform or transaction mapping needed to create a complete architectural design is

### Select Correct Option

- ☐ Entity relationship diagrams
- ☐ The data dictionary
- ☐ **Processing narratives for each module**
- ☐ Test cases for each module

### Question No 30

A prototype is not the real product. It is rather just a real looking\_\_\_\_\_of what would be eventually delivered and might not do anything useful.

- ☐ **S**
- ☐ **e**
- ☐ **l**
- ☐ **e**
- ☐ **c**
- ☐ **t**

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Ad-hoc  
Design  
Structur  
e

## CS 504-Software Engineering-I

Grand Quiz December

2020 By: MCS of

Virtuallians

**IF there is any correction needed do it accordingly**

1. Whole part structure is also called\_\_\_\_\_
  - a. Generalization
  - b. Aggregation
  - c. Specialization

- d. Association
2. A prototype is not the real product. It is rather just a real looking\_\_\_\_\_of what would be eventually delivered and might not do anything useful.
- Mock-up page 68
  - Ad-hoc
  - Design
  - Structure
3. In "Railway ticket reservation system" the roles such as inquiry, reservation, ticketing and cancellation are to be performed by the user called:-
- Passenger
  - System analyst
  - System developer
  - System designer
4. A useful technique for evaluating the overall complexity of a proposed architecture is to look at the component's\_\_\_\_\_.
- number and size of components
  - flow dependencies and sharing dependencies
  - size and cost
  - function points
5. \_\_\_\_\_ relationship is concerned with classes not with the class instantiates.
- Association
  - Inheritance
  - Aggregation
  - Composition
6. Software development is a step-by-step process, and in\_\_\_\_\_phase of software development Business Objective of an organization gets cleared.
- Maintenance
  - Development
  - Definition
  - Vision
7. The data on which the program operates is also considered as part of the\_\_\_\_\_.
- Important Data
  - Software page 01
  - Logical Data
  - Utility Software
8. From the following which is/are not the example(s) of illegal data flow in Data Flow Diagram (DFD)
- External Agents directly communicating with each other
  - External Agent updating information in a DataStore
  - External Agent accessing information from a Data Store
  - There must be an intermediate process which should transform data received from one external entity and then send the transformed data to the other external entity
9. In sequence diagram, the objects are organized in a\_\_\_\_\_line and the events in a \_\_\_\_\_time line.
- Horizontal, straight
  - Vertical, straight

- c. Horizontal, vertical
  - d. Vertical, horizontal
10. Consider the following piece of code:
- ```
public class Square extends Shape {
// some code
}
```
- The above code is an example of:
- a. Part-Whole relationship
 - b. Generalization/Specialization
 - c. Data Sharing
 - d. Data encapsulation
11. Requirement engineering mainly deals with the _____ of the system
- a. Vision Phase
 - b. Definition Phase page 16
 - c. Development Phase
 - d. Maintenance Phase
12. A _____ relationship indicates that one entity is composed of one or more parts which are themselves instances of that or another entity.
- a. Inheritance
 - b. Whole-part
 - c. Generalization
 - d. Specialization
13. In UML Object Model Notation _____
- a. C++ language-specific programs are constructed
 - b. mathematical problems are visualised
 - c. relationships among classes and sub-classes are expressed page 92
 - d. Graphs and tables are used to explain Software Engineering principles
14. In the case of _____, intra component linkages are stronger while inter component linkages are weak.
- a. high cohesion
 - b. low coupling page 73
 - c. low cohesion
 - d. high coupling
15. Transactions are the _____ that must be remembered through time.
- a. Events page 93
 - b. Actions
 - c. Triggers
 - d. Methods
16. The modules interacting with each other through message passing have _____ between them.
- a. low cohesion
 - b. high cohesion
 - c. low coupling page 74
 - d. high coupling
17. In UML based Object Oriented model of a system, the inheritance relation between two classes is shown by a _____ sign towards the super-class side on the association

line between the two.

- a. A filled diamond
- b. An unfilled diamond
- c. A half arrowhead

d. An arrowhead **page 92**

18. Which statement is not according to the software engineering principles? Software engineering is a(n)_____.

- a. Balancing act
- b. Disciplined approach
- c. **Unsystematic approach** **page 5**
- d. Quantifiable approach

19.A _____ is not the real product but just a real looking mock-up of what would be eventually delivered.

- a. Software
- b. Program
- c. **Prototype(page 68)**
- d. Test Case

20. The best way to conduct a requirements validation review is to

- a. examine the system model for errors
- b. have the customer look over the requirements
- c. send them to the design team and see if they have any concerns
- d. **use a checklist of questions to examine each requirement**

21. More powerful hardware resulted into the development of _____powerful and _____**software.**

- a. less, complex

b. more, complex **page 4**

- c. more, simple
- d. less, simple

22. _____ pointed out the elegant conceptual integrity exhibited by layered organization of software systems, with the resulting gains in development and maintenance ease.

- a. David Parnas
- b. Shaw and Garlan
- c. Barry Boehm

d. EdsgerDijkstra **page 115**

23. The architecture components for product engineering are

- a. **data, hardware, software, people**
- b. data, documentation, hardware, software
- c. data, hardware, software, procedures
- d. documentation, hardware, people, procedures

24. In Object oriented design, combining the services offered by an object with the attributes they work on, results in:

- a) Lower coupling and stronger cohesion
 - b) Lower cohesion and stronger coupling
 - c) Increased likelihood of reuse
 - d) Decreases the modularity of the system
- a. a only

- b. band c
 - c. **a and c**
 - d. b and d
25. Architectural design process involves performing a number of activities which includes system structuring, control modeling and_____.
- a. System Analysis
 - b. Modular Decomposition** **page 120**
 - c. Testing
 - d. Graphical User Interface
26. In use case diagram, the scope of the system is defined by:
- a. Actor
 - b. Entity
 - c. **System Boundary**
 - d. "Extends"
- relationship 27.Specialization means
- a. Calling the same method with object of child object
 - b. Hiding the data
 - c. **Creating new subclasses from an existing class** **page 34 ,**
 - d. Abstraction**
28. Which of the following sentence is true regarding user interface design?
- a. GUI interfaces are good for all tasks which a user needs to perform at an interface.
 - b. The higher the response time, the better is the interface
 - c. **The simpler the interface, the efficient is the system**
 - d. Command-line interfaces are faster for some tasks which the user needs to perform
29. In the architecture trade-off analysis method the architectural style should be described using the
- a. data Flow view and process view
 - b. data Flow view and module view
 - c. module view and process view
 - d. **data Flow view, module view and process view** **page**
30. In Data Flow Diagram, the entity or system, outside the boundary of this system is called
- a. Process
 - b. Data Flow
 - c. **External Agent**
 - d. Data Store
- 31.A cohesive Class is one which emphasizes on_____unit of functionality
- a. Single
 - b. Multiple** **page 76**
 - c. Static
 - d. Both Single and Multiple
32. The goal of_____is to translate the customer's desire for a set of defined capabilities into a working product.
- a. Electrical Engineering

- b. Product Engineering
 - c. Hardware Engineering
 - d. Mechanical Engineering
33. In Object Oriented Design, _____ layer contains the details that enable each object to communicate with its collaborators
- a. Subsystem
 - b. Responsibility
 - c. Message page 89
 - d. Object
- 34.A DFD is normally leveled (adding more levels of abstraction) as
- a. it is a good idea in design
 - b. it is recommended by many experts
 - c. it is easy to do it
 - d. it is easier to read and understand a number of smaller DFDs than one large DFD
35. As per Peter Coad's methodology, which of the following may NOT be a perfect candidate for being an object?
- a. Zone
 - b. Recipient
 - c. Garage
 - d. Password page 93 94
36. Which of the given component of software engineering framework provides different techniques that can be used to perform a particular task?
- a. Processes
 - b. Tools
 - c. Methods page 12
 - d. Quality Focus
37. To determine the architectural style or combination of styles that best fits the Proposed system requirements engineering is used to uncover
- a. algorithmic complexity
 - b. characteristics and constraints page 126
 - c. control and data
 - d. design patterns
38. In _____ relationship, a class shares the structure and behavior defined in another class.
- a. Aggregation
 - b. Composition
 - c. Inheritance page 86
 - d. Uses
39. Prototyping is used when there is regarding requirements.
- a. Uncertainty (page 68)
 - b. Confirmation
 - c. Conflict
 - d. Consensus
40. The process of utilizing our knowledge of computer science in effective production of software systems is called
- a. Chemical Engineering

- b. Electrical Engineering
- c. Computer Engineering

d. Software Engineering **page 2**

41. In Abbot's Textual Analysis technique, different parts of speech are identified within the text of the specification and these parts are modeled using different_____

- a. Event
- b. Process
- c. Operations

d. Components **page 90**

42. To help separate an object's external behavior from its implementation, the technique used is called_____

- a. Generalization
- b. Association
- c. Composition

d. Abstraction **page 86**

43.A Process in Data Flow Diagram (DFD) represents

- a. Flow of data

b. Transformation of data **(page 49)**

- c. Storage of data
- d. An external agent

44. In data flow diagram (DFD). Create, Update, Delete and Read operations are normally called:

a. CRUD operations **page 53**

- b. DURC operations
- c. RUDC operations
- d. CDUR operation

45. Collaboration diagram can show_____

- a) Binary messages
- b) Asynchronous messages
- c) Synchronous messages
- a. a only
- b. b only
- c. c only

d. both b and c **page 111**

46. Data cannot flow from one external entity to other external entity because

a. It will get corrupted

b. It is not allowed in DFD **page 59**

- c. An external entity has no mechanism to read or write
- d. Both are outside the context of the system

47. In_____Phase of software development, Requirement Engineer focuses on realizing the Business Object of an under developed product.

a. Maintenance

b. Development **page 16**

- c. Definition
- d. Vision

48. Which of the following is not part of software architecture?

- a. Databases
- b. data design
- c. program structure
- d. **algorithm details**

49. Selecting objects in a domain) include:

- a. **Actors, Participants and Places**
- b. Only Participants
- c. Only Actors
- d. Only Actors and Places

50. Defining the services of an object means:

- a. **What it does? Page 96**
- b. What it knows?
- c. Who knows it?

51. Whom it knows In sequence diagram, the solid lines depict:

- a. Objects (or classes)
- b. **Messages being sent from one object to the other page**
- c. Life-time of an object
- d. state of the

object 52. Sequence diagrams:

- a. Provide the static behavior
- b. Provide Data Flow
- c. **Provide a time-based view 106**
- d. **Provide parallel data flow**

53. _____ requirements cause frequent modifications in user interface

- a. Functional
- b. Non-functional
- c. **Unstable page 62**
- d. User

54. , security and maintainability are the types of _____ requirements.

- a. **Non-functional page 120**
- b. Domain
- c. Functional
- d. Business

55. A change becomes _____ because of close presence of data and functions

- a. Accessible
- b. Global
- c. Private
- d. **Localized page 81**

56. System _____ are built to allow the system engineer to evaluate the system components in relationship to one another.

- a. Requirements
- b. Documents
- c. **Models page 42**
- d. Test cases

57. In _____ phase of software development, requirement analyst focuses on possible design of the proposed solution.
- Maintenance
 - Development (page 16)**
 - Definition
 - Vision
58. The focus of sequence diagrams is:
- On objects/classes and messages exchanged among them (page**
 - On static Model of system
 - On object constraints
 - On the flow of Control
59. In Data Flow Diagram (DFD), data flow can:
- Only originate from an external entity
 - Only terminate in an external entity
 - Originate and terminate in an external entity**
 - Either originate or terminate in an external entity but not both
60. In _____ the analyst determines all the sources of requirements and where do these requirements consume
- Data Flow Analysis
 - Source and Sink Analysis (page 40)**
 - Down Parsing
 - Up Parsing
61. In a top-down system analysis, a/an _____ required to develop high level view of the system at first.
- Analyst page 54**
 - Designer
 - Tester
 - Developer
62. The Use case diagram shows that which _____ interact with each use case.
- Use case
 - Actor page 32**
 - Component
 - Relation
63. The context diagram is used as the top level abstraction in a _____ develop
-
- Dataflow diagram (page 31)**
- d according to principles of structured analysis.**
- Activity Diagram
 - State Transition Diagram
 - Use Case Diagram
64. Different messages in sequence diagrams includes:
- Simple
 - Asynchronous
 - Notify
 - Both Simple and Asynchronous page 108**
65. A software requirement document describes all the _____ provided by the system along with the constraints under which it must operate.

- a. Conditions
- b. **Services** page
- c. Events
- d. Processes

66. In case of a _____ message, the called routine that handles the message is completed before the caller resumes execution.

- a) Synchronous
- b) Asynchronous
- c) Bidirectional

a. a only page 108

- b. b only
- c. c only
- d. both b and c

67. In object oriented design, _____ layer contains the data structures and algorithmic design for all attributes and operations for each object.

- a. Subsystem
- b. Class and Object
- c. Message

d. **Responsibility** page

89 68. Asynchronous messages:

- a. are implemented as operation call
- b. These block caller before response

c. occurs in multi-threaded applications (page 109)

d. are shown by dotted line

69. In UML based object oriented model of a system, a composition relation between two objects is shown by a _____ sign on the whole side of a relation line.

- a. An unfilled diamond
- b. **A filled diamond**
- c. A half diamond
- d. A dot

70. In UML based Object Oriented Model of a system, the diamond sign is used to depict _____ **relations between two objects/classes.**

- a. Aggregation and Association
- b. Inheritance and Association
- c. **Composition and Aggregation**
- d. Composition, Aggregation and Association

71. A "register" in "Point of sale system" is an example of:

- a. Actor
- b. Participant
- c. **Tangible thing** page
- d. **Transaction**

72. A use case represents

- a. A class, its attributes and operation
- b. An operation's interfaces and signature
- c. **The role a user plays when interacting with the system** page
- d. **The system's functionality for a particular purpose**

73. In the case of _____ approach, data is decomposed according to functionality

requirements.

- a. Object-oriented
 - b. Action-oriented page 80
 - c. Event-oriented
 - d. Process-oriented
74. To construct a system model the engineer should consider one of the following restraining factors?
- a. Assumptions and constraints
 - b. Budget and expenses
 - c. Data object and operation
 - d. Schedule and milestone
75. When two components of a system are using the same global data area, they are related as
- a. Data coupling
 - b. Content coupling
 - c. Common coupling google
 - d. External coupling
76. _____ Structure represents the internal organization of the various data and control items.
- a. Data
 - b. Value
 - c. Information
 - d. Conceptual
77. The method of dividing and assigning different portions of a large system to different groups for construction is called _____
- a. Work Breakdown Structure (page
 - b. Working Boundary Structure
 - c. Work Basic Structure
 - d. Work Breakdown System
78. "A car is made up of a body, three or four wheels, a steering mechanism. a breaking mechanism and a power-engine"
- The above statement is example of :**
- a. Whole-Part relationship page 95
 - b. Inheritance
 - c. Specialization
 - d. Generalization
79. In software engineering paradigm, any engineering approach must be founded on organizational commitment to _____.
- a. Cost
 - b. Scheduling
 - c. Quality (page 15)
 - d. Performance
80. _____ is a technique that can be used to reduce customer dissatisfaction at requirement stage.
- a. Analysis
 - b. Negotiation
 - c. Prototyping (page 71)
 - d. GUI
81. Which of the following activities are included in the design process of a software

architecture_____?

- a. System Development and Deployment
- b. Architectural Analysis and Testing
- c. Requirement Specifications of the system
- d. System Structuring and Modular Decomposition

82. In UML based Object Oriented model of a system, the association relation between two objects is depicted by_____.

- a. **A straight line**
- b. A filled diamond sign on the whole side of the line
- c. An unfilled diamond sign on the whole side of the line
- d. Any arrowhead sign on one side of the line

83. _____are kind of umbrella activities that are used to smoothly and successfully perform the construction activities

- a. Designee activities
- b. Management activities (page 14)**
- c. Testing activities
- d. Maintenance activities

84. Software Design discusses_____aspect of software development

- a. What
- b. **How**
- c. Who
- d. When

85. Include and extend relationship is used in UML notation of a/an_____.

- a. Activity Diagram
- b. Data Flow Diagram
- c. Entity Relationship Diagram
- d. **Use Case**

Model 86.External Entity
may be

- a. source of input data only
- b. **source of input data or destination of results**
- c. destination of results only
- d. repository of data

87. An object model of a system captures the_____structure of a system.

- a. Static page 93**
- b. dynamic
- c. iterative
- d. Hierarchical

88. The criteria used to assess the quality of an architectural design should be based on system

- a. accessibility and reliability
- b. **data and control**
- c. functionality
- d. implementation details

89. In case of_____approach, decomposition of a problem revolves around data

- a. Object-oriented

- b. **Action-oriented** page 80
 - c. Event-oriented
 - d. Process-oriented
90. How can we implement generalization in Object Oriented programming languages?
- a. Polymorphism
 - b. Encapsulation
 - c. Abstraction
 - d. **Inheritance**
91. Which of the following is NOT among one of the four layers of the Object Oriented (OO) design pyramid
- a. The subsystem layer
 - b. The class and object layer
 - c. **The Abstract layer** page 89
 - d. **The message layer**
92. _____ is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.
- a. Inheritance
 - b. Polymorphism
 - c. Aggregation
 - d. **Abstraction** page 79
93. The project manager would need _____ document to monitor and track the progress of the project.
- a. Design
 - b. Project
 - c. **Requirement** page 18
 - d. **Planning**
94. In the case of action-oriented approach data is decomposed according to:
- a. Object requirements
 - b. **Functionality requirements**
 - c. Corresponding domain model
 - d. Compatibility with object interface
95. An architectural style encompasses which of the following elements?
- a. **Constraints, Set of Components and Semantic Models** page
 - b. Set of Components and Semantic Models
 - c. Semantic Models and Constraints
 - d. Set of Components and Constraints
96. In order to determine the role and responsibilities of the identified objects, we need to consider which of the following step(s):
- a) Who I am?
 - b) What I know?
 - c) Who I know?
 - d) What I do?
- a. a only
 - b. a and b
 - c. **b, cand d** page 102
 - d. cand d
97. In sequence diagram, the boxes denote:
- a. **Objects(or classes)** page 106

- b. Messages, sent from one object to other
 - c. Life-time of Objects
 - d. Relationships
98. _____ is a system component that provides services to other components but would not normally be considered as a separate system.
- a. Method
 - b. Module** **page 121**
 - c. Message
 - d. Relationship
- 99.A tangible entity in the real life is called _____.
- a. Functions
 - b. Object** **page 85**
 - c. Class
 - d. Sub-Class
100. Sequence of messages can be present in:
- a) Use case diagram
 - b) Sequence diagram
 - c) Collaboration diagram
- a. a only
 - b. b only
 - c. c only
 - d. **b and c**
101. The system model template contains which of the following elements
- a. Input
 - b. Output
 - c. System Out
 - d. **Input/Output**
102. Identify the TRUE statement:
- a. **Normally Object Oriented design is more maintainable than functional oriented.**
 - b. Software with Functional oriented design does not fulfill non functional requirements.
 - c. Object Oriented design cannot implement "Separation of concerns" strategy
 - d. Function Oriented design does not lead to an efficient product
103. An external entity that interacts with a system is called a(n):
- a. Use case
 - b. **Actor**
 - c. Stakeholder
 - d. Association
104. By levelling a DFD (adding more levels of abstraction) we mean
- a. Splitting it into different levels
 - b. Make its structure uniform
 - c. **Expanding a process into one with more sub-processes giving more detail**
 - d. Summarizing a DFD to specify only the essentials
105. Software Engineering is the combination of tools, techniques and _____.
- a. Testing**
 - b. Processes** **page 6**

- c. Maintenance
 - d. Design
106. _____ is concerned with decomposing a system into interacting sub-systems.
- a. System Structuring
 - b. Control Modeling
 - c. Modular Decomposition** **page 121**
 - d. Work Breakdown Structure**
107. There are _____ most important characteristics of an object.
- a. Six
 - b. Four
 - c. Two
 - d. Three**
108. “System should maintain transaction log of every transaction”
The above statement is an example of
- a. Functional requirement**
 - b. Non-functional requirement
 - c. Pseudo requirement
 - d. Both Non-functional and Pseudo requirements
109. In object oriented approach, _____ are the people and organizations that take part in the system under consideration.
- a. Actors**
 - b. Places
 - c. Participants
 - d. Tangible things
110. The modules that interact with each other through message passing have _____.
- a. Low coupling** **page 73**
 - b. High coupling
 - c. Low cohesion
 - d. High cohesion
111. Identifying Whole-part structures (Aggregations) means what are my _____.
- a. Components** **page 94**
 - b. Attributes
 - c. Methods
 - d. Messages
112. In “point of sale system “ the term payment represents
- a. Actor
 - b. Participant
 - c. Transaction** **(page 99)**
 - d. Container**
113. Return values in synchronous messages are:
- a. Compulsory
 - b. May not used when response is obvious** **(page 109)**
 - c. Not used at all

- d. Represented by solid lines
- 114. GUI stand for
 - a. Generic user interface
 - b. **Graphic user interface**
 - c. Generic user interaction
 - d. Graphical user interaction
- 115. Which of the following is not the object model principle?
 - a. Abstraction
 - b. Encapsulation
 - c. Hierarchy or inheritance
 - d. **Exposure** page 86
- 116. The system specification describes the
 - a. function and behavior of a computer-based system
 - b. **implementation of each allocated system element**
 - c. algorithmic detail and data structures
 - d. time required for system simulation
- 117. The_____provides the software engineer with a view of the system as a whole.
 - a. Process Model
 - b. **Architectural Model**
 - c. Business model
 - d. Requirements Model
- 118. Requirement engineering focuses on_____aspect of the software development process.
 - a. Both what and how
 - b. What
 - c. How
 - d. why and how
- 119. An arrow in Data Flow Diagram (DFD) represents
 - a. **Direction of flow of data**
 - b. Processing of data
 - c. External agent
 - d. Internal agent
- 120. The_____relationship is kind of a generalization specialization relationship.
 - a. Bit-Byte
 - b. Uses
 - c. Binary
 - d. **Extends**
- 121. In the case of_____ in a system, module boundaries are not well defined.
 - a. low cohesion
 - b. **high coupling**
 - c. low coupling
 - d. high cohesion
- 122. _____Component of software engineering framework provides automated or semi-automated support in a software development.
 - a. Processes

- b. Methods
 - c. Quality Focus
 - d. Tools
123. In Data Flow Diagram (DFD), one data store cannot directly copy the data from another_____.
- a. Agent
 - b. Process
 - c. Data store
 - d. Flow
124. All the documents related to the software are also considered as part of the _____.
- a. Physical Document
 - b. Logical Document
 - c. Relational Database
 - d. Software
125. Which elements of business processing engineering are the responsibilities of the software engineer?
- a. business area analysis
 - b. business system design
 - c. product planning
 - d. information strategy planning
126. A class will be cohesive if:
- a. Class does not implement Complex interfaces
 - b. Class does not have complex methods
 - c. If most of the methods do not use most of the data members most of the time
 - d. If most of the methods use most of the data members most of the time
127. A context diagram is used
- a. As the first step in developing a detailed DFD of a system
 - b. In systems analysis of very complex systems
 - c. As an aid to system design
 - d. As an aid to programmers
128. _____ is an extremely powerful technique for dealing with complexity.
- a. Aggregation
 - b. Abstraction
 - c. Inheritance
 - d. Association
129. In multiprocessing applications, different execution threads may pass information to one another by sending _____ to each other.
- a. Interrupt calls
 - b. Synchronous messages
 - c. Asynchronous messages
 - d. System calls
130. Normally a system is more easy to modify if its modules have
- a. High coupling and high cohesion
 - b. High coupling and low cohesion
 - c. Low coupling and high cohesion
 - d. Low coupling and Low cohesion

131. UML is a language for
- High-level Programming
 - Low-level Programming
 - Modeling and Design
 - Creating diagrams only
132. _____ is not the part of Peter Coad methodology
- Select actors
 - Select participants
 - Select places
 - Select actions
133. Which of the following is the external quality of software product?
- Correctness
 - Concision
 - cohesion
 - Low coupling
134. While establishing the services for an object, the goal is to keep data and action together for _____ coupling and _____ cohesion
- Lower, Higher
 - Higher, Lower
 - Lower, Lower
 - Higher, Higher
135. A necessary supplement to transform or transaction mapping needed to create a complete architectural design is
- entity relationship diagrams
 - the data dictionary
 - processing narratives for each module
 - test cases for each module
136. Construction activities are directly related to software_____.
- Management
 - Planning
 - Quality Assurance
 - Development
137. OOD results in a design that achieves a number of different levels of _____.
- Operation
 - Event page 89
 - Modularity
 - Process
138. At which stage of software development loop, results are delivered?
- Problem definition
 - Solution integration
 - Technical development
 - Status quo
139. _____ Requirements are often called product features.
- Functional
 - Non-functional
 - Developer
 - User

140. Strong cohesion implies that
- All parts of component have a class logical relationship with each other
 - All part of component do not have a close logical relationship with each other
 - Component is dynamic in nature
 - Component is static in nature
141. _____ Diagram does not capture control flow information; it just shows the flow of the data in a system.
- Sequence
 - Data Flow
 - Activity
 - Class
142. A life line represents the object's life during the interaction in a sequence diagram while its notation is depicted by _____
- Solid Lines
 - Dotted Lines
 - Full Arrow
 - Curved Lines
143. An attribute that varies over time, e.g. price of an item, should be replaced by a/an _____ with an effective date and value
- Additional Class
 - Additional Method/Function
 - Interface
 - Structure
144. Which of the given component of software engineering framework demands rational and Timely development of a software?
- Tools
 - Methods
 - Quality Focus
 - Processes
145. A context diagram
- describes detailed design of a system
 - is a DFD which gives an overview of the system
 - is a detailed description of a system
 - is not used in drawing a detailed DFD
146. Which one is not a part of Software Development phase?
- Construction
 - Scope
 - Project Vision
 - Definition
147. Software architecture must address _____ requirement of a software system
- Functional
 - Non Functional
 - User interface Requirements
 - Both Functional and Non Functional
148. If Cat is derived from Mammal Class, and Mammal is derived from Animal Class, then:
- Cat will inherit Animal's functions and data

- b. Cat will not inherit Animal's functions and data
 - c. Cat will not be able to access any class
 - d. Cat is allowed to access only the Mammal's Class
149. Which of the following is not supported by a maintainable design?
- a. Change
 - b. debugging
 - c. Adding new features
 - d. Higher maintenance cost
150. The condition that must be met before the use case can be invoked, is called:
- a. Pre-Condition
 - b. Post-Condition
 - c. Pre-Assertion
 - d. Post-Assertion
- 151.

CS504 GRAND QUIZ MCS FAMILY GROUP

TEAM

BY: ARSAL SHANI & AINA MALIK

- 1) Modules with high cohesion and low coupling can be treated and analysed as
 - i. White boxes
 - ii. black boxes
 - iii. grey boxes
 - iv. none of these
- 2) While establishing the services for an object, the following fundamental questions should be asked:
 - i) Why does the system need this object anyway?
 - ii) What useful questions can it answer? What useful action can it perform?
 - iii) All of the given options.
- 3) is a role that each actor plays in the system under consideration.
 - i) An act
 - ii) A participant
 - iii) A function
 - iv) None of the given
- 4) Any Engineering approach must be founded on organizational commitment to ---- -.
 - i. Cost
 - ii. Scheduling
 - iii. Quality
 - iv. Performance
- 5) Return values in synchronous messages are:
 - i) Compulsory
 - ii) May not used when response is obvious
 - iii) Not used at all
 - iv) Represented by solid lines
- 6) According to Caper Jhones analysis of project activities, coding only has affect part in system development. i.

13-

14% ii.

36-

40%

iii. 50-60%

iv. 70-80%

- 7) Normally a system is more easy to modify if its modules have
- High coupling and high cohesion
 - High coupling and low cohesion
 - Low coupling and high cohesion
 - Low coupling and Low cohesion
- 8) In multi-threaded or multiprocessing applications where different execution threads may pass information to one another by sending-----to each other.
- Interrupt calls
 - Synchronous messages
 - Asynchronous messages
 - System Calls
- 9) Which of the following is not among the four layers of the object-oriented pyramid?
- The subsystem layers
 - The class and object layer
 - The abstract layer
 - The message layers
- 10) System models include:
- User business processes
 - User activities for conducting the business process
 - Processes that need to be automated
 - All of the given options
- 11) In the architecture trade-off analysis method the architectural style should be described using the
- Data flow view
 - Module view
 - Process view
 - All of the given
- 12) is concerned with decomposing the system into interacting sub-systems.
- iv)
- System structuring
 - Control Modelling
 - Molecular Decomposition
 - None of the given
- 13) A use case represents:
- A class, its attributes and operations
 - An operation's interface and signature
 - The role a user plays when interacting with the system
 - The system's functionality for a particular purpose
- 14) External entity may be:
- Source of input data only
 - Source of input data and destination of results

iii. Destination of results

- iv. Repository of data
- 15) The process of utilizing our knowledge of computer science in effective production of
-.....-.
- i. Chemical Engineering
 - ii. Electrical Engineering
 - iii. Computer Engineering
 - iv. System Engineering
- 16) Coupling is a measure of----- of a component.
- i. Independence
 - ii. Dependence
 - iii. Aggregation
 - iv. Composition
- 17) has become a standard notation for object-oriented system modelling:
- i. UML
 - ii. C++
 - iii. OCL(Object Constraint Language)
 - iv. None of the given option
- 18) An arrow in data flow diagram represents:
- i. Direction of flow of data
 - ii. Processing of data
 - iii. External agent
 - iv. Internal Agent
- 19) diagrams does not capture control flow information, it just shows the flow of data
in a system.
- i. Sequence
 - ii. Data Flow
 - iii. Activity
 - iv. Class
- 20) In----- the analyst determines the source of requirements and where do these
requirements consume:
- a) Data flow analysis
 - b) Source and sink analysis
 - c) Down parsing
 - d) Up parsing
- 21) Data cannot flow from one external entity to other external entity because:
- a) It will get corrupted
 - b) It is not allowed in DFD
 - c) An external entity has no mechanism to read or write
 - d) Both are outside the context of the system
- 22) In the functional design, the structure of the system resolves around:
- a) Objects
 - b) Properties
 - c) Functions
 - d) All of the given options
- 23) is one of the techniques to document domain knowledge

- a) State transition diagram
 - b) Feasibility matrix
 - c) System matrix
 - d) None of the given options
- 24) In case of -----approach , decomposition of a problem revolves around data.
- a) Object-Oriented
 - b) Action-Oriented
 - c) Event-Oriented
 - d) Process-Oriented
- 25) The -----relationship is a kind of a generalization specialization relationship:
- a) Bit-Byte
 - b) Uses
 - c) Binary **d)**
Extends
- 26) Which statement is not true about system software?
- a) Change request has direct impact on software
 - b) Passes through a constant process of evolution
 - c) Change requests have direct impact in the form of defects
 - d) None of the given
- 27) Strong cohesion implies that:
- a) **All parts of a component have a close logical relationship with each other**
 - b) All parts of a component don't have a close relationship with each other
 - c) Component is dynamic in nature
 - d) Component is static in nature
- 28) The intent of Object-Oriented Analysis(OOA) is to define:
- a) All classes
 - b) Relationships among classes
 - c) Behaviour of classes
 - d) **All of the given options**
- 29) Requirement engineering focuses on ----- aspect of the software development process.
- a) **Both what and how** **b)**
What
 - c) How
 - d) Why and how
- 30) relationship is concerned with classes, not with class instantiates.
- a) **Association** **b)**
Inheritance
 - c) Aggregation
 - d) Composition
- 31) Which of the following statements are true in context of the object model deviation through the Coad methodology?
- a) A place is also a contains
 - b) Every container needs to be a place
 - c) Same person may play different times in the system.
 - i. A only

- ii. A and b
- iii. A and c
- iv. All of the given

32) The goal of -----is to translate the customer's desire for a set of defined capabilities into a **working product**.

- i. Electrical engineering
- ii. Product engineering
- iii. Hardware engineering
- iv. Mechanical engineering

33) In case of a ----- message, the called routine that handles the message is completed before the **caller resumes execution**.

- a) Synchronous
- b) Asynchronous
- c) Bidirectional

A only

- ii. B only
- iii. C only
- iv. All of the given

34) In multiprocessing applications, different execution threads may pass information to one another by sending to each other.

- i. Interrupt calls
- ii. Synchronous messages
- iii. Asynchronous messages
- iv. System calls

35) A car is made up of a body, three or four wheels, a steering mechanism, a breaking mechanism, and a power engine"

The above statement is example of:

- i. Whole-part relationship
- ii. Inheritance
- iii. Specialization
- iv. Generalization

36) To help separate an object's external behaviour from its implementation, the technique used is called -----

- i. Generalization
- ii. Association
- iii. Composition
- iv. Abstraction

37) Sequences of messages can be present in:

- (a) Use case diagram
- (b) Sequence diagram
- (c) Collaboration diagram
- 1) a only
- 2) b only
- 3) c only
- 4) b and c

38) Which of the following strategies lead to good software design:

- i. Separation of concerns
 - ii. Modularity
 - iii. Divide-and-conquer
 - iv. All of the given options
- 39) Data flow model:
- i. Captures the flow of data in a system
 - ii. Helps in developing an understanding of system's functionality
 - iii. Describes data origination, transformations and consumption in a system
 - iv. All of the given options
- 40) requirements are often called product features.
- i. Functional
 - ii. **Non-functional**
 - iii. Developer
 - iv. User
- 41) In the functional design, the structure of the system revolves around.
- i. Objects
 - ii. Properties
 - iii. **Functions**
 - iv. All of the given option
- 42) The first step in any OOA process model is to
- i. **Build an object-relationship model.**
 - ii. Define collaborations between objects.
 - iii. Elicit customer requirements
 - iv. Select a representation language
- 43) The -----relationship is kind of a generalization specialization relationship.
- i. Bit-byte
 - ii. Uses
 - iii. Binary
 - iv. **Extends**
- 44) Regarding data flow model, which of the following statement(s) is true:
- i. It captures the transformation of data between processes/functions of a system
 - ii. **Processes on a data flow can operate in parallel**
 - iii. Only those processes are represented which we need to automate
 - iv. All of the given option
- 45) is an extremely powerful technique for dealing with complexity.
- i. Aggregation
 - ii. **Abstraction**
 - iii. Inheritance
 - iv. Association
- 46) In "point of sale system". the term "payment" represents
- i. Actor
 - ii. Participant
 - iii. **Transaction**
 - iv. **Container**
- 47) The architecture components for product engineering are
- i. **Data, hardware, software, people**

- ii. Data, documentation, hardware, software
 - iii. Data, hardware, software, procedures
 - iv. Documentation, hardware, people, procedures
- 48) An object model encompasses the principle(s) of
- i. Abstraction
 - ii. Encapsulation
 - iii. Hierarchy or inheritance
 - iv. All of the given option
- 49) Prototyping is used when there is-----regarding requirements.
- i. Uncertainty
 - ii. Confirmation
 - iii. Conflict
 - iv. Consensus
- 50) In-----phase of software development, requirement analyst focuses on possible design of the
- i. proposed solution. Maintenance
 - ii. Development
 - iii. Definition
 - iv. Vision
- 1) At which stage of software development loop, results are delivered?
- i. Problem definition
 - ii. Solution integration
 - iii. Technical development
 - iv. Status quo
- 2) A class will be cohesive if:
- i. Class does not implement complex interfaces
 - ii. Class does not have complex methods
 - iii. If most of the methods do not use most of the data members most of the time
 - iv. If most of the methods use most of the data members most of the time.
- 3) A DFD is normally levelled (adding more levels of abstraction) as
- i. It is a good idea in design
 - ii. It is recommended by many experts
 - iii. It is easy to do it
 - iv. It is easier to read and understand a number of smaller DFDs than one large DFD
- 4) Identify the true statement(s)
- i. An attribute that may have a number of values should be replaced by a new class and an object connection
 - ii. An attribute that varies over time, e.g. price of an item, should be replaced by an additional class with an affective data and value
 - iii. Replace “yes/no” type attribute with “status” type attributes for flexibility
 - iv. All of given option
- 5) Is a technique in which we construct a model of an entity based upon its essential characteristics and ignore the inessential details.
- i. Inheritance
 - ii. Polymorphism
 - iii. Aggregation
 - iv. Abstraction

- 6) A structure is a manner of an organization which expresses a strong organization within the problem domain.
 - i. **Semantically**
 - ii. Syntactically
 - iii. Graphically
 - iv. None of the given
- 7) An object model of a system captures the ---structure of a system.
 - i. **Static**
 - ii. Dynamic
 - iii. Iterative
 - iv. Hierarchical
- 8) To determine the architectural style or combination of styles that best fits the proposed system, requirements engineering is used to uncover
 - i. Algorithmic complexity
 - ii. **Characteristics and constraints**
 - iii. Control and data
 - iv. Design patterns
- 9) Which statement is not according to the software engineering principles? Software engineering is a(n) ____
 - i. Balancing act
 - ii. Disciplined approach
 - iii. **Unsystematic approach**
 - iv. **Quantifiable approach**
- 10) In abbot's textual analysis technique, different part of speech is identified within the text of the specification and these part are modelled using different ____
 - i. Event
 - ii. Process
 - iii. Operations
 - iv. Components
- 11) Quantitative methods for assessing the quality of proposed architectural designs are readily available.
 - i. True
 - ii. False
- 12) In order to determine the role and responsibilities of the identified objects, we need to consider which of the following step(s):
 - a) Who I am?
 - b) What I know?
 - c) Who I know?
 - d) What I do?
 - i. A only
 - ii. A and b
 - iii. B ,c and D
 - iv. All of the given
- 13) In object oriented design ____ layer contains the details that enable each object to communicates with its collaborators.
 - i. Subsystem

- ii. Responsibility
 - iii. Message
 - iv. Object
- 14) In sequence diagram, the boxes denote:
- i. Objects (or classes)
 - ii. Messages, sent from one object to other
 - iii. Life-time of objects
 - iv. None of the given option
- 15) In “ railway tickit reservation system” the roles such as enquiry. Reservation and ticketing and cancellation are to be performed by the user called:
- i. **Passenger**
 - ii. System analyst
 - iii. System developer
 - iv. System designer
- 16) Class responsibilities are defined by _____
- i. Its attributes only
 - ii. Its collaborators
 - iii. Its operations only
 - iv. Both its attributes and operations
- 17) Requirement engineering mainly deals with the ___ of the system
- i. Vision phase
 - ii. **Definition phase**
 - iii. Development phase
 - iv. Maintenance phase
- 18) In UML based object oriented model of a system, a composition relation between two objects is shown by a ___ sign on the whole side of a relation line.
- i. An unfilled diamond
 - ii. A filled diamond
 - iii. A half diamond
 - iv. A dot
- 19) _____ analysis educates the analyst on business domain complexity and shows a way to deal with it.
- i. Domain
 - ii. Use case
 - iii. Object collaboration
 - iv. None of the given options
- 20) In this case of _____ intra component linkages are stronger while inter component linkages are weak.
- i. High cohesion
 - ii. Low coupling
 - iii. Low cohesion
 - iv. High coupling
- 21) An architectural style encompasses which of the following elements?
- i. Constraints
 - ii. Set of components
 - iii. Semantic models

- iv. All of the given
- 22) In software engineering paradigm, any engineering approach must be founded on organizational commitment to
- i. Cost
 - ii. Scheduling
 - iii. Quality
 - iv. Performance
- 23) Identify the true statement:
- i. **Normally object oriented design is more maintainable than functional oriented.**
 - ii. Software with functional oriented design does not fulfil non functional requirements.
 - iii. Object oriented design can not implement “separation of concerns” strategy
 - iv. Function oriented design does not lead to an efficient product
- 24) A process in data flow diagram (DFD) represents
- i. Flow of data
 - ii. **Transformation of data**
 - iii. Storage of data
 - iv. An external agent
- 25) A maintainable design is a design , which supports
- i. Change
 - ii. Debugging
 - iii. Adding new features
 - iv. **All of the given**
- 26) Whole part structure is also called _____
- i. Generalization
 - ii. **Aggregation**
 - iii. Specialization
 - iv. Association
- 27) _____are kind of umbrella activities that are used to smoothly and successfully perform the construction activities.
- i. Design activities
 - ii. Management activities
 - iii. Testing activities
 - iv. Maintenance activities
- 28) When you encounter both transform flow in the same DFD the flow is partitioned and the appropriate mapping technique is used on each part of the DFD.
- i. True
 - ii. False
- 29) Software architecture must address -----requirements of a software system.
- i. Functional
 - ii. Non-functional
 - iii. User Interface Requirements
 - iv. **Both functional and non-functional.**
- 30) To construct a system model the engineer should consider one of the following restraining factors.
- i. **Assumptions and constraints**
 - ii. Budget and expenses

- iii. Data objects and operations
 - iv. Schedule and milestones
- 31) A cohesive class is one which emphasizes on----- unit of functionality.
- i. Single
 - ii. **Multiple**
 - iii. Static
 - iv. None
- 32) The best way to conduct a requirement validation review is to
- i. Examine the system model for errors
 - ii. have the customer look over the requirements
 - iii. Send them to the design team and see if they have any concerns
 - iv. **Use a checklist of the questions to examine each requirement**
- 33) Defining the services of an object means:
- i. What it does?
 - ii. **What it knows?**
 - iii. **Who knows it?**
- 34) Which one of the following is the external quality of a software product?
- i. Correctness
 - ii. Concision
 - iii. Cohesion
 - iv. **Low coupling**
- 35) In Data Flow Diagram, the entity or system, outside the boundary of this system is called:
- i. Process
 - ii. Data flow
 - iii. **External agent**
 - iv. Data store
- 36) GUI stands for:
- i. Genaric user Interface
 - ii. Graphical user interface
 - iii. Genaric user interaction
 - iv. Graphical user interaction
- 37) Specialization means:
- i. Calling the same method with the object of child object
 - ii. Hiding the data
 - iii. **Creating new subclasses for an existing class**
 - iv. None of the given options
- 38) In a use case diagram, an ellipse signifies a(n):
- i. Actor
 - ii. Class
 - iii. **Use case**
 - iv. System boundary
- 39) Software development is a step-by-step process, and in ----phase of software development
- Business objective of an organization get cleared**
- i. Maintenance
 - ii. Development

iii. Definition

iv. Vision

40) If you try to make software more user-friendly then the ---- may suffer.

i. Reliability

ii. Software

iii. Efficiency

iv. Cost

41) In object-oriented design, the structure of the system revolves around. i.

ii. Properties

iii. Methods

iv. All of the given option

42) In ----- relationship, a class shares the structure and behaviour defined in another class:

i. Aggregation

ii. Composition

iii. Inheritance

iv. Uses

43) In Object Oriented Design, combining the services offered by an object with the attributes they work on, results in:

(a) Lower coupling and strong cohesion

(b) Lower cohesion and strong coupling

(c) Increased likelihood of reuse

(d) Decrease the modularity of the system

i. A only

ii. B and c only

iii. A and c only

44) A change becomes -----because of close presence of data and functions.

i. Accessible

ii. Global

iii. Private

iv. Localized

45) Software engineering is a ----- approach.

i. Systematic

ii. Disciplined

iii. Scheduled

iv. All of the given options

46) An external entity that interacts with the system is called a(n):

i. Use case

ii. Actor

iii. Stakeholder

iv. Association

47) More powerful hardware resulted into the development of -----powerful and -----software.

i. Less, complex

ii. More , complex

iii. More, simple

- iv. Less, simple
- 48) A context diagram is used:
- i. As a first step in developing a detailed DFD of a system
 - ii. In systems analysis of very complex systems
 - iii. As an aid to system design
 - iv. As an aid to programmers
- 49) “System should maintain transaction log of every system”
The above statement is an example of:
- i. Functional requirement
 - ii. Non-functional requirement
 - iii. Pseudo requirement
 - iv. None of these
- 50) The architectural model provides the software engineer with the view of the system as a whole:
- i. True
 - ii. False
- 51) As per Peter Coad’s methodology , which of the following may not be a perfect candidate for
being an object?
- i. Zone
 - ii. Recipient
 - iii. Garage
 - iv. **Password**
- 52) In the case of ----- approach , data is decomposed according to functionality
requirements.
- i. Object-oriented
 - ii. **Action-oriented**
 - iii. Event-oriented
 - iv. Process-oriented
- 53) In UML based Object Oriented Model of a system, the diamond sign is used to depict -----

--- **relations between two objects/classes.**
- i. Aggregation and Association
 - ii. Inheritance and Association
 - iii. **Composition and Aggregation**
 - iv. **Composition, Aggregation and Association**
- 54) The system specification describes the:
- i. **Function and behavior of a computer-based system**
 - ii. Implementation of each allocated system element
 - iii. Algorithmic detail and data structures
 - iv. Time required for system simulation
- 55) In object oriented approach, -----are the people and organizations that take part in the
system under consideration: i.
Actors
- ii. Places
 - iii. Participants

- 56) Software Design discusses ----- aspect of software development.
- What **ii.**
 - How**
 - Who
 - When
- 57) ----- requirements cause frequent modifications in user interface. The ----- on which program operates is also considered as part of the software.
- Data**
 - Information
 - Program
 - None of the given
- 58) ----- diagram provides a time-based view and collaboration diagrams which provide an organization based view of the system's dynamics.
- Data flow diagram
 - Entity relationship diagram
 - Class diagram
 - Sequence diagram
- 59) Synchronous messages are "call events" and are denoted by -----
- Full arrow
 - Half arrow
 - <<create>>
 - <<destroy>>
- 60) Which of the following are the components of system engineering software?
- Process
 - Methods
 - Tools
 - All of the given
- 61) Identifying system features include -----
- Log important information
 - Conduct business
 - Analyze business results
 - All of the above
- 62) ----- is yet another technique that is used to reduce customer dissatisfaction at the requirement stage.
- Study of similar systems
 - Site visits
 - Prototyping
 - All of the above
- 63) Data store notation in DFD represents:
- Data input
 - Data output
 - Data input and data output**
 - None of the above
- 64) The process of defining attributes is called -----
- Who know me? **ii.**

What I know?

- iii. Whom I know?
 - iv. All of the above
- 65) The output of the design process is a description of the:
- i. **Software architecture**
 - ii. Software Code
 - iii. Software
 - iv. All of the above
- 66)
- i. Functional
 - ii. Non-functional
 - iii. **Unstable**
 - iv. User
- 67) By levelling a DFD (adding more levels of abstraction) we mean:
- i. **Splitting it into different levels**
 - ii. Make its structure uniform
 - iii. Expanding a process into one with more sub-processes giving more detail
 - iv. Summarizing a DFD to specify only these essentials
- 68) A “register” in “Point of Sale system” is an example of:
- i. Actor
 - ii. Participant
 - iii. **Tangible thing**
- 69) Transaction Which of the following are the levels of software requirements?
- i. Business requirements
 - ii. User requirements
 - iii. Functional requirements
 - iv. **All of the above**
- 70) Given below are some statements associated with data flow diagrams. Identify the correct statement among them.
- i. Data flow is made used of to model what systems do
 - ii. Flows of data can take place from a process to a sink
 - iii. **Context diagrams shows the major system processes**
 - iv. **All processes have to be labelled or decomposed**
- 71) In which of the following diagram the actors and attributes are represented with system boundary?
- i. Data flow diagram
 - ii. Entity relationship diagram
 - iii. Class diagram
 - iv. **Use case diagram**
- 72) ___is real looking mock_up of what would be eventually delivered and might not do anything useful.
- i. Study of similar system
 - ii. Site visits
 - iii. **Prototyping**
 - iv. **All of the above**

- 73) _____ is blueprint for software construction.
- Object oriented design
 - Sequence design
 - Software design
 - All of the above
- 74) _____ requirements lead to ill-spent time and rework.
- Unacceptable
 - Ambiguous**
 - Dissatisfaction of customer
 - None of the above
- 75) Which type of diagram is used to depict the dynamic behaviour of a system.
- ERD diagram
 - DFD diagram
 - Class diagram
 - Collaborations diagram**
- 76) What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?
- Security
 - Reliability
 - Easy usability
 - Accuracy**
- 77) A better design has an objective achieve
- High cohesion
 - Low cohesion
 - Low coupling
 - High cohesion and low coupling**
- 78) Which of the following are the components of software engineering framework is combine the three remaining components?
- Process
 - Method
 - Tools**
 - All of the above**
- 79) In sequence diagrams the time required by the receiver object to process the message is denoted by an _____
- Activation box**
 - Message line
 - Life line
 - All of the above
- 80) How many types of OOD modes have____.
- One
 - Two**
 - Three
 - Four
- 81) Which notation is used to represent the process of the system in DFD model.
- Process**
 - External agent

- iii. Data flow
 - iv. Data store
- 82) Requirement engineering mainly deals with the_____of the system
- i. Process
 - ii. Maintenance
 - iii. Development phase
 - iv. Definition phase
- 83) Insufficient user involvement leads to____products. i.
- Unacceptable**
- ii. Ambiguous
 - iii. Dissatisfaction of customer
 - iv. None of the above
- 84) Collaboration diagrams have basically two types of components: objects and _
- i. Messages
 - ii. Method
 - iii. Classes
 - iv. None of the above
- 85) In object-oriented analysis how many number of tasks must occurs .
- i. 1
 - ii. 2
 - iii. 3
 - iv. None of the above
- 86) State transition diagram is helpful in determining_____.
- i. Data store
 - ii. Process flow
 - iii. Business understanding
 - iv. None of the above
- 87) In sequence diagram events are organized in a____time life line. i.
- Vertical**
- ii. Horizontal
 - iii. Both A and B
 - iv. All of the above
- 88) Asynchronous messages are “signals,” denoted by_____.
- i. Full arrow
 - ii. Half arrow
 - iii. <<create>>
 - iv. <<destroy>>
- 89) When we write a program for computer and then we named it as_____.
- i. Data
 - ii. Information
 - iii. Software
 - iv. None of the given
- 90) Context level diagram present in which of the following document. i. SRS-software requirement specification
- ii. Design document
 - iii. Test phase

- iv. All of the above
- 91) ___ is diagram in which objects are interact with each other and these are arranged in a sequence.
- i. ERD diagrams
 - ii. Inheritance diagrams
 - iii. Class diagrams
 - iv. **Sequence diagrams**
- 92) Which of the following layers are include in object-oriented design?
- i. The subsystem layers
 - ii. The class and object layer
 - iii. **All of the above**
 - iv. **The message layers**
- 93) Which notation is used to represent the boundary of the system in DFD model?
- i. Process
 - ii. **External agent**
 - iii. Data flow
 - iv. Data store
- 94) Identifying whole-part structures (aggregations) means what are my **i. Components**
- ii. Structures
 - iii. Class
 - iv. Object
- 95) An object or class may further be classified on the basis of__.
- i. Behaviour driven attributes
 - ii. Data driven attributes
 - iii. Responsibility driven attributes
 - iv. **All of the above**
- 96) DFD notation contains
- i. Process
 - ii. External agent
 - iii. Data flow
 - iv. **All of the above**
- 97) The dotted lines in sequence diagram are called **i. Life line**
- ii. Message line
 - iii. Entities line
 - iv. All of the above
- 98) An object may create another object via a___message .
- i. Full arrow
 - ii. half arrow
 - iii. <<create>>
 - iv. <<destroy>>
- 99) How many levels of software requirements are___?
- i. One
 - ii. Two
 - iii. Three

- iv. Four
- 100) Which of the following diagram has iterative activities?
 - i. Data flow diagram
 - ii. Entity relationship diagram
 - iii. Class diagram
 - iv. Use case diagram
- 101) Which of the items listed below is not one of the software engineering layers?
 - i. Tools
 - ii. Manufacturing
 - iii. Process
 - iv. Methods
- 102) Coupling is a measure of ____ a module or component.
 - i. Independent
 - ii. Dependent
 - iii. Closeness
 - iv. All of the given
- 103) Software maintenance phase involves
 - i. Debugging
 - ii. Adding new features
 - iii. Making changes
 - iv. **iv. All of the given**
- 104) The hardest single part of building a software system is deciding precisely to build.
 - i. When **ii. What**
 - ii. **What**
 - iii. Why
 - iv. All of the given
- 105) Interaction diagrams depict the ____ behaviour of the system.
 - i. Static
 - ii. Active **iii. Dynamic**
 - iii. **Dynamic**
 - iv. **None of the given**
- 106) A ____ can be used to describe the dynamic behaviour of an object-oriented system.
 - i. ERD diagrams
 - ii. Inheritance diagrams
 - iii. Class diagrams
 - iv. Series diagrams
- 107) The Use case diagram shows that which interact with each use case.
 - i. Use case
 - ii. Actor
 - iii. Component
 - iv. Relation
- 108) Transactions are the ____ that must be remembered through time.
 - i. Events
 - ii. Action

- iii. Triggers
 - iv. Methods
- 109) A necessary supplement to transform or transaction mapping needed to create a complete architectural design is
- i. Entity relationship diagrams
 - ii. The data dictionary
 - iii. Processing narratives for each module
 - iv. Best cases for each module

**CS504 Quiz by Aarsal Shani and Aina
Malik MCS Family Group**

1. Vertical partitioning is also known as
 - a. Balancing
 - b. Mutating
 - c. Parallelizing
 - d. Factoring
2. It ensures that a class only has one instance and provides a global point of access to it.
 - a. Singleton Pattern
 - b. Observer Pattern
 - c. Real Pattern
 - d. None of the given above
3. Fat Client Model is one of the configurations of _____ Model.
 - a. Data-centered
 - b. Layered
 - c. Reference
 - d. Client Server
4. N-tier architecture stems from the struggle to find a _____ between the fat-client architecture and the thin-client architecture.
 - a. Concurrency
 - b. Distribution point
 - c. Middle ground
 - d. Similarity
5. Which of these represents the Krutchen's 4+1 architectural view model?
 - a. Logical view, Process view, Physical view, Development view, Use case view
 - b. Logical view, Dynamic view, Physical view, Development view, User case view
 - c. Logical view, Process view, Physical view, Development view, Sequence view
 - d. Dynamic view, Process view, Physical view, Development view, Use case view
6. Thin Client Model is one of the configuration of _____ Model
 - a. Data-centered
 - b. Client Server
 - c. Layered
 - d. Reference
7. In the N-Tire Architecture the idea is to enhance scalability and by distributing both

the data and the application using multiple server machines.

- a. Usability
 - b. **Performance**
 - c. Interoperability
 - d. Integrity
8. Floating point constants should always be written with decimal point and at least
- a. **One decimal**
 - b. Two decimal
 - c. Three decimal
 - d. None of the given
9. Which of the following statements are same in output: 1) $a = a >> 2$
2) $a = a / 4$
3) $a = a * 2$
- a. (1) and (3) only
 - b. (2) and (3) only
 - c. **(1) and (2) only**

d. All produce the same result

10. Dynamic process model shows the process _____ of the system.
- a. Components
 - b. Objects
 - c. **Structure**

d. Linkage

11. There are _____ layers in OSI reference model.
- a. 5
 - b. 6
 - c. **7**
 - d. 8
12. Complex expressions:
- a. Make the code easy to modify
 - b. **Make the code difficult to modify**
 - c. Make the code easy to understand
 - d. Does not affect understand ability
13. Vertical participation divides the architecture application from a _____ **making perspective. a.**
- a. **Decision**
 - b. Design
 - c. Conclusion
 - d. Move
14. Issues like system performance, availability, scalability, and security are analyzed in _____.
- a. Logical view
 - b. **Physical view**
 - c. Code view
 - d. Concurrency view

15. Class variables should never be declared public. Public variable violate which of the following:
- Information hiding
 - Encapsulation
 - Information hiding and encapsulation**
 - None of the given
16. Variable should be initialized where they are _____ and they should be declared in the _____ scope possible.
- Defined and smallest
 - Declared and medium
 - Defined and medium
 - Declared and smallest**
17. Patterns are devices that allows program to share knowledge about their _____.
- Code
 - Design**
 - Analysis
 - Implementation
18. Identifier names also play a significant role in enhancing the _____ of the program.
- Write ability
 - Read ability**
 - Reliability
 - All of the given choices
19. Goto statements violates the idea of
- Object oriented code
 - Structured code**
 - Control structure
 - Repetition structure
20. “Is” prefix should be used for _____ variables and methods.
- General
 - Boolean**
 - Constant
 - None of the given
21. development view in Krutchen’s 4+1 architectural view model captures
- object model of design
 - concurrency and synchronization aspects of the design
 - Mapping(s) of the software onto the hardware and reflects its distributed aspect
 - Static organization of the software in its development**
22. A self-documenting code is a code that explain its self without the need of comments extraneous documentation like
- flow chart
 - UML diagrams
 - process flow state diagram
 - d. all**

of the given choices

23. using proper parenthesis normally makes the code
- a. easy to read
 - b. easy to understand
 - c. less ambiguous
 - d. **all of the given options**
24. ____ was the first pure Object Oriented language in which observe pattern was used in implementing its model view controller pattern
- a. Smalltalk
 - b. PASCAL
 - c. JAVA
 - d. C++
25. the form for (;;) should be used for
- a. nested loop
 - b. empty loop
 - c. more than 1000 iteration
 - d. less than 100 iteration
26. client server is a _____ system model
- a. integrated
 - b. **distributed**
 - c. heterogeneous
 - d. Homogeneous
27. ____ architectural model is widely is used in mainframe applications.
- a. **client-server model**
 - b. Repository model
 - c. filter model
 - d. layered model
28. In client server model the application is modeled as a set of ____ **that are provided by servers.**
- a. requests
 - b. protocol
 - c. services
 - d. Requirements
29. Which one is correct?
- a. **double total=0.5;**
 - b. double total =5:

- c. double total =50:
 - d. all of the given
30. Charles Simonyi first discussed Hungarian notation.

He was of _____

- a. Microsoft
- b. IBM
- c. Dell
- d. Cisco

31. In case of header file construction is to avoid _____ error.

The construction should appear in the top of the file (before the file header).

- a. compilation
- b. run time
- c. logical
- d. all of the given

32. In case of a file servers, Client requests selected records from a _____ **and the server transmits records to client over the network.**

- a. Local memory
- b. Network
- c. Database
- d. File

33. Which of the following is used for multi-level commenting?

- a. //Comment
- b. /*Comment*/
- c. (Comment)
- d. *(Comment)*

34. Maintain ability and portability issues are discussed in _____.

- a. Functional view
- b. Physical view
- c. Code view

d. Logical view

35. Names representing methods and functions should be-----and **written in mixed case starting ----- case.**

- a. Noun ----lower
- b. Verb ----lower
- c. Noun ----upper
- d. Verb ----upper

36. C++ header files should have the extension ----- Source files can **have the extension ----**

- a. .c and .h
 - b. .cc and .h
 - c. .h and .cpp
 - d. All of the given
37. Distributing the responsibilities of different subsystems so that we get a software system which is easy to maintain, is calledthe architecture.
- a. Subtracting
 - b. Partitioning
 - c. Cloning
 - d. Balancing
38. _____ architecture model is widely used in mainframe application.
- a. Client-server Model
 - b. Repository
 - c. Filter Model
 - d. Layered Model
43. Global variables in C++ should always be referred to by using the
- a. ::operator
 - b. :operator
 - c. Without an operator
 - d. None of the given
44. Static structural model shows the major systems
- a. Leaks
 - b. Components
 - c. Activities

Objectiv

MIDTERM EXAMINATION 2010

By following modern system engineering practices simulation of reactive systems is no longer necessary

- ▶ True
- ▶ False

The state transition diagram

- ▶ depicts relationships between data objects
- ▶ depicts functions that transform the data flow
- ▶ indicates how data are transformed by the system
- ▶ indicates system reactions to external events

Control flow diagrams are

- ▶ needed to model event driven systems.
- ▶ required for all systems
- ▶ used in place of data flow diagrams
- ▶ useful for modeling user interfaces

A change becomes-----because of close presence of data and functions

- ▶ Localized (page 81)

- ▶ Private
- ▶ Global
- ▶ Accessible

A complex System evolves from a

- ▶ **smaller system** page 83
- ▶ medium sysetem
- ▶ bigger system
- ▶ non of the given

Coupling is a measure of ----- of a module or component.

- ▶ **Independent** (Page 71)
- ▶ Dependent
- ▶ Closeness
- ▶ All of the given

Software Maintenance

phase involves

- ▶ Debugging
- ▶ Adding new features
- ▶ making changes
- ▶ **all of the given**
- ▶

The hardest single part of building a software system is deciding precisely ----- to build.

- ▶ **what** (Page 17)
- ▶ How
- ▶ When
- ▶ Why

Interaction Diagrams depict the ----- Behavior of the system

- ▶ Static
- ▶ **Dynamic** (Page 106)
- ▶ Active
- ▶ Non of the above

In Sequence Diagrams the time required by the receiver object to process the message is denoted by an---

- ▶ **Activation Box** (Page 108)
- ▶ Simple Box
- ▶ Arrow
- ▶ Non of the above

A poorly designed interface can cause a user to make catastrophic errors is one of the motivations for GUI.

- ▶ **True** (Page 62)
- ▶ False

Identifying System Features include -

- ▶ log important information
- ▶ conduct business
- ▶ analyze business results
- ▶ **all of the above** (Page 98)

Establishing responsibilities

for objects includes

- ▶ Generalization Relationships
- ▶ Specialization Relationships
- ▶ **all of the above** (Page 86)
- ▶ identifying Association relationships

Which of the following is a fact finding method?

- ▶ Site visits
- ▶ **Prototyping**
- ▶ Study of similar systems
- ▶ All of given

Seven common fact-finding methods (Sampling, Research, Observation, Questionnaires, Interviews, Prototyping, Joint Requirements Planning) are introduced as a means to discover requirements.

Windows mobile is a popular mobile operating system which seen commonly on PDAs.

Which of the following category pair(s) best describe(s) it?

- ▶ **Application software, embedded software**
- ▶ system software, web-based software
- ▶ application software, scientific software
- ▶ system software, embedded software

The Object Oriented approach

- ▶ **improves the reusability of codes.** (Page 83)
- ▶ makes objects less independent.
- ▶ increases testing time.
- ▶ complicates mapping of a system model to an implementation model.

The object-oriented approach improves the maintainability, reusability, and extensibility of the software

MIDTERM EXAMINATION 2010

Most software continues to be custom built because

- ▶ Software is easier to build without using someone else's components.
- ▶ **Off the shelf software components are not commonly available**
- ▶ Component reuse is common in the software world
- ▶ Reusable components are too expensive to use

Which of the items listed below is not one of the software engineering layers?

- ▶ Tools
- ▶ Process
- ▶ **Manufacturing**
- ▶ Methods

Which of these people would not be likely to part of the FAST team?

- ▶ hardware and software engineers
- ▶ manufacturing representative
- ▶ marketing representatives
- ▶ **senior financial officers**
(Page 305)

A PRACTITIONER'S APPROACH

An state transition can only occur when triggered by a(n)_____

- ▶ actor
- ▶ collaboration attempt
- ▶ **event**
- ▶ none of the given

The evolution of the system depends on a sequence of state transitions triggered by a set of coming events.

A change becomes-----because of close presence of data and functions

- ▶ **Localized (page 81)**
- ▶ Private
- ▶ Global
- ▶ Accessible

A process which does not take input is called

- ▶ **miracle process** **Miracle": Process has output, but no input**
- ▶ core process
- ▶ secondary process
- ▶ all of the above

Data Store notation in DFDs presents.

- ▶ Data input
- ▶ Data output
- ▶ **Data input and Data output** **(Page 54)**
- ▶ non of the given

Coupling is a measure of ----- of a module or component.

- ▶ **Independent (Page 71)**
- ▶ Dependent

- ▶ Closeness
- ▶ All of the given

If a problem is identified and fixed at a-----stage in the software development process, it will cost much more than if it was fixed at and ----- stage.

- ▶ eralier & lator
- ▶ after & before
- ▶ none of the above
- ▶ later and earlier (Page 17)

Once these requirements have been finalized, the ----- process starts.

- ▶ Construction (Page 18)
- ▶ Development
- ▶ Maintenance phase
- ▶ Construction and Development

Identifying Whole-Part structures (Aggregations) means what are my -----

- ▶ Components (Page 95)
- ▶ Structures
- ▶ Objecs
- ▶ Classes

Data design actually begins during the creation of the analysis model, not the architectural model.

- ▶ True
- ▶ False

System Architecture is important to consider because it helps in making -----

- ▶ Mutual communication.
- ▶ Early design decisions.
- ▶ Reusable abstraction of a system.
- ▶ All of the above (Page 118)

Software architecture is "the -----of the components of a program/system, their interrelationships, and principles and guidelines governing their design and evolution over time."

- ▶ combination
- ▶ collection
- ▶ structure (Page 117)
- ▶ unification

Cohesion is about making sure each component does___thing and does it well.

- ▶ three
- ▶ two
- ▶ infinite
- ▶ one

Following are some statements associated with data flow diagrams. Identify the correct statement from among them.

- ▶ DFDs are used to model complex interfaces.

▶ DFDs are used to represent only functional processing, data stores and data movements between functions. DFD over flow charts(Page 2)

▶ DFDs depict only processes which can be decomposed.

▶ DFDs do not show external data sources and external data sinks

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UML (unified modeling language) analysis modeling focuses on the_____.

▶ behavioral model and environment model.

▶ behavioral model and implementation model.

▶ user model and environmental model

▶ user model and structural model A PRACTITIONER'S APPROACH
(Page 604)

The data flow diagram must be augmented by descriptive text in order to describe the functional requirements for a software product.

▶ True

▶ False

It is not possible to use ordinary data flow diagrams to model the functional requirements of real-time systems.

▶ True

▶ False [Click here for detail](#)

An object or class may further be classified on the basis of

▶ Behavior Driven attributes

▶ Data Driven attributes

▶ Responsibility Driven attributes

▶ All of the above (Page 85)

A public Interface provides a way for_____with other Classes.

▶ Communication

▶ Accessibility

▶ Reaching

▶ All of the above

DFD Notation contains

▶ Data Store

▶ External Agents

▶ Processes

▶ All of the given (Page 51)

Question No: 7 (Marks: 1)

- Please choose one A complex

System evolves from a

▶ smaller system (Page 69)

▶ medium sysetem

- ▶ bigger system
- ▶ non of the given

Question No: 8 (Marks: 1) - Please choose one
Requirement engineering mainly deals with the----- of the system

- ▶ definition phase (Page 16)
- ▶ development phase
- ▶ maintenance
- ▶ non of the above

Question No: 9 (Marks: 1) - Please choose one
Insufficient user involvement leads to products.

- ▶ unacceptable (Page 19)
- ▶ ambiguous
- ▶ dissatisfaction of customer to
- ▶ non of the above

Question No: 10 (Marks: 1) - Please choose one
Which one is not the purpose of Interaction Diagrams ?

- ▶ Model interactions between objects
- ▶ Assist in understanding how a system (a use case) actually works
- ▶ Identify responsibilities/operations and assign them to classes
- ▶ Identify dependencies among objects (Page 106)

Question No: 11 (Marks: 1) - Please choose one
Collaboration diagrams have basically two types of components: objects and -----

- ▶ messages (Page 111)
- ▶ methods
- ▶ classes
- ▶ non of the above

Question No: 12 (Marks: 1) - Please choose one
The scope description establishes the ----- between the system we are developing and everything else in the universe

- ▶ Boundary (Page 31)
- ▶ Balance
- ▶ Constraint
- ▶ None of the given

Question No: 13 (Marks: 1) - Please choose one
Which one is not a type of messages which Sequence Diagrams Depict

- ▶ Synchronous
- ▶ Asynchronous
- ▶ Create
- ▶ Update (Page 108)

Question No: 14 (Marks: 1) - Please choose one

What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?

- ▶ Security
- ▶ Reliability
- ▶ Usability
- ▶ Accuracy (not sure)

Question No: 15 (Marks: 1) - Please choose one

Given below are some statements associated with data flow diagrams. Identify the correct statement from among them.

- ▶ Data flow is made use of to model what systems do.
- ▶ Flows of data can take place from a process to a sink.
- ▶ All processes have to be levelled or decomposed.
- ▶ Context diagram shows the major system processes. (Page 54)

Question No: 16 (Marks: 1) - Please choose one

What would be the most suitable architecture to develop a commercial web page to do business transactions over the internet?

- ▶ Client server model (p129)
- ▶ Island model
- ▶ RAD model
- ▶ Repository model

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Question No: 1 (Marks: 1) - Please choose one

The first step in any OOA process model is to _____

- ▶ build an object-relationship model (p90)
- ▶ define collaborations between objects
- ▶ elicit customer requirements
- ▶ select a representation language

Question No: 2 (Marks: 1) - Please choose one

Dynamic components of an OOA model are _____

- ▶ not reusable
- ▶ sensitive to timing and event processing
- ▶ stable throughout the operational life of an application
- ▶ structural in nature

Dynamic components:

Focus on control and are sensitive to timing and event processing.

**Dynamic view of Communication,
Dynamic view of Control and**

Time.

Please choose one Events
occur whenever a(n)___

- ▶ actor and the OO system exchange information **A P R A C T I T I O N E R ' S A P P R O A C H (Page 622)**
- ▶ class operation is invoked
- ▶ messages are passed between objects
- ▶ all of the given

By following modern system engineering practices simulation of reactive systems is no longer necessary

- ▶ True
- ▶ False

Information flow represents the manner in which data and control
are related to one another.

- ▶ **change as each moves through the system. A P R A C T I T I O N E R ' S A P P R O A C H (Page 312)**
- ▶ will be implemented in the final design.
- ▶ none of the given

Which view should be consider first during software requirements analysis?

- ▶ actor view
- ▶ data view
- ▶ essential view
- ▶ **implementation view A P R A C T I T I O N E R ' S A P P R O A C H (Page**

Please choose one State Transition
Diagram is helpful in
determining

- ▶ **Business Understanding (Page 52)**
- ▶ Process Flow
- ▶ Data store
- ▶ Non of the given

Please choose one A better Design
has an objective achieve

- ▶ High Cohesion
- ▶ Low Cohesion
- ▶ Low Coupling
- ▶ **High cohesion and Low coupling A P R A C T I T I O N E R ' S A P P R O A C H (Page 316)**

Please choose one The process of
defining attributes is called

- ▶ **What I know (Page 95)**
- ▶ Who know me
- ▶ Whom I know
- ▶ Non of the above

Which of the following is not an objective for building an analysis model?

- ▶ define set of software requirements

- ▶ describe customer requirements
- ▶ **develop an abbreviated solution for the problem** **A P R A C T I T I O N E R ' S A**
P P R O A C H (Page 329)
- ▶ establish basis for software design

Please choose one The output of this design process is a description of the

- ▶ **Software Architecture** (Page 115)
- ▶ Software Code
- ▶ Software
- ▶ Non of the above

Which of these characteristics are true of a data warehouse, but not a typical database?

- ▶ business level orientation and large size
- ▶ **currency and correctness of information**
- ▶ integration and non volatility
- ▶ all of the given

Please choose one Identifying System Features include -----

- ▶ log important information
- ▶ conduct business
- ▶ analyze business results
- ▶ **all of the above** (Page 98)

The two main documents produced during this phase are Requirement Statement and Requirement pecification. They are also called Requirement Definition and -----

- ▶ **Functional Specification** (Page 25)
- ▶ Mathematical specification
- ▶ System Specification
- ▶ None of the given

Project ----- defines the concept and range of the proposed solution, and limitations identify certain capabilities that the product will not include

- ▶ **Scope** (Page 30)

In sequence Diagram events are organized in a -----time line

- ▶ **vertical** (Page 106)
- ▶ horizontal
- ▶ Vertical and Horizontal
- ▶ Non of above

choose one Asynchronous messages are denoted

- ▶ **Half Arrow** (Page 109)
- ▶ Simple Line
- ▶ Full Arrow
- ▶ Non of above

Software crisis came in 1960 what is the main reason to for the crisis

- ▶ **Software development technique (Page 4)**
- ▶ Hardware
- ▶ Software
- ▶ Non of above

central role of the software requirement Document in entire development process is related to

- ▶ Project planning
- ▶ Project tracing
- ▶ Change control
- ▶ **All of above (Page 19)**

The scope description establishes the-----between the system we are developing and everything else in the universe. Project planning

- ▶ **Boundary (Page 31)**
- ▶ System
- ▶ limitation
- ▶ None of above

which one is not the type of message which sequence diagram depict

- ▶ Synchronous
- ▶ asynchronous
- ▶ Create
- ▶ **Update (Page 108)**

cohesion is about making sure each component does -----thing and does it well

- ▶ Three
- ▶ infinite
- ▶ **one A PRACTITIONER'S APPROACH (Page 381)**
- ▶ two

what is the most important software quality of a web based transaction

- ▶ efficiency
- ▶ Modularity
- ▶ **usability**
- ▶ Security

**CS504 Midterm Solved McQs Papers By Waqar Sidhu
(File 1)**

Question No 1

What is the most crucial non-functional requirement of a system to control radiation dosages that are emitted as treatment for cancer?

Select Correct Option

Accuracy	Accuracy.....
Security	Security

Reliability
Usability

Question NO 2

Which of the following is NOT an objective for building an analysis model?

Select Correct Option

- ☐ Define of software requirements
- ☐ Describe customer requirements
- ☒ Develop an abbreviated solution for the problem...
- ☐ Establish basis for software design

Question NO 3

CRUD operations do not include_____.

Select Correct Option

- ☐ Create
- ☐ Read
- ☐ Update
- ☐ Flow

Flow.....

Question NO 4

Which of the following is not considered as benefit of prototyping?

Select Correct Option

- ☐ Provide rapid feedback from the user to the designer Help validate requirements with fewer errors
- ☐ Enhance designer and user understanding of what system should accomplish
- ☒ Prototyping is a technique that can be used to reduce customer satisfaction

Question NO 5

“isAuthorized” , “assess Performance”, are the examples of

Select Correct Option

☐ Properties

- ▶ Agreement
- ▶ Plan
- ▶ None of the given

Windows mobile is a popular mobile operating system which seen commonly on PDAs.
Which of the following category pair(s) best describe(s) it?

▶ **application software, embedded software (not sure)**

▶ system software, web-based software

▶ application software, scientific software

- ▶ system software, embedded software

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choose one Control flow diagrams are.

- ▶ Needed to model event driven systems.
- ▶ Required for all systems.
- ▶ Used in place of data flow diagrams.
- ▶ Useful for modeling user interfaces.

The Data dictionary contain description of each software.

- ▶ Configuration item.
- ▶ **Data object** **A P R A C T I T I O N E R ' S A P P R O A C H (Page 356)**
- ▶ Diagram
- ▶ Notation

A cohesion class is one which emphasize ----- unit of----.

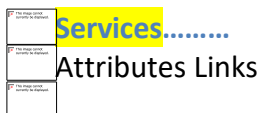
- ▶ Single and multiple.
- ▶ Multiple and functionality
- ▶ Functional and single
- ▶ **Single and functional (Page 72)**

choose one Flow charts represent.

- ▶ **Sequence. (Page 50)**
- ▶ Random
- ▶ Parallel
- ▶ Non of above

----- gives OO the ability to handle essential

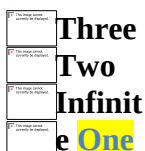
- ▶ **Decentralization (Page 80)**
- ▶ Centralization
- ▶ Decentralization and Centralization
- ▶ Non of above



Question NO 6

Cohesion is about making sure that each component does ____thing(s) and does it well.

Select Correct Option



Question NO 7

Static components of an OOA model are_____.

Select Correct Option

- ☐ Focus on control
- ☐ Not reusable
- ☐ Sensitive to timing and event processing
- ☒ Structural in nature

Question NO 8

OOA is intended to define-----, their relationship, and their behavior.

Select Correct Option

- ☐ Variables
- ☐ Classes
- ☒ Objects
- ☐
- ☐ Subjects

Question NO 9

Which of the following is not part of software architecture?

Select Correct Option

- ☒ Algorithm details
- ☐ Databases
- ☐ Data design
- ☐ Program structure

Question NO 10

The vision statement should reflect a _____ that will satisfy the need of diverse customers.

Select Correct Option

- ☒ Balanced View
- ☐ System View
- ☐ Functional View
- ☐ Non Functional View

Question NO 11

Use cases describe the system mas a _____ and hide the internal details from the user.



Black Box

White Box

Select Correct Option

Question NO 12

Software Engineering is the combination of tools. Techniques and_____.

Select Correct Option

- ☐ Testing
- ☒ Processes
- ☐ Maintenance
- ☐ Design

Question NO 13

The project manager world need_____to monitor and track the progress of the project.

Select Correct Option

- ☐ Feasibility report
- ☒ Requirement document
- ☐ Design document
- ☐ User manual

Question NO 14

A software requirement document describes all the___provided by the system along with the constraints under which it must operate?

Select Correct Option

- ☐ Conditions
- ☒ Services
- ☐ Tasks
- ☐ Actions

Question NO 15

Agreement, assignment, authorization, contract, deposit, incident etc are examples of___.

Select Correct Option

- ☐ Object
- ☐ Participant
- ☐ Tangible
- ☒ Transaction

Question NO 16

In sequence diagram, a message is represented by a/an _____

Select Correct Option

consists of three pieces of interrelated information. Which of the following is not a piece of interrelated information?

- ☐ Attributes
- ☐ Data objects
- ☐ Relationships
- ☒ Controls

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—
.

- ☒ **Arrow**
- ☐ Straight line

- ☐ Dotted line
- ☐ Box

Question NO 19

The first most important step in developing any system is to identify _____

ion NO 17

**Q
u
e
s
t**

Private Global

Accessible

Question NO 2

“A website developed to sell movies over the internet”, is_____.

Select Correct Option

☐

Scientific software

☐

Safety critical
software

☐ ☐ ☐ ☐

Web based software

Embedded software

Question NO 3

Given below are some statements associated with prototyping. Which one is correct statement?

Select Correct Option

☐ ☐ ☐ ☐

The scope and the complexity of the system can quickly expand beyond original plans. Prototyping decrease creativity since it does not allow quick feedback.

Prototypes negate the requirement for the system analysis phase.

Prototyping is a technique that can be used to reduce customer satisfaction

Question NO 4

Which of the following is not a fact finding method?

Select Correct Option

☐ ☐ ☐ ☐

Site visits

Prototyping

Study of similar system

Modulation

Question NO 5

The nature of software application can be characterized by their information_____.

Select Correct Option

☐ ☐ ☐ ☐

Complexity

Content

Determinacy

Content and determinacy

Question NO 6

A project is considered successful if:

Select Correct Option

☐ ☐ ☐

The system was delivered in time and with in budget

The system meets at least some of the customer's requirements

The system development process has a maximum impact on the ongoing business process.

☐ Minimum time was spent for requirement gathering and designing

Question NO 7

Cohesion is an internal property of a module whereas coupling is its -----with other modules.

Select Correct Option

☐	Relationship
☒	Dependence
☐	Independence
☐	Separation

Question NO 8

As per Peter Coad's methodology the following may NOT be a perfect candidate for being an object.

Select Correct
Option
Folder

☐	Bin
☐	Student
☐	s Size

Question NO 9

_____ requirement are often called product features.

Select Correct Option

☒	Functional
☐	Business
☐	User
☐	Non-functional

Question NO 10

Function oriented methodology is also known as_____.

Select Correct Option

☐	Object oriented
☐	Structure Oriented
☒	Action Orient
☐	Module Oriented

Question NO 11

Software crisis appeared in_____.

Select Correct Option

☒	Early 50s
☐	Early 60s
☐	Early 70s
☐	Early 80s

70s
Early
80s

Question NO 12

_____ is another technique to document domain knowledge.

Select Correct Option

- ☐ Business model
- ☐ Activity diagram
- ☐ Use-case model
- ☐ State transition diagram

State transition diagram

Question NO 13

Agreement, assignment, authorization, contract, deposit, incident etc are examples of ____.

Select Correct Option

- ☐ Object
- ☐ Participant
- ☐ Tangible
- ☐ Transaction

Question NO 14

Oval shape in state transition diagram represent a _____.

Select Correct Option

- ☐ State
- ☐ Input
- ☐ Output
- ☐ Data

Question NO 15

In a DFD, communication between two external agents without an intermediate process is ____.

Select Correct Option

- ☐ Prohibited
- ☐ Must
- ☐ Possible
- ☐ Avoided

Question NO 16

UML provides _____ different mechanisms to document the dynamic behavior of the system.

Select Correct Option

☒	Two
☐	Three
☐	Four
☐	Five

Question NO 17

Diagrams used to capture dynamic behavior are called _____ diagrams.

Select Correct Option

☒	Interaction
☐	Flow
☐	Sequence
☐	Collaboration

Question NO 18

In sequence diagram, a message is represented by a/an _____.

Select Correct

☐	Option
☐	Arrow
☐	Straight
☐	line Dotted
☐	line Box

Question NO 19

_____ is NOT a type of messages of sequence diagram.

Select Correct Option

☐	Synchronous
☐	Asynchronous
☐	Create
☒	Update

Question NO 20

UML(Unified Modeling Language) is among different.....which are used for documenting the object oriented design.

Select Correct Option

☐	Document
☐	Notations
☐	SRS
☐	Reports

110) is a set of processes and tools to develop software.

- i. **Software engineering**
- ii. Information
- iii. Software
- iv. None of the given

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